

# ALTERATION

## JOHNSTOWN, PENNSYLVANIA

GENERAL BUILDING DESCRIPTION:

- BUILDING TYPE: COMMERCIAL MIXED-USE

SCOPE OF WORK:

- ELECTRICAL
- PLUMBING
- MECHANICAL
- FIRE ALARM

PROJECT: COMMERCIAL MIXED-USE ALTERATION  
ADDRESS: 508 MAIN STREET, JOHNSTOWN, PA 15901

APPLICABLE CODES

- 2021 INTERNATIONAL BUILDING CODE, AS ADOPTED/AMENDED BY PA UCC.
- 2021 INTERNATIONAL EXISTING BUILDING CODE, AS ADOPTED/AMENDED BY PA UCC.
- 2021 INTERNATIONAL PLUMBING CODE, AS ADOPTED/AMENDED BY PA UCC.
- 2021 INTERNATIONAL MECHANICAL CODE, AS ADOPTED/AMENDED BY PA UCC.
- 2021 INTERNATIONAL FUEL GAS CODE, AS ADOPTED/AMENDED BY PA UCC.
- 2021 INTERNATIONAL FIRE CODE, AS REFERENCED/ADOPTED BY PA UCC.
- 2021 INTERNATIONAL ENERGY CONSERVATION CODE, AS ADOPTED/AMENDED BY PA UCC.
- 2020 NATIONAL ELECTRICAL CODE, NFPA 70, AS REFERENCED BY THE ADOPTED PA UCC CODE CYCLE.
- 2018 IBC CHAPTER 11 / 2009 ICC A117.1 ACCESSIBILITY PROVISIONS, WHERE APPLICABLE UNDER PA UCC ACCESSIBILITY ENFORCEMENT.
- CITY OF JOHNSTOWN FLOODPLAIN ORDINANCE AND LOCAL AMENDMENTS.

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OWNER SIGNATURE

REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
| 01  |      |             |          |
| 02  |      |             |          |
| 03  |      |             |          |
| 04  |      |             |          |

CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE:

ELECTRICAL

PROJECT TITLE:

COMMERCIAL MIXED-USE

DRAWING TITLE:

COVER SHEET

|                  |                    |                     |
|------------------|--------------------|---------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026 |
| JOB NO           | SHT NO<br>JS-EL-00 | SCALE:<br>NTS @A1   |



BASIC ELECTRICAL REQUIREMENTS:

1. GENERAL: ALL ELECTRICAL WORK SHALL CONFORM TO THE REQUIREMENTS OF THE NATIONAL ELECTRICAL CODE (NEC) AND NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA) STANDARDS UNLESS MORE STRINGENT REQUIREMENTS ARE INDICATED. ALL ELECTRICAL WORK SHALL COMPLY WITH ADA RECOMMENDATIONS. ALL ELECTRICAL WORK SHALL BE PERFORMED IN A NEAT AND WORKMAN LIKE MANNER BY A LICENSED ELECTRICIAN, OR A CERTIFIED APPRENTICE WORKING UNDER THE DIRECT SUPERVISION OF A LICENSED ELECTRICIAN, USING THE BEST METHODS KNOWN TO THE TRADE AND SHALL PRESENT A NEAT AND PROFESSIONAL APPEARANCE WHEN COMPLETED. THE OWNER RESERVES THE RIGHT TO CHANGE, WITHOUT ADDITIONAL COST, THE LOCATION OF ANY APPARATUS OR OUTLET, PROVIDED SUCH CHANGE DOES NOT ADD MORE THAN 10 FT TO THE FEEDER AND IS ORDERED BEFORE THE AFFECTED PORTION OF THE WORK IS COMMENCED. THE CONTRACTOR SHALL BRING ALL CONFLICTS ON THE DRAWINGS TO THE OWNERS ATTENTION FOR HIS RESOLUTION PRIOR TO PERFORMING THAT WORK.
2. THE CONTRACTOR SHALL SURVEY THE PROJECT SITE PRIOR TO THE BID TO ASSESS ACTUAL FIELD CONDITIONS, FAILURE TO PERFORM THIS INSPECTION BINDS THE CONTRACTOR TO PERFORM THE WORK WITHOUT EXTRA CHARGES DESPITE THE IGNORANCE OF REASONABLY ANTICIPATED CONDITIONS.
3. ROUGH-INS: THE CONTRACTOR SHALL VERIFY AND COORDINATE THE ROUGH-IN REQUIREMENTS OF EACH ITEM OF EQUIPMENT WITH THE CONTRACTOR SUPPLYING THE EQUIPMENT.
4. INSTALLATION: THE ELECTRICAL DRAWINGS INDICATE THE EXTENT AND GENERAL LOCATION AND ARRANGEMENT OF EQUIPMENT AND MATERIALS. THE CONTRACTOR SHALL BECOME FAMILIAR WITH ALL DETAILS OF THE WORK AND VERIFY ALL DIMENSIONS IN THE FIELD SO THAT EQUIPMENTS AND MATERIALS WILL BE PROPERLY LOCATED AND READILY ACCESSIBLE. THE CONTRACTOR SHALL SEQUENCE, COORDINATE, AND INTEGRATE THE VARIOUS ELEMENTS OF ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS, COMPLY WITH THE FOLLOWING REQUIREMENTS:
- A. VERIFY ALL DIMENSIONS BY FIELD MEASUREMENTS.
- B. COORDINATE ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS INSTALLATION WITH OTHER BUILDING COMPONENTS AND TRADES.
- C. ARRANGE FOR CHASES, SLOTS, AND OPENINGS IN OTHER BUILDING COMPONENTS DURING PROGRESS OF CONSTRUCTION, TO ALLOW FOR INSTALLATION OF ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS.
- D. SEQUENCE, COORDINATE, AND INTEGRATE INSTALLATION OF ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS FOR EFFICIENT FLOW OF THE WORK.
- E. WHERE MOUNTING HEIGHTS ARE NOT INDICATED, INSTALL ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS TO PROVIDE MAXIMUM HEADROOM POSSIBLE.

- F. INSTALL ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS TO CONFORM WITH APPROVED SUBMITTAL DATA TO THE GREATEST EXTENT POSSIBLE. CONFORM TO THE ARRANGEMENTS INDICATED ON THE ELECTRICAL DRAWINGS, RECOGNIZING THAT PORTIONS OF THE WORK ARE SHOWN ONLY IN DIAGRAMATIC FORM. WHERE COORDINATION REQUIREMENTS CONFLICT WITH INDIVIDUAL SYSTEM REQUIREMENTS, REFER CONFLICT TO THE ARCHITECT/ENGINEER OR OWNER'S REPRESENTATIVE FOR RESOLUTION.
- G. IN GENERAL, INSTALL ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS LEVEL AND PLUMB, PARALLEL AND PERPENDICULAR TO BUILDING LINES AND/OR FEATURES AND/OR OTHER BUILDING SYSTEMS.

- H. INSTALL ELECTRICAL EQUIPMENT TO FACILITATE SERVICING, MAINTENANCE, AND REPAIR OR REPLACEMENT OF EQUIPMENT AND COMPONENT PARTS, AS MUCH AS PRACTICAL, CONNECT EQUIPMENT FOR EASE OF DISCONNECTING, WITH MINIMUM OF INTERFERENCE WITH OTHER INSTALLATIONS.

- I. INSTALL ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS GIVING RIGHT-OF-WAY PRIORITY TO SYSTEMS REQUIRED TO BE INSTALLED AT A SPECIFIC SLOPE (INCL. SPRINKLER SYSTEMS).

5. CUTTING AND PATCHING: ALL ELECTRICAL WORK SHALL BE CAREFULLY LAID OUT IN ADVANCE, AND WHERE CUTTING, CHANNELING, CHASING, OR DRILLING OF FLOORS, WALLS, PARTITIONS, CEILINGS, OR OTHER SURFACES IS NECESSARY FOR THE PROPER INSTALLATION, SUPPORT, OR ANCHORAGE OF CONDUIT OR OTHER ELECTRICAL WORK, THIS WORK SHALL BE CAREFULLY DONE. ANY RESULTING DAMAGE TO THE BUILDING OR OTHER SYSTEMS, EQUIPMENT, OR MATERIALS SHALL BE REQUIRED BY SKILLED MECHANICS OF THE TRADES INVOLVED, AT NO ADDITIONAL COST TO THE OWNER.

6. PRODUCTS: SYSTEMS, EQUIPMENT, AND MATERIALS DESCRIBED ON THE ELECTRICAL DRAWINGS ESTABLISH THE MINIMUM STANDARDS FOR QUALITY AND STYLE AND SHALL BE THE BASIS OF THE BID. ALL SYSTEMS, EQUIPMENT, AND MATERIALS SHALL BE NEW AND SHALL BEAR THE UL LABEL OR BE UL LISTED, WHERE APPLICABLE, AND SHALL BE INSTALLED IN ACCORDANCE WITH THE NEC AND NEMA STANDARDS.

7. SUBSTITUTIONS: WHERE SYSTEMS, EQUIPMENT, OR MATERIALS ARE SPECIFIED BY MANUFACTURER OR BRAND NAME AND CATALOG NUMBER, SUCH SPECIFICATION SHALL ESTABLISH THE MINIMUM STANDARDS FOR QUALITY AND STYLE AND SHALL BE THE BASIS OF THE BID. SYSTEMS, EQUIPMENT, AND MATERIALS SO SPECIFIED SHALL BE FURNISHED UNDER THE CONTRACT UNLESS CHANGED BY WRITTEN AGREEMENT. SHOULD THE CONTRACTOR PROPOSE TO FURNISH PRODUCTS OTHER THAN THOSE SPECIFIED, AS PERMITTED BY THE APPROVED EQUIPMENT CLAUSES, HE SHALL SUBMIT A WRITTEN REQUEST FOR SAID SUBSTITUTIONS THROUGH APPROPRIATE CHANNELS TO THE ARCHITECT/ENGINEER FOR HIS REVIEW. SUCH REQUEST SHALL BE ACCOMPANIED WITH COMPLETE DESCRIPTIVE LITERATURE, INCLUDING, BUT NOT LIMITED TO, CATALOG CUT SHEETS, BROCHURES, CIRCULARS, SPECIFICATIONS, PERFORMANCE DATA, INSTALLATION INSTRUCTIONS, SHOP DRAWINGS, AND OTHER PRINTED INFORMATION IN SUFFICIENT DETAIL AND SCOPE TO VERIFY COMPLIANCE WITH THE REQUIREMENTS OF THE CONTRACT. DESCRIPTIVE LITERATURE ON PROPOSED SUBSTITUTIONS SHALL BE CLEAR, CONCISE, AND LOGICALLY ARRANGED. ALL DATA WHICH IS, AND IS NOT, APPLICABLE SHALL BE CLEARLY IDENTIFIED AS SUCH. IF REQUESTED, THE CONTRACTOR SHALL SUBMIT SAMPLES OF BOTH SPECIFIED AND PROPOSED ITEMS FOR INSPECTION. DESCRIPTIVE LITERATURE ON PROPOSED SUBSTITUTIONS SHALL BE RETURNED WITHOUT REVIEW IF NOT PROPERLY PREPARED. ACCEPTANCE OR REJECTION OF PROPOSED SUBSTITUTIONS SHALL BE UP TO THE DESCRIPTION OF THE ARCHITECT/ENGINEER AND/OR THE OWNER.

8. SUBMITTALS: THE CONTRACTOR SHALL FOLLOW THE GENERAL PROVISIONS OF THE CONTRACT AND ESTABLISHED PROCEDURES. SUBMITTALS SHALL CONSIST OF COMPLETE DESCRIPTIVE LITERATURE INCLUDING, BUT NOT LIMITED TO, CATALOG CUT SHEETS, BROCHURES, CIRCULARS, SPECIFICATIONS, PERFORMANCE DATA, INSTALLATION INSTRUCTIONS, SHOP DRAWINGS, AND OTHER PRINTED INFORMATION IN SUFFICIENT DETAIL AND SCOPE TO VERIFY COMPLIANCE WITH THE REQUIREMENTS OF THE CONTRACT. DESCRIPTIVE LITERATURE SHALL BE CLEAR, CONCISE, AND LOGICALLY ARRANGED. ALL DATA WHICH IS, AND IS NOT, APPLICABLE SHALL BE CLEARLY IDENTIFIED AS SUCH. IF REQUESTED, THE CONTRACTOR SHALL SUBMIT SAMPLES OF SPECIFIED ITEMS FOR INSPECTION. DESCRIPTIVE LITERATURE SHALL BE RETURNED WITHOUT REVIEW IF NOT PROPERLY PREPARED. THE FOLLOWING SYSTEMS, EQUIPMENT, AND MATERIALS, AS A MINIMUM, REQUIRE SUBMITTALS:

- A. ANY PROPOSED SUBSTITUTIONS:
- B. WIRING DEVICES.
- C. PANELBOARDS.
- D. DISCONNECT SWITCHES.
- E. CIRCUIT BREAKERS.
- F. LIGHTING FIXTURES INCLUDING BALLASTS.

9. RECORD DRAWINGS: THE CONTRACTOR SHALL MAINTAIN AT THE SITE A CLEAN, UNDAMAGED SET OF BLUE- OR BLACK-LINE WHITE PRINTS OF CONTRACT DRAWINGS. THIS RECORD SET OF CONTRACT DRAWINGS SHALL BE MARKED TO SHOW THE ACTUAL INSTALLATION AND WHERE THE ACTUAL INSTALLATION VARIES SUBSTANTIALLY FROM THE ELECTRICAL WORK AS ORIGINALLY SHOWN, MARK WHICHEVER DRAWINGS ARE MOST CAPABLE OF SHOWING CONDITIONS FULLY AND ACCURATELY, GIVE PARTICULAR ATTENTION TO CONCEALED ELEMENTS THAT WOULD BE DIFFICULT TO MEASURE, AND RECORD AT A LATER DATE. MARK RECORD DRAWINGS WITH A RED ERASABLE PENCIL; USE OTHER COLORS TO DISTINGUISH BETWEEN VARIATIONS IN SEPARATE PARTS OF THE ELECTRICAL WORK. NOTE CONTRACT MODIFICATIONS AND APPROVED SUBSTITUTIONS WHERE APPLICABLE.

10. PROTECTION OF INSTALLED SYSTEMS, EQUIPMENT, AND MATERIALS: PROTECT INSTALLED SYSTEMS, EQUIPMENT, AND MATERIALS FROM DAMAGE UNTIL FINAL ACCEPTANCE BY THE OWNER. REPAIR OR REPLACE, AT NO ADDITIONAL COST TO THE OWNER, DAMAGED SYSTEMS, EQUIPMENT, AND MATERIALS TO THE SATISFACTION OF THE ARCHITECT/ENGINEER AND/OR OWNER.

11. CLEANING: UPON COMPLETION OF INSTALLATION, INSPECT INTERIOR AND EXTERIOR OF ALL ELECTRICAL EQUIPMENT, REMOVE PAINT SPATTERS AND OTHER SPOTS, DIRT, AND DEBRIS. TOUCH-UP SCRATCHES AND MARKS OF FINISH TO MATCH ORIGINAL FINISH.

12. CERTIFICATIONS: THE FOLLOWING SHALL BE OBTAINED AND SUBMITTED TO THE OWNER PRIOR TO FINAL PAYMENT:

- A. ELECTRICAL SYSTEMS: A CERTIFICATE OF FINAL INSPECTION AND APPROVAL BY THE AUTHORITIES HAVING JURISDICTION.

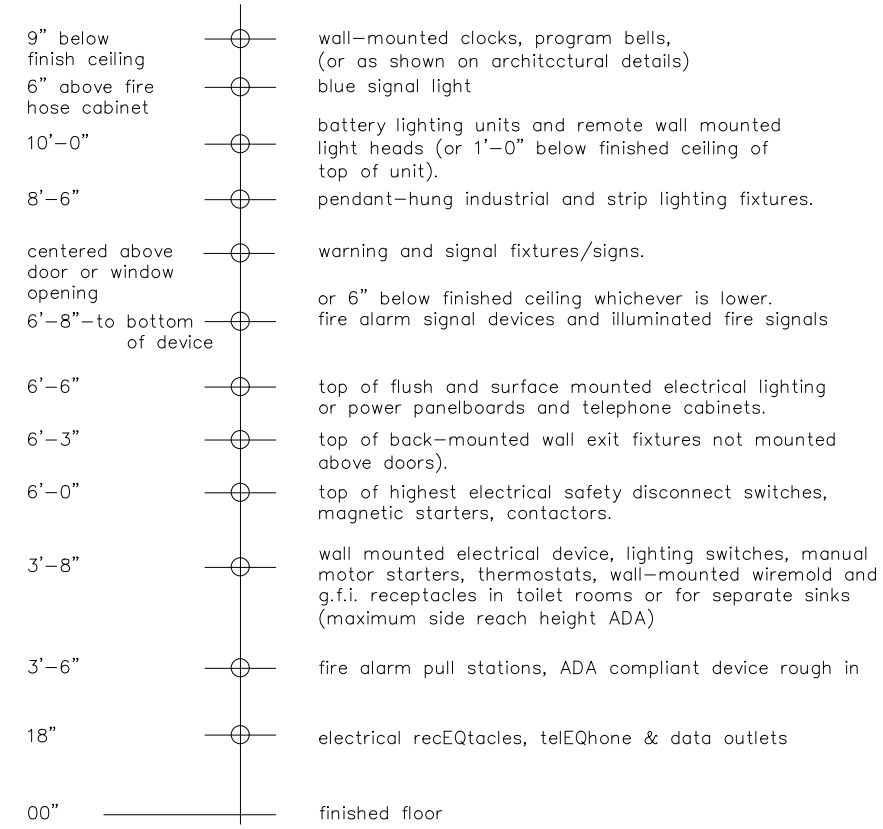
13. GUARANTEE: THE CONTRACTOR SHALL SUBMIT A WRITTEN GUARANTEE TO THE OWNER, PRIOR TO FINAL PAYMENT, THAT WARRANTS THE INSTALLATION SHALL REMAIN FREE OF DEFECTS FOR A PERIOD OF ONE YEAR AFTER FINAL ACCEPTANCE BY THE OWNER. THE GUARANTEE SHALL STATE THAT THE OWNER IS NOT LIABLE FOR PARTS AND LABOR COSTS INCURRED BY THE CONTRACTOR IN THE REPAIR OF ACTUAL PRODUCT OR INSTALLATION DEFECTS. THE GUARANTEE SHALL STATE THAT THE ON-SITE RESPONSE FOR ASSISTANCE REQUESTS WILL BE 24 HOURS FOR NON-EMERGENCY CONDITIONS AND 2 HOURS FOR EMERGENCY CONDITIONS.

ELECTRICAL MATERIALS:

1. RACEWAYS:
- A. RIGID GALVANIZED STEEL (RGS) CONDUIT: ANSI C80.1.
- B. ELECTRICAL METALLIC TUBING (EMT) AND FITTINGS: ANSI C80.3 WITH COMPRESSION TYPE FITTINGS.
- C. LIQUIDTIGHT FLEXIBLE METAL CONDUIT: UL 360. FLEXIBLE STEEL CONDUIT WITH PVC JACKET.
- D. NONMETALLIC CONDUIT AND TUBING (ENT): NEMA TC 13.
2. BOXES:
- A. SHEET METAL: NEMA OS 1.
- B. CAST METAL: NEMA FB 1, TYPE FD, CAST FRALLY BOX WITH CASKETED COVER.
- C. RINGED COVER ENCLOSURES: NEMA 250. GALVANIZED STEEL ENCLOSURE WITH CONTINUOUS HINGE COVER, QUICK RELEASE TYPE LATCHES, REMOVABLE INTERIOR PANEL, AND MANUFACTURER'S STANDARD GRAY ENAMEL INSIDE AND OUT.
3. WIRES AND CABLES:
- A. CONDUCTOR MATERIAL: ANNEALED COPPER.
- B. INSULATION: THHN/THWN CONFORMING TO WC 5.
- C. CONDUCTORS #10 AWG AND SMALLER SHALL BE SOLID; CONDUCTORS #14 AWG AND LARGER SHALL BE STRANDED. MC CABLE IS PERMITTED FOR BRANCH CIRCUITRY, AND SHALL UTILIZE SOLID CONDUCTORS.
- D. GROUND CONDUCTORS #10 AWG AND SMALLER SHALL HAVE GREEN THHN/THWN INSULATION.
4. WIRING DEVICES:
- A. GENERAL: COMPLY WITH NEMA WD 1, "GENERAL PURPOSE WIRING DEVICES."
- B. COLOR: WHITE, BLACK, GRAY, IVORY, OR BROWN TO BE AS SELECTED BY THE ARCHITECT/ENGINEER AND/OR THE OWNER.
- C. RECEPTACLES: COMPLY WITH UL 498, "ELECTRICAL ATTACHMENT PLUGS AND RECEPTACLES."
- D. "HEAVY DUTY SPECIFICATION GRADE EXCEPT AS OTHERWISE INDICATED, GROUND FAULT CIRCUIT INTERRUPTER (GFCI) TYPE RECEPTACLES SHALL COMPLY WITH UL 943, "GROUND FAULT CIRCUIT INTERRUPTERS," WITH INTEGRAL NEMA 5-20R DUPLEX RECEPTACLE DESIGNED FOR INSTALLATION IN A 2-3/4" DEEQ DEVICE BOX WITHOUT ADAPTER.
- E. COMPLY WITH UL 20, "GENERAL USE SNAP SWITCHES," SINGLE-POLE, TWO-POLE, THREE-WAY, AND FOUR-WAY AS INDICATED AND/OR REQUIRED.
- F. CORRESPONDING WIRING DEVICES: SINGLE AND COMBINATION TYPES WHICH MATE AND MATCH WITH CORRESPONDING WIRING DEVICES.
- G. MANUFACTURER: SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE PRODUCTS AS MANUFACTURED BY HUBBELL INC. OR APPROVED EQUAL BY LEVITON.
5. GROUNDING:
- A. GROUNDING AND BONDING PRODUCTS: OF TYPES INDICATED AND/OR OF SIZES AND RATINGS TO COMPLY WITH THE NEC, WHERE TYPES, SIZES, RATINGS, AND QUANTITIES ARE IN EXCESS OF NEC REQUIREMENTS, THE MORE STRINGENT REQUIREMENTS AND THE GREATER SIZE, RATINGS, AND QUANTITY INDICATIONS SHALL GOVERN. SMOOTH MATCHING NYLON IN ALL AREAS.
- B. CONDUCTOR MATERIAL: COPPER.
- C. WIRE AND CABLE CONDUCTORS: CONFIRM TO NEC TABLE B, EXCEPT AS OTHERWISE INDICATED, FOR CONDUCTOR PROPERTIES, INCLUDING STRANDING.
- D. MATERIALS USED: CONNECTOR PRODUCTS: UL LISTED AND LABELED AS GROUNDING CONNECTORS FOR THE CONNECTOR PRODUCTS.
6. PANELBOARDS:
- A. CIRCUIT BREAKERS: PROVIDE TYPE, RATING, AND FEATURES INDICATED, BOLT-ON EXCEPT WHERE PLUG-IN FOR USE ON EXISTING PANELBOARDS (NOT BEING UPGRADED) TANDEM CIRCUIT BREAKERS SHALL NOT BE USED. MULTIPLE CIRCUIT BREAKERS SHALL HAVE AN INTERNAL COMMON TRIP AND A SINGLE HANDLE.
- B. ENCLOSURES: NEMA TYPE 1, UNLESS OTHERWISE INDICATED.
- C. FRONT: SECURED TO BOX WITH CONCEALED THUMB CLAMPS EXCEPT AS INDICATED, FRONT FOR.
- D. SURFACE MOUNTED PANELBOARDS SHALL BE SAME DIMENSIONS AS BOX.
- E. DIRECTORY FRAME: METAL WITH CLEAR PLASTIC COVER MOUNTED ON INSIDE OF PANELBOARD DOOR.
- F. BUS WORK: HARD DRAWN COPPER OF 95% CONDUCTIVITY.
- G. MAIN AND NEUTRAL PLUGS: COMPRESSION TYPE.
- H. EQUIPMENT GROUND BUS: ADEQUATE FOR FEEDER AND BRANCH CIRCUIT EQUIPMENT GROUND.
- I. CONDUCTORS: BONDED TO BOX.
- J. PROVISIONS FOR FUTURE DEVICES: EQUIP WITH MOUNTING BREAKETS, BUS CONNECTION, AND NECESSARY APPURTENANCES, FOR THE CIRCUIT BREAKER AMPERE RATING INDICATED FOR FUTURE INSTALLATION OF DEVICES.
- K. MAIN AND SUFFERED LUGS: PROVIDE WHERE INDICATED.
- L. NAMEPLATE: CUSTOM ENGRAVED PLASTIC LAMINATE, WHITE LETTERS ON BLACK FIELD, FOR EACH PANELBOARD MOUNTED WITH EPOXY OR INDUSTRIAL CEMENT OR ADHESIVE.
- M. MANUFACTURER: SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE PRODUCTS AS MANUFACTURED BY SQUARE-D, SIEMENS OR CUTLER HAMMER CO.
7. DISCONNECTS AND CIRCUIT BREAKERS:
- A. FUSIBLE SWITCH, 800A AND SMALLER: NEMA KS 1, TYPE HD, CLIPS TO ACCOMMODATE SPECIFIED.
- B. FUSES, ENCLOSURE SUITABLE FOR THE ENVIRONMENT WHERE INSTALLED, HANDLE LOCKABLE WITH TWO (2) PAD LOCKS, AND INTERLOCKED WITH COVER IN "CLOSED" POSITION. ENCLOSURES SHALL COMPLY WITH NEMA KS 1, TYPE 1 INDOOR DRY LOCATIONS.
- C. MOLDED CASE CIRCUIT BREAKER: NEWMAN AB 1, HANDLE LOCKABLE WITH TWO (2) PADLOCKS.
- D. FRAME SIZE, TRIP RATING NUMBER OF POLES, AND AUXILIARY DEVICES AS INDICATED.
- E. ENCLOSURES SHALL COMPLY WITH NEWMAN AB 1, TYPE 1 INDOOR DRY LOCATIONS.
- F. CIRCUIT BREAKERS SHALL HAVE A MINIMUM INTERRUPTING CAPACITY AS FOLLOWS:
- G. MANUFACTURER: SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE PRODUCTS AS SUITABLE FOR INSTALLATION IN EXISTING PANELS. DISCONNECTS SHALL BE BY SQUARE-D.

STANDARD MOUNTING HEIGHTS

(UNLESS OTHERWISE NOTED)



MOUNTING HEIGHT NOTES:

1. STANDARD MOUNTING HEIGHTS: (COORDINATE WITH ARCH DRAWINGS)
- ALL MOUNTING HEIGHTS SHALL BE AS INDICATED BY ARCHITECT, IF NOT INDICATED BY ARCHITECT THEN PROVIDE AS NOTED ABOVE.
2. MOUNTING HEIGHTS TO CENTER OF OUTLETS UNLESS OTHERWISE NOTED. IN MASONRY CONSTRUCTION THE ABOVE MOUNTING HEIGHTS SHALL BE USED FOR REFERENCE TO NEAREST BLOCK OR BRICK COURSEING.
3. THE ABOVE MOUNTING HEIGHT SHALL BE ADHERED TO UNLESS SPECIFICALLY NOTED OR DETAILED OTHERWISE ON THE DRAWING OR SPECIFICATIONS.
4. INDICATION (+) NEXT TO A DEVICE INDICATES THAT DEVICE IS MOUNTED ABOVE A COUNTER OR CASEWORK. COORDINATE WITH ARCHITECTURAL DETAIL AND CASEWORK CONTRACTOR.
5. 3"-6" FOR ADA COMPLIANT DEVICES VERIFY EXACT HEIGHT PRIOR TO ROUGH IN.

ELECTRICAL NOTES:

1. WIRING METHODS: WIRING SHALL CONSIST NO MORE (NON-METALLIC) OR CABLES AND WIRES INSTALLED IN RGS CONDUIT, EMT, LIQUIDTIGHT FLEXIBLE METAL CONDUIT AND MC CABLES. RACEWAYS SHALL BE CONCEALED WITHIN FINISHED WALLS AND CEILINGS UNLESS OTHERWISE INDICATED. OTHER THAN ROOF PENETRATIONS REQUIRED TO FEED ROOF MOUNTED EQUIPMENT, RACEWAYS WILL NOT BE ROUTED EXPOSED OUTSIDE OF THE BUILDING.
2. RACEWAYS: RACEWAYS SHALL BE PROVIDED WHERE INDICATED AND REQUIRED AND SHALL BE INSTALLED AS SPECIFIED BELOW, UNLESS OTHERWISE INDICATED. MINIMUM RACEWAY SIZE SHALL BE 3/4 IN. RGS CONDUIT SHALL BE USED FOR ALL OUTDOOR INSTALLATIONS. EMT SHALL BE USED FOR ALL INDOOR INSTALLATIONS. LIQUIDTIGHT FLEXIBLE METAL CONDUIT, 6 FT MAXIMUM LENGTH, SHALL BE USED FOR ALL CONDUIT TERMINATIONS AT EQUIPMENT SUBJECT TO VIBRATION, BUSHINGS, MANUFACTURED FITTINGS, OR BOXES PROVIDING EQUIVALENT MEANS OF PROTECTION SHALL BE INSTALLED ON THE ENDS OF ALL CONDUITS AND SHALL BE OF THE INSULATING TYPE WHERE REQUIRED BY THE N.E.C. ONLY LISTED ADAPTERS SHALL BE USED TO CONNECT EMT TO RGS CONDUIT AND CAST METAL BOXES AND CONDUIT BODIES. PENETRATIONS OF SLABS AND FIRE RATED WALLS SHALL BE FIRESTONE.
- KEEP RACEWAYS AT LEAST 6 IN. AWAY FROM PARALLEL RUNS OF FLUES AND STEAM OR HOT WATER PIPING. INSTALL HORIZONTAL RACEWAY RUNS HIGHER THAN WATER AND STEAM PIPING. RACEWAYS CROSSING STRUCTURAL EXPANSION JOINT SHALL BE PROVIDED WITH SUITABLE EXPANSION FITTINGS OR OTHER SUITABLE MEANS TO COMPENSATE FOR THE BUILDING EXPANSION AND CONTRACTION. RACEWAYS SHALL BE INSTALLED PARALLEL OR PERPENDICULAR TO WALLS, STRUCTURAL MEMBERS AND FEATURES, MECHANICAL DUCT AND PIPING SYSTEMS, OR INTERSECTIONS OF VERTICAL PLANES AND CEILINGS. CHANGES IN DIRECTION OF RUNS SHALL BE ACCOMPISHED WITH SYMMETRICAL BENDS OR CAST METAL FITTINGS. FLEED-MADE ELBOWS AND OFFSETS SHALL BE MADE WITH AN APPROVED HICKEY OR CONDUIT BENDING MACHINE. CRUSHED OR DEFORMED RACEWAY SHALL NOT BE INSTALLED. CARE SHALL BE TAKEN TO PREVENT THE LODGMENT OF DIRT, AND CONSTRUCTION MATERIALS AND DEBRIS IN RACEWAYS. DURING THE COURSE OF CONSTRUCTION, CLOSED RACEWAYS SHALL BE ENTIRELY FREED OF OBSTRUCTIONS OR SHALL BE REPLACED.

- RGS CONDUIT AND EMT SHALL BE SECURELY AND RIGIDLY FASTENED IN PLACE AT INTERVALS OF NOT MORE THAN 4 FT WITH APPROVED WIRE STRAPS. BOXES FOR METALLIC RACEWAYS, RACEWAYS, CONDUIT CLAMPS, CONDUIT HANGERS, THREADED C-CLAMPS, OR CEILING TRAPEZE, C-CLAMPS AND BUSHINGS SHALL BE INSTALLED IN ACCORDANCE WITH THE N.E.C. AND THE FOLLOWING: RACEWAYS SHALL BE COORDINATED WITH SUPPORTING STRUCTURES TO PREVENT DEFORMATION OR DAMAGE TO STRUCTURES, BUT NO LOAD SHALL BE APPLIED TO JOIST BRIGING. FASTENINGS SHALL BE BY WOOD SCREWS OR SCREW TYPE NAILS TO WOOD, BY TOGGLE BOLTS ON HOLLOW DUAL, EXPANSION BOLTS ON CONCRETE OR BRICK, AND BY MACHINE SCREWS, WELDED THREADED STUDS, HEAT TREATED OR SPRING STEEL TENSION CLAMPS ON STEEL WORK. NAIL TYPE NYLON ANCHORS OR THREADED STUDS DRIVEN INTO A POWDER CHARGE AND PROVIDED WITH LOCK WASHERS AND NUTS MAY BE USED IN LIEU OF EXPANSION BOLTS OR MACHINE SCREWS. RACEWAYS OR PIPE STRAPS SHALL NOT BE WELDED TO STEEL STRUCTURES. IN PARTITIONS OF LIGHT STEEL CONSTRUCTION, SHEET METAL SCREWS MAY BE USED. CONDUIT SHALL NOT BE SUPPORTED USING WIRE OR NYLON TIES. RACEWAYS SHALL BE INSTALLED AS A COMPLETE SYSTEM AND BE INDEPENDENTLY SUPPORTED FROM THE STRUCTURE. UPPER RACEWAYS SHALL NOT BE THE SUPPORT OF LOWER RACEWAYS. SUPPORTING MANS WILL NOT BE SHARED BETWEEN ELECTRICAL RACEWAYS AND MECHANICAL DUCTS OR PIPING. MOUNTING HARDWARE SHALL NOT PRESENT SHARP EDGES WHERE PERSONNEL CONTACT IS POSSIBLE. IN MECHANICAL SPACES, "MINERALS" SUPPORTS SHALL NOT BE USED BELOW 10' A.F.F.

3. BOXES: BOXES SHALL BE PROVIDE IN RACEWAY SYSTEMS WHEREVER REQUIRED FOR PULLING OF WIRES, MAKING CONNECTIONS, AND MOUNTING OF DEVICES OR LIGHTING FIXTURES. IN GENERAL, BOXES SHALL BE CONSTRUCTED OF NOT-DIPPED GALVANIZED FINISHED SHEET METAL. BOXES FOR METALLIC RACEWAYS, 4 IN. BY 4 IN. NOMINAL SIZE AND SMALLER, SHALL BE OF THE CAST METAL HUB TYPE AND CASKETING WHEN LOCATED OUTSIDE OF THE BUILDING. BOXES SHALL BE LISTED AS SUITABLE FOR THE ENVIRONMENTAL CONDITIONS OF THE LOCATION THEY ARE INSTALLED. BOXES FOR MOUNTING OF LIGHTING FIXTURES SHALL BE NOT LESS THAN 4 IN. SQUARE EXCEPT SMALLER BOXES SHALL BE INSTALLED WHERE REQUIRED BY FUTURE CONFIGURATION. UNLESS OTHERWISE INDICATED, DEVICE BOXES FOR RECEPTACLES SHALL BE MOUNTED WITH THE CENTER OF THE DEVICE BOX APPROXIMATELY 18 IN. AFF.
- INDICATED, DEVICE BOXES FOR TOGGLE SWITCHES SHALL BE MOUNTED WITH THE CENTER OF THE DEVICE BOX APPROXIMATELY 48 IN. AFF. BOXES AND BOX SUPPORTS SHALL BE FASTENED TO WOOD WITH WOOD SCREWS OR SCREW TYPE NAILS OF EQUAL HOLDING STRENGTH, WITH BOLTS AND METAL EXPANSION SHELDOS ON CONCRETE AND BRICK, WITH TOGGLE BOLTS ON HOLLOW DUAL, AND WITH MACHINE SCREWS OR WELDED STUDS ON STEEL WORK. THREADED STUDS DRIVEN IN BY POWDER CHARGE AND PROVIDED WITH LOCKWASHERS AND NUTS, OR NAIL TYPE NYLON ANCHORS MAY BE USED IN LIEU OF EXPANSION SHELDOS OR MACHINE SCREWS. HANGERS SHALL NOT BE FASTENED OR SUPPORTED FROM JOIST BRIDGING.

4. WIRES AND CABLES: EXAMINE RACEWAYS AND BUILDING FINISHES TO RECEIVE WIRES AND CABLES FOR COMPLIANCE WITH INSTALLATION TOLERANCES AND OTHER CONDITIONS. DO NOT PROCEED WITH INSTALLATION UNTIL UNSATISFACTORY CONDITIONS HAVE BEEN CORRECTED. PULL WIRES AND CABLES INTO RACEWAY SIMULTANEOUSLY WHERE MORE THAN ONE IS BEING INSTALLED IN THE SAME RACEWAY. USE PULLING COMPOUND OR LUBRICANT WHERE NECESSARY. COMPOUND USED MUST NOT DEGRADATE CONDUCTORS OR INSULATION. USE PULLING MEANS, INCLUDING FISH TAPE, CABLE, ROPE, AND BASKET-WEAVE WIRE/CABLE DRIPS THAT WILL NOT DAMAGE WIRES/CABLES OR RACEWAY. INSTALL EXPOSED CABLE, PARALLEL AND PERPENDICULAR TO WALLS, STRUCTURAL MEMBERS AND FEATURES, MECHANICAL DUCT AND PIPING SYSTEMS, OR INTERSECTIONS OF VERTICAL PLANES AND CEILING. HORIZONTAL RUNS OF MC CABLE SHALL BE SUPPORTED ON 3 FT MAXIMUM CENTERS. VERTICAL RUNS OF MC CABLES SHALL BE SUPPORTED ON 6 FT MAXIMUM CENTERS EXPOSED PLENUM CABLE SHALL BE SUPPORTED ON 3 FT MAXIMUM CENTERS. THE NUMBER OF SPLICES SHALL BE KEPT TO AN ABSOLUTE MINIMUM. WIRING AT EACH OUTLET SHALL BE INSTALLED WITH AT LEAST 8 IN. SLACK.

5. WIRING DEVICES: INSTALL WIRING DEVICES WHERE INDICATED IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS. THE NEC, USED RECOGNIZED INDUSTRY PRACTICES. INSTALL WIRING DEVICES IN DEVICE BOXES WHICH ARE CLEAN AND FREE FROM DIRT AND CONSTRUCTION MATERIALS AND DEBRIS. INSTALL WIRING DEVICES AFTER WIRING WORK IS COMPLETE. INSTALL DEVICE PLATES AFTER PAINTING WORK IS COMPLETE. TIGHTEN ELECTRICAL CONNECTORS AND TERMINALS, INCLUDING SCREWS AND BOLTS, IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED TORQUE-TIGHTENING VALUES. PRIOR TO EMERGENCY RESPONSE, TEST WIRING FOR ELECTRICAL CONTINUITY AND SHORTS. ENSURE PROPER POLARITY OF CONNECTIONS IS MAINTAINED. SUBSEQUENT TO EMERGING, TEST WIRING DEVICES AND DEMONSTRATE COMPLIANCE WITH REQUIREMENTS.

6. GROUNDING: ELECTRICAL SYSTEMS AND EQUIPMENT, METALLIC RACEWAYS AND BOXES, CABLE SHIELDS, METALLIC CABLE SHEATHS AND ARMOR, AND OTHER NON-CURRENT CARRYING METALLIC PARTS OR EQUIPMENT SHALL BE GROUNDED IN CONFORMANCE WITH THE NEC. EQUIPMENT GROUND CONDUCTORS SHALL COMPLY WITH NEC ARTICLE 250 FOR SIZES AND QUANTITIES, EXCEPT WHERE LARGER SIZES AND/OR MORE CONDUCTORS ARE INDICATED. PROVIDE SEPARATE INSULATED GROUND CONDUCTOR IN ALL RACEWAYS, REGARDLESS OF RACEWAY TYPE. SEPARATELY DERIVED SYSTEMS AS DEFINED BY THE NEC SHALL BE GROUNDED IN CONFORMANCE WITH NEC ARTICLE 250 PARA. 26. TERMINATE EQUIPMENT GROUND WIRES FOR FEEDERS AND BRANCH CIRCUITS WITH PRESSURE TYPE GROUND LUGS. WHERE METALLIC CONDUITS TERMINATE AT METALLIC HOUSINGS WITHOUT MECHANICAL AND ELECTRICAL CONNECTION TO HOUSING, TERMINATE EACH CONDUIT WITH A GROUNDING BUSHING. CONNECT GROUNDING BUSHINGS WITH A GROUND WIRE TO THE GROUND BUS IN THE HOUSING. BOND ELECTRICALLY NONCONTIGUOUS CONDUITS AT BOTH ENTRANCES AND EXITS WITH GROUNDING BUSHINGS AND GROUND WIRES. TIGHTEN GROUNDING AND BONDING CONNECTORS AND TERMINALS, INCLUDING SCREWS AND BOLTS, IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED TORQUE-TIGHTENING VALUES.

7. PANELBOARDS: INSTALL PANELBOARDS AND ACCESSORY ITEMS IN ACCORDANCE WITH NEMA PB 1.1, "GENERAL INSTRUCTIONS FOR PROPER INSTALLATION, OPERATION, AND MAINTENANCE OF PANELBOARDS RATED 600 VOLTS OR LESS," AND MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS. MOUNT PANELBOARDS PLUMB AND RIGID WITHOUT DISTORTION OF BOX AND WITH THE TOP OF THE TRIM AT 78 IN. AFF UNLESS OTHERWISE INDICATED. PROVIDE NEATLY TYPED AND ACCURATE CIRCUIT DIRECTORIES IN EACH PANELBOARD. REFLECTIVE OF FINAL CIRCUIT CONFIGURATION. PROVIDE FILLER PLATES IN ALL UNUSED SPACES. TRAIN WIRES THROUGH THE PANELBOARD. PROVIDE WIRE BUNDLES AND WIRE TRAYS WITH WIRE TIES. GROUND PANELBOARD IN CONFORMANCE WITH THE NEC. TIGHTEN ELECTRICAL CONNECTORS AND TERMINALS, INCLUDING GROUNDING CONNECTIONS, IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED TORQUE-TIGHTENING VALUES. PERFORM RESISTANCE TESTS OF PANELBOARD BUSES, COMPONENTS, AND FEEDER AND BRANCH CIRCUIT WIRING; INSULATION RESISTANCE LESS THAN 100 MEGOHM IS UNACCEPTABLE.

8. DISCONNECTS AND CIRCUIT BREAKERS: PROVIDE DISCONNECTS AND CIRCUIT BREAKERS WHERE INDICATED ON THE ELECTRICAL DRAWINGS AND/OR WHERE REQUIRED BY THE NEC, WHETHER INDICATED ON THE ELECTRICAL DRAWINGS OR NOT. INSTALL DISCONNECTS AND CIRCUIT BREAKERS PLUMB AND LEVEL AND IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS. UPON COMPLETION OF INSTALLATION OF DISCONNECTS AND CIRCUIT BREAKERS, ENERGIZE CIRCUITS AND DEMONSTRATE CAPABILITY AND COMPLIANCE WITH REQUIREMENTS. EXCEPT AS OTHERWISE INDICATED, DO NOT DEMONSTRATE DISCONNECTS AND CIRCUIT BREAKERS BY OPERATING THEM UNDER LOAD; HOWEVER, DEMONSTRATE DISCONNECT AND CIRCUIT BREAKERS OPERATION THROUGH SIX OPENING/CLOSING CYCLES WITH CIRCUIT UNLOADED. OPEN DISCONNECT AND CIRCUIT BREAKER ENCLOSURES FOR INSPECTION OF INTERIOR, MECHANICAL AND ELECTRICAL CONNECTIONS, FUSE INSTALLATION IF APPLICABLE, AND FOR VERIFICATION OF TYPE AND RATING OF FUSES INSTALLED IN APPLICABLE. CORRECT DEFICIENCIES THEN RETEST TO DEMONSTRATE COMPLIANCE WITH REQUIREMENTS. REMOVE AND REPLACE DEFECTIVE UNITS WITH NEW UNITS AND RETEST.

9. LIGHTING FIXTURES: INSTALL FIXTURES WHERE, AND AT HEIGHTS, INDICATED IN ACCORDANCE WITH FIXTURE MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS, REQUIREMENTS OF THE NEC, AND RECOGNIZED INDUSTRY PRACTICES. TIGHTEN ELECTRICAL CONNECTORS AND TERMINALS, INCLUDING GROUND CONNECTIONS, IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED TORQUE-TIGHTENING VALUES. RECESSED OR SEMI RECESSED FIXTURES MAY BE SUPPORTED BY CEILING SUPPORT SYSTEM. INSTALL CEILING SUPPORT SYSTEM RODS OR WIRES AT A MINIMUM OF FOUR (4) RODS OR WIRES PER FIXTURE LOCATED NOT MORE THAN 6 IN. FROM FIXTURE CORNERS. FIXTURES WHICH ARE SMALLER THAN CEILING GRID SHALL BE CENTERED IN ACOUSTICAL CEILING PANEL AND SUPPORTED BY AT LEAST TWO (2) 3/4 IN. METAL CHANNELS SPANNING AND SECURED TO CEILING SYSTEM GRID Tees. FIXTURES WHICH LAY-IN CEILING GRID SYSTEM SHALL BE SECURED IN PLACE BY INSTALLATION OF CLIPS WHICH SECURELY FASTEN FIXTURE TO CEILING GRID Tees. SUPPORT SURFACE MOUNT FIXTURES GREATER THAN 2 FT IN LENGTH AT A POINT IN ADDITION TO THE OUTLET BOX FIXTURE STUD. UPON COMPLETION OF INSTALLATION AND JUST PRIOR TO DEMONSTRATION CLEAN AND RELAMP FIXTURES. LAMP FIXTURES WITH SPECIFIED LAMPS IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED INSTRUCTIONS. UPON COMPLETION OF INSTALLATION, DEMONSTRATE CAPABILITY AND COMPLIANCE WITH REQUIREMENTS. WHERE POSSIBLE, CORRECT MALFUNCTIONING FIXTURES AT THE SITE, THEN RETEST TO DEMONSTRATE COMPLIANCE; OTHERWISE, REMOVE MALFUNCTIONING UNITS AND REPLACE WITH NEW UNITS AND PROCEED WITH RETESTING.

10. FLOOD-RESISTANT MEP MATERIAL, EQUIPMENT, WHERE MEP/FA COMPONENTS ARE INSTALLED BELOW BFE/DFE/RFE, MATERIALS SHALL BE FLOOD-DAMAGE-RESISTANT AND SUITABLE FOR WET LOCATION EXPOSURE. ELECTRICAL/FA RACEWAYS SHALL BE RMC, IMC, OR SCH 40S BE PVC WITH WET-LOCATION FITTINGS AND THWN-2/THHN-2 CONDUCTORS. PLUMBING PIPING SHALL BE PVC DWV, SERVICE-WEIGHT CAST IRON, TYPE L COPPER, PEX, CPVC, OR OTHER APPROVED NON-CORROSIVE MATERIAL. PLUMBWORK SHALL BE GALVANIZED SHEET METAL, ALUMINUM, OR STAINLESS STEEL. SUPPORTS AND FASTENERS SHALL BE GALVANIZED, STAINLESS STEEL, OR ALUMINUM ABSORBENT INSULATION, FLEX DUCT, DUCTBOARD, NM CABLE, STANDARD DRY-LOCATION BOXES, AND UNPROTECTED FERROUS SUPPORTS SHALL NOT BE USED BELOW FLOOD ELEVATION.

OWNER SIGNATURE

REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
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| 00  |      |             |          |
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| 02  |      |             |          |
| 03  |      |             |          |
| 04  |      |             |          |

CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:

TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE:

ELECTRICAL

PROJECT TITLE:

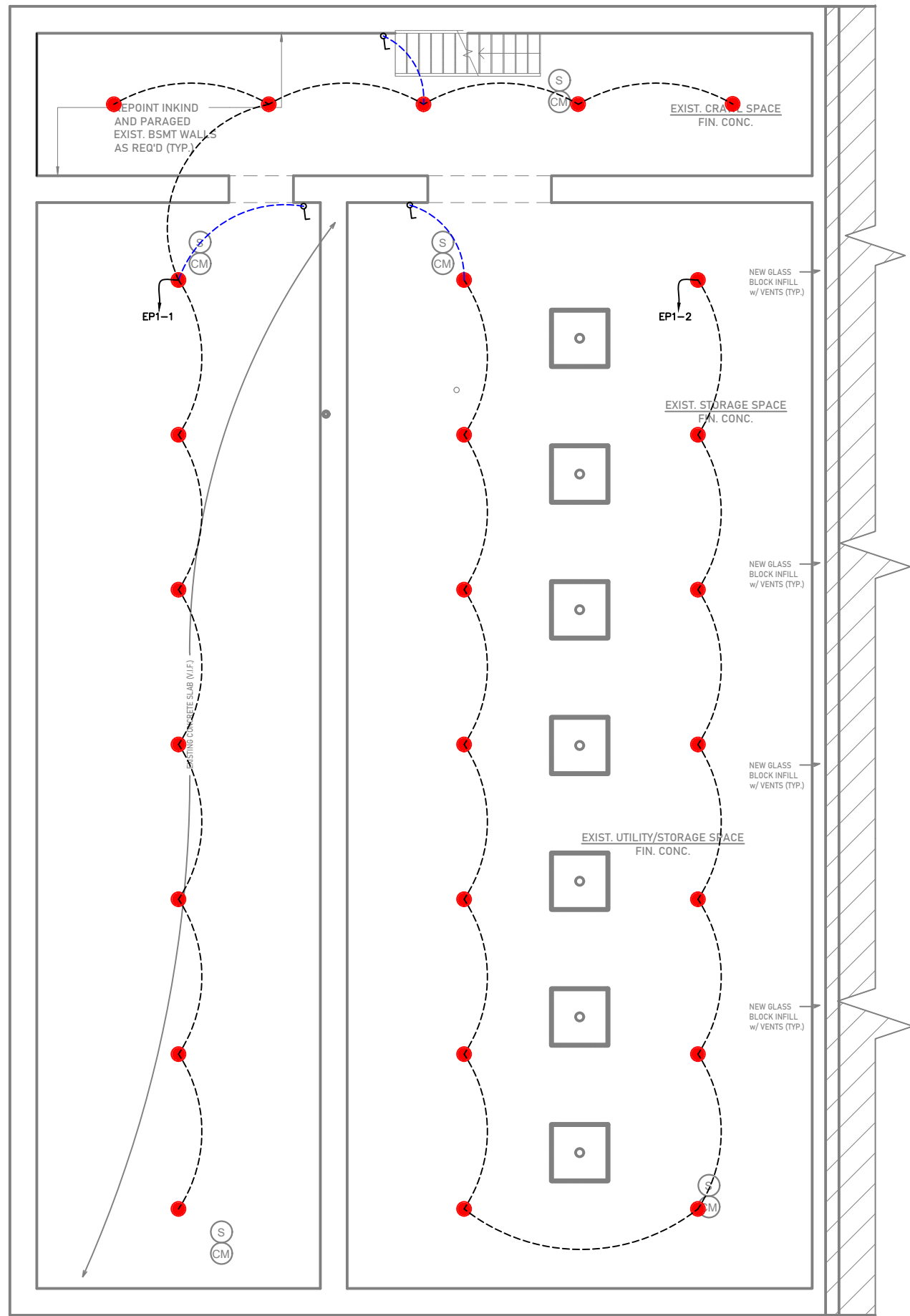
COMMERCIAL MIXED-USE

DRAWING TITLE:

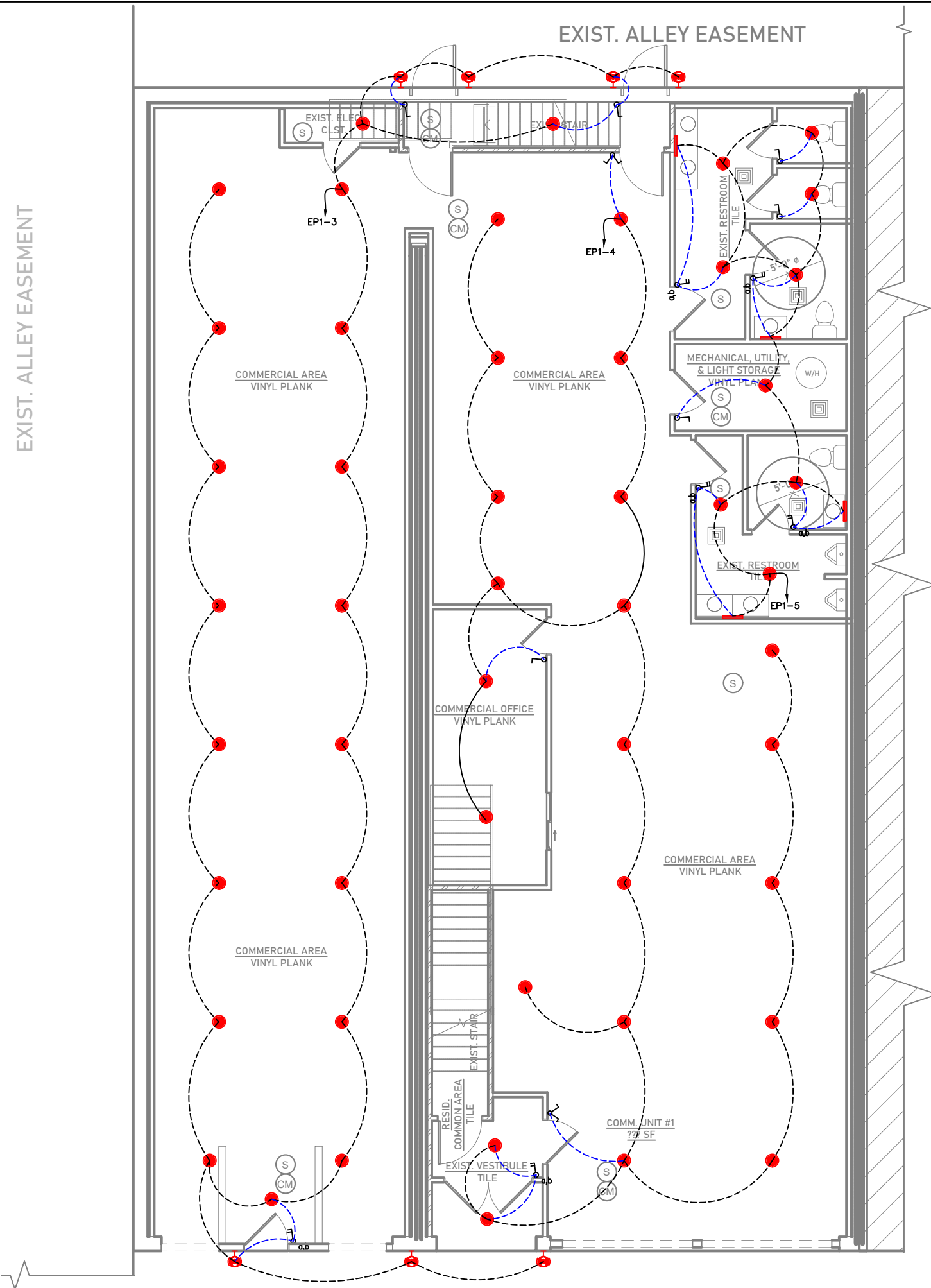
NOTES

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| DRAWN BY: | CHECKED BY: | DATE:      |
| G.M       | N.B         | 11/05/2026 |
| JOB NO    | SHT NO      | SCALE:     |
|           | JS-EL-01    | NTS @A1    |

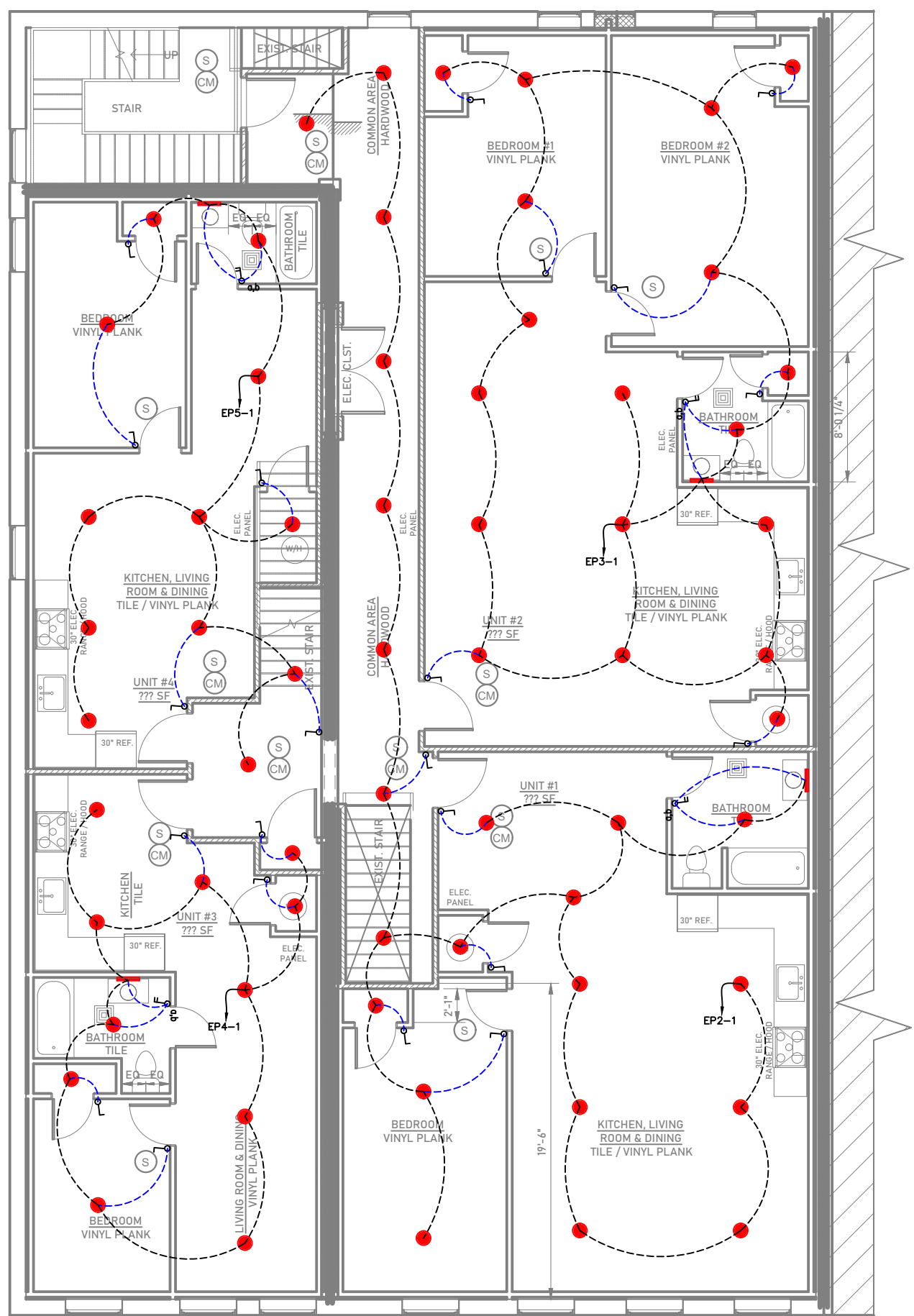




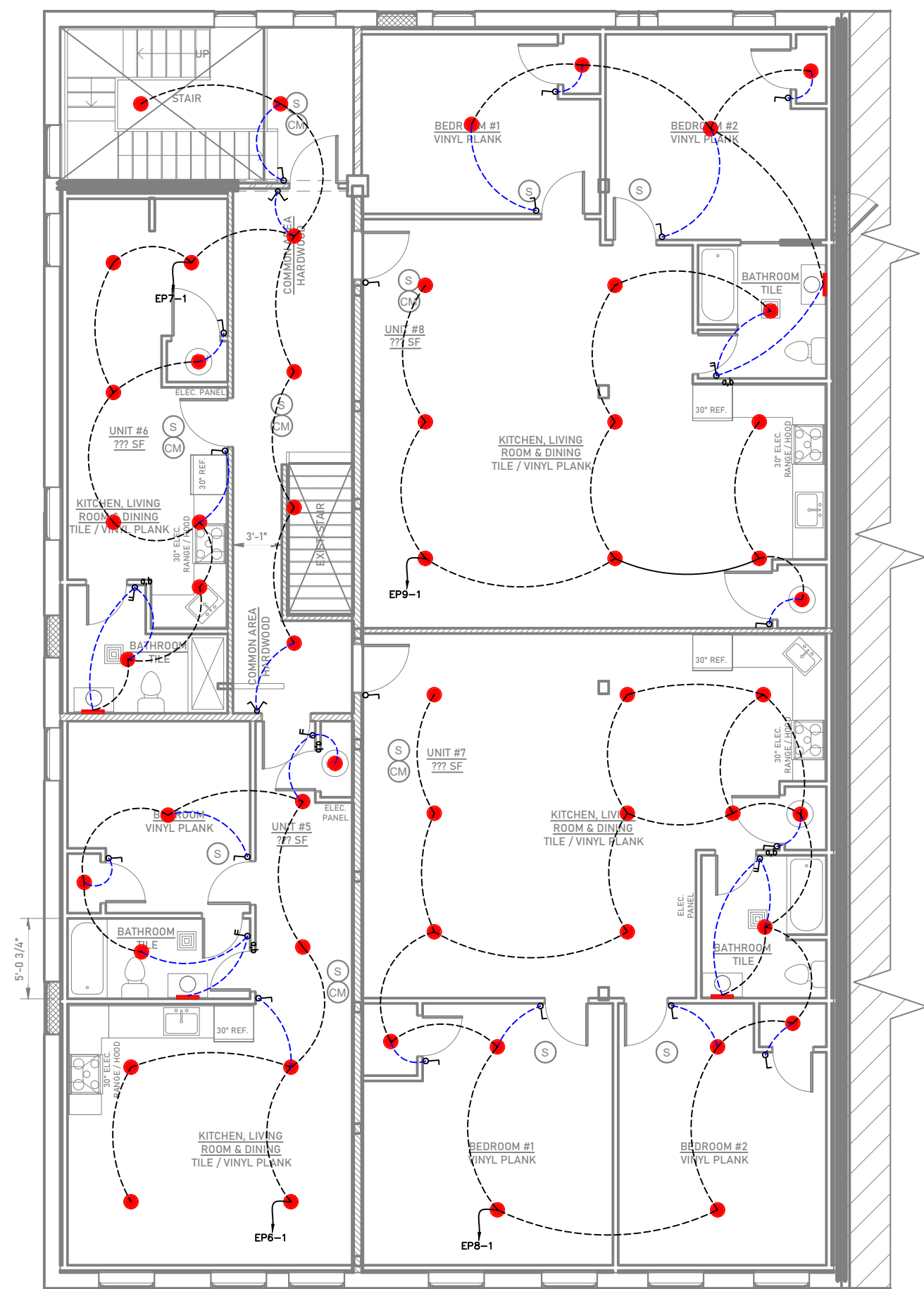
**BASEMENT FLOOR LIGHTING PLAN**  
SCALE 3" = 1'-0"



**FIRST FLOOR LIGHTING PLAN**  
SCALE 3" = 1'-0"



**SECOND FLOOR LIGHTING PLAN**  
SCALE 3" = 1'-0"



**THIRD FLOOR LIGHTING PLAN**  
SCALE 3" = 1'-0"

**GENERAL ELECTRICAL NOTES:**  
1- ELECTRICAL CONTRACTOR TO VERIFY ALL EXISTING SITE CONDITIONS & MAKE ANY NECESSARY ADJUSTMENTS  
2- SMALLEST WIRE & SIZE TO BE 14-2  
3- ALL WORK SHALL COMPLY WITH THE NATIONAL ELECTRIC CODE & ANY OTHER APPLICABLE REGULATIONS OR CODES, AT THE SITE  
4- CONTRACTOR TO LOCATE PHONE, CABLE, ETC. AS PER OWNER'S SPECIFICATIONS.  
**LEGEND NOTES:**  
+ VERIFY ALL MOUNTING HEIGHTS OF WALL MOUNTED LIGHT FIXTURES  
○ TYPICAL MOUNTING HEIGHTS FOR OUTLETS - CENTERLINE 18" AFF (EXCEPT WHERE NOTED OTHERWISE)  
\$ MAXIMUM HEIGHT FOR ALL SWITCHES IS 48" AFF  
\$ ROCKER STYLE SWITCHES  
ALL SWITCHES / OUTLETS / COVER PLATES - WHITE FINISH  
♦ VENT FANS SIZED FOR ROOM VOLUME  
FAN SOUND LEVEL NOT TO EXCEED 1.0 SONES

| : LEGEND : |                                                   |
|------------|---------------------------------------------------|
| SYMBOL     | DESCRIPTION                                       |
|            | MIRROR LIGHT 1X18W FLUORESCENT LIGHT              |
|            | 40W BEAM DIRECT/INDIRECT G2 SERIES (OFFICES)      |
|            | 15W SURFACE MOUNTED LIGHT IN CORRIDOR             |
|            | 20W WATERPROOF WALL MOUNTED EXTERIOR LIGHT        |
|            | 20W WATERPROOF DOWN LIGHTS EXTERIOR LIGHT         |
|            | EMERGENCY LIGHT                                   |
|            | 20W BATHROOM/OUTDOOR WALL MOUNTED EMERGENCY LIGHT |
|            | 12W 4" LED ROUND SPOT RECESSED                    |
|            | 15W SURFACE MOUNTED LIGHT 2X4 ON LEVELS/BALCONY   |
|            | 20W PENDANT LIGHTS OVER DISHWASHER AND SINK       |
|            | ONE WAY ONE GANG SWITCH                           |
|            | ONE WAY TWO GANG SWITCH                           |
|            | ONE WAY THREE GANG SWITCH                         |
|            | TWO WAY ONE GANG SWITCH                           |
|            | TWO WAY TWO GANG SWITCH                           |
|            | TWO WAY THREE GANG SWITCH                         |
|            | SPAN ELECTRICAL PANEL, TYPE INDICATED.            |

OWNER SIGNATURE

REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
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| 04  |      |             |          |

CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL: NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE:

ELECTRICAL

PROJECT TITLE:

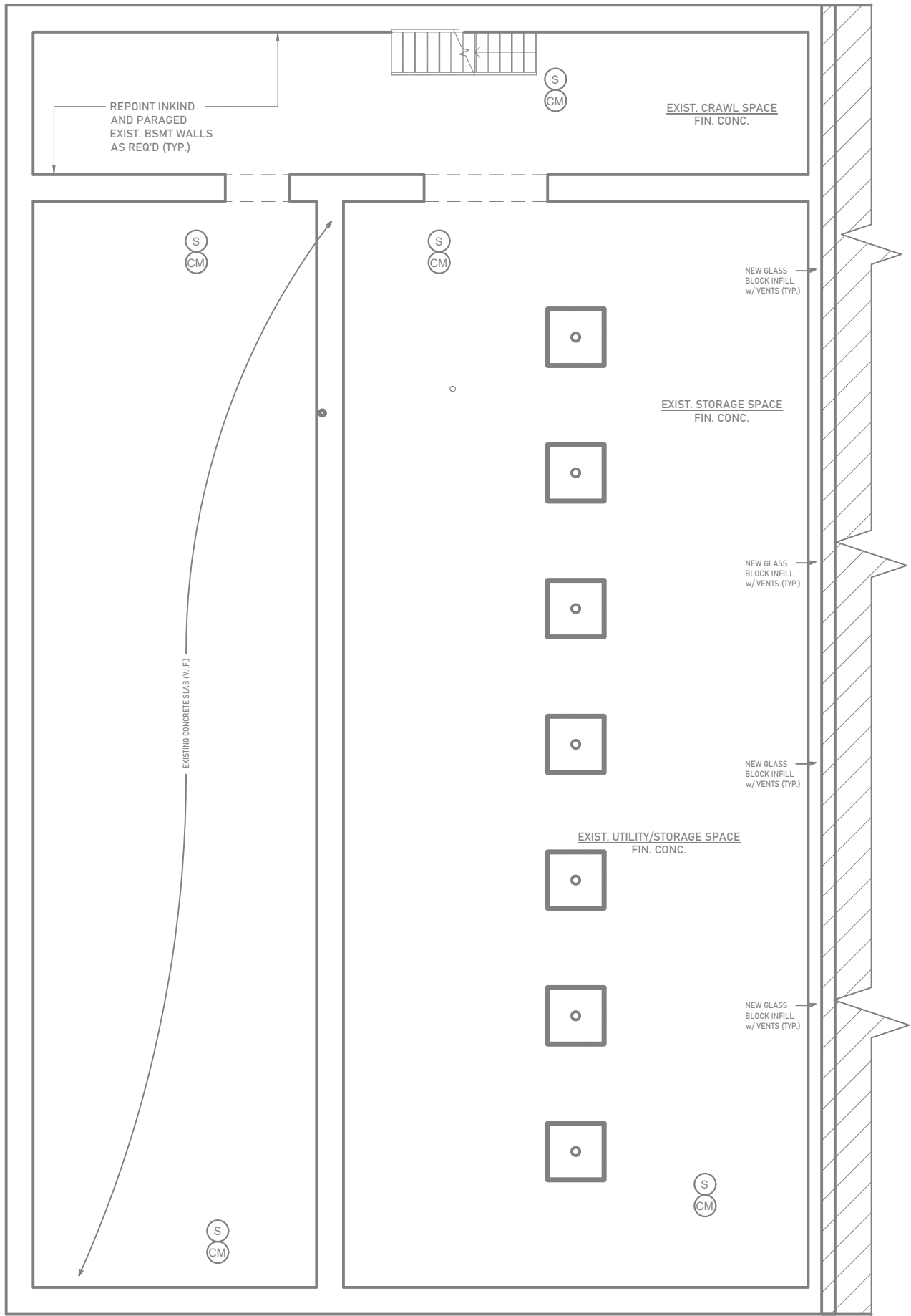
COMMERCIAL MIXED-USE

DRAWING TITLE:

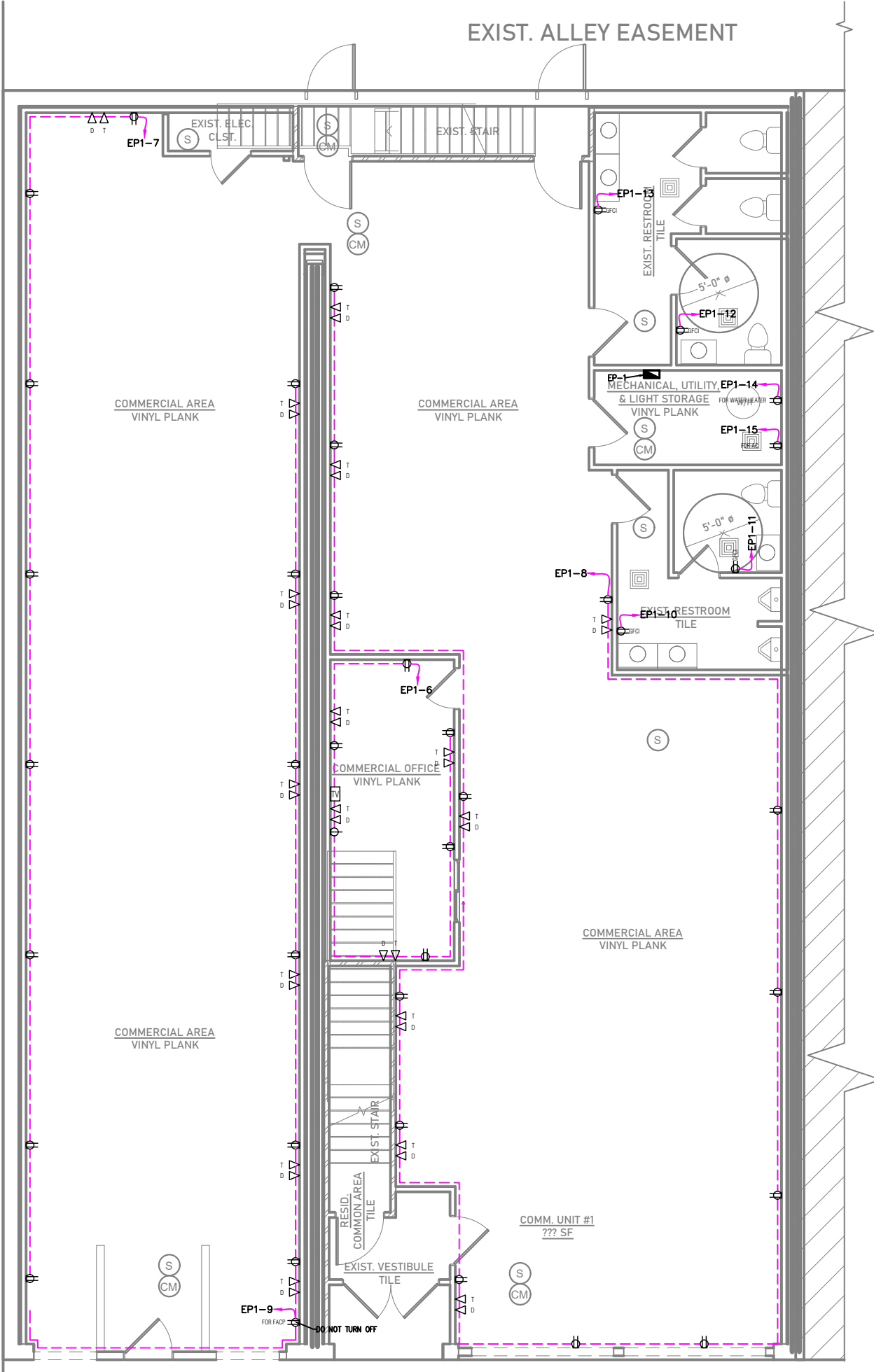
BASEMENT, FIRST, SECOND & THIRD  
FLOOR LIGHTING PLANS

|                  |                    |                        |
|------------------|--------------------|------------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026    |
| JOB NO           | SHT NO<br>JS-EL-02 | SCALE:<br>AS SHOWN @A1 |

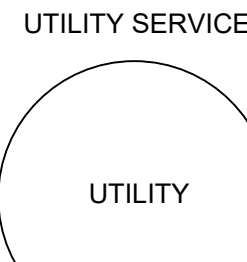




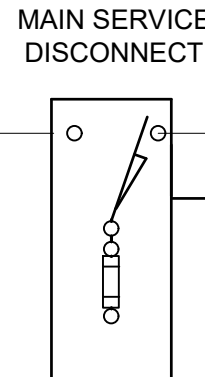
EL 82 / SCALE  
BASEMENT FLOOR POWER PLAN  
3" = 1'-0"



EL 83 / SCALE  
FIRST FLOOR POWER PLAN  
3" = 1'-0"

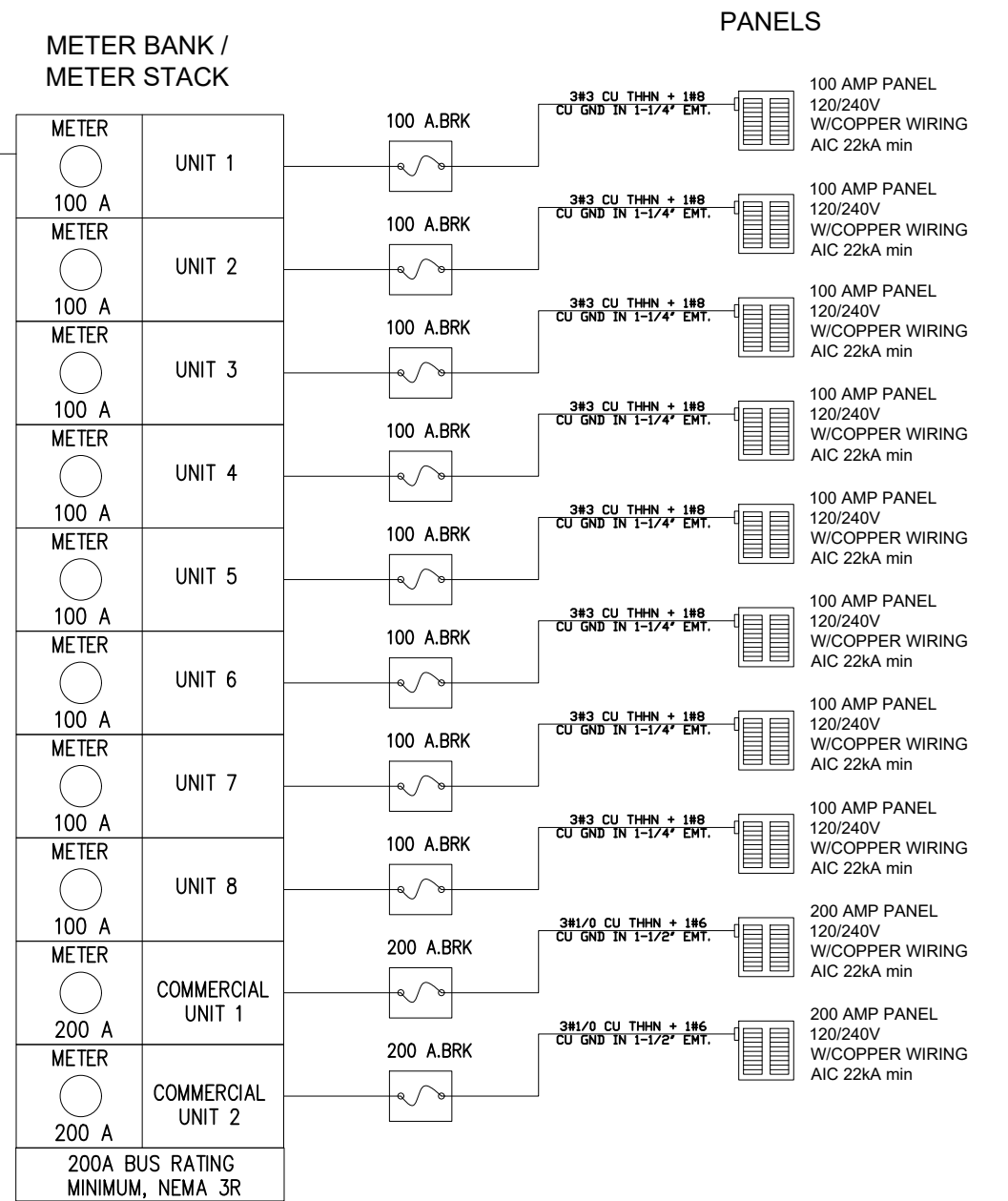


MAIN SERVICE FEEDER  
30 FEET OF 3/4" O.D.  
1/2" CU THIN W/ 1/2" O.D.  
1/2" CU THIN W/ 1/2" O.D.  
1/2" CU THIN W/ 1/2" O.D.



2 GROUNDING RODS  
BOND TO WATER PIPE  
BOND TO BUILDING STEEL  
GEC. 2M CU MINIMUM

GROUNDING PER NEC 250



EL 03 / SCALE  
SINGLE LINE DIAGRAM  
NTS

- GENERAL ELECTRICAL NOTES
- 1- ELECTRICAL CONTRACTOR TO VERIFY ALL EXISTING SITE CONDITIONS & MAKE ANY NECESSARY ADJUSTMENTS
  - 2- SMALLEST WIRE & SIZE TO BE 14-2
  - 3- ALL WORK SHALL COMPLY WITH THE NATIONAL ELECTRIC CODE & ANY OTHER APPLICABLE REGULATIONS OR CODES, AT THE SITE.
  - 4- CONTRACTOR TO LOCATE: PHONE, CABLE, ETC. AS PER OWNER'S SPECIFICATIONS.
- LEGEND NOTES
- ↑ VERIFY ALL MOUNTING HEIGHTS
  - ◇ OF WALL MOUNTED LIGHT FIXTURES
- TYPICAL MOUNTING HEIGHTS FOR OUTLETS- CENTERLINE 18" AFF (EXCEPT WHERE NOTED OTHERWISE)
- MAXIMUM HEIGHT FOR ALL SWITCHES IS 48" AFF  
ROCKER STYLE SWITCHES  
ALL SWITCHES/OUTLETS COVER PLATES-WHITE FINISH
- VENT FANS SIZED FOR ROOM VOLUME  
FAN SOUND LEVEL NOT TO EXCEED 1.0 SONES

| : LEGEND : |                                                                                                                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SYMBOL     | DESCRIPTION                                                                                                                                                                               |
| ⊕          | SINGLE SOCKET OUTLET, 13A, 230V, GROUNDING TYPE. MOUNT AT 42" A.F.F. UNLESS NOTED OTHERWISE.                                                                                              |
| ⊕          | DUPLEX SOCKET OUTLET, 13A, 230V, GROUNDING TYPE. MOUNT AT 18" A.F.F. UNLESS NOTED OTHERWISE.                                                                                              |
| ⊕          | GFCI DUPLEX SOCKET OUTLET, WEATHER PROOF, 13A, 230V, GROUNDING TYPE. MOUNT AT 42" A.F.F. UNLESS NOTED OTHERWISE.                                                                          |
| ⊕          | POINT FOR KITCHEN HOT PLATE, 32A, GROUNDING TYPE. MOUNT AT 42" A.F.F. UNLESS NOTED OTHERWISE.                                                                                             |
| TV         | TV SOCKET HEIGHT AS PER LCD ON LOCATION                                                                                                                                                   |
| ↑          | TELECOMMUNICATION OUTLET. PROVIDE SINGLE GANG JUNCTION BOX WITH PLASTER RING. STUB UP 1" E.C. WITH PULL STRING INTO ACCESSIBLE CEILING SPACE. MOUNT AT 18" A.F.F. UNLESS NOTED OTHERWISE. |
| ◇          | DATA OUTLET. PROVIDE SINGLE GANG JUNCTION BOX WITH PLASTER RING. STUB UP 1" E.C. WITH PULL STRING INTO ACCESSIBLE CEILING SPACE. MOUNT AT 18" A.F.F. UNLESS NOTED OTHERWISE.              |
| EP         | ELECTRICAL PANEL                                                                                                                                                                          |

OWNER SIGNATURE

| REVISION STATUS |      |             |          |
|-----------------|------|-------------|----------|
| Rev             | DATE | DESCRIPTION | DRAWN BY |
| 00              |      |             |          |
| 01              |      |             |          |
| 02              |      |             |          |
| 03              |      |             |          |
| 04              |      |             |          |

| CLIENT/CONSULTANT APPROVAL STATUS |     |
|-----------------------------------|-----|
| APPROVED                          | (A) |
| APPROVED WITH NOTES               | (B) |
| RESUBMIT                          | (C) |
| DISAPPROVED                       | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL: NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING  
DISCIPLINE: ELECTRICAL

PROJECT TITLE:  
COMMERCIAL MIXED-USE

DRAWING TITLE:  
BASEMENT, FIRST, SECOND, THIRD  
FLOOR POWER PLANS & SINGLE LINE  
DIAGRAM

|                  |                    |                        |
|------------------|--------------------|------------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026    |
| JOB NO           | SHT NO<br>JS-EL-03 | SCALE:<br>AS SHOWN @A1 |

NOTES:

FE-1: LOWER-LEVEL CIRCUITS SHALL BE ROUTED FROM PANELS ABOVE BFE/DFE/RFE AND DROPPED FROM ABOVE. USE RMC, IMC, OR SCH. 80 PVC WITH WET-LOCATION FITTINGS BELOW FLOOD ELEVATION. PROVIDE THWN-2/XHHW-2 COPPER CONDUCTORS.

FE-2: DEVICES BELOW BFE/DFE/RFE SHALL BE MINIMIZED. WHERE REQUIRED, PROVIDE WEATHER-RESISTANT GFCI RECEPTACLES, EXTRA-DUTY IN-USE COVERS, CORROSION-RESISTANT BOXES, AND WET-LOCATION-RATED FITTINGS.

FE-3: PANELBOARDS, SERVICE DISCONNECTS, METER BANKS, FIRE ALARM POWER SUPPLIES, AND CONTROL EQUIPMENT SHALL BE INSTALLED ABOVE BFE/DFE/RFE. DO NOT LOCATE ELECTRICAL DISTRIBUTION EQUIPMENT BELOW FLOOD ELEVATION UNLESS INSTALLED IN AHJ-APPROVED WATERPROOFED ENCLOSURE.



| PANEL BOARD SCHEDULE         |                                                                                                                                                                                    |                              |         |           |        |                                 |                        |   |   |   |                        |                  |        |           |         |              |                                  |                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------|-----------|--------|---------------------------------|------------------------|---|---|---|------------------------|------------------|--------|-----------|---------|--------------|----------------------------------|------------------------------|
| PANEL TAG:                   |                                                                                                                                                                                    | EP1                          |         | MAINS:    |        | 200A/T/250AF, 120/240V, 1P,60HZ |                        |   |   |   |                        |                  |        |           |         | NEUTRAL: YES |                                  |                              |
| LOCATION:                    |                                                                                                                                                                                    | BASEMENT & FIRST FLOOR PLANS |         | MOUNTING: |        | SURFACE MOUNTED                 |                        |   |   |   |                        |                  |        |           |         | GROUND: YES  |                                  |                              |
| ENCLOSURE:                   |                                                                                                                                                                                    | IP42                         |         | VOLTAGE:  |        | 120/240V, 1P,3W                 |                        |   |   |   |                        |                  |        |           |         |              |                                  |                              |
| FEEDER ENTRANCE:             |                                                                                                                                                                                    | TOP                          |         |           |        |                                 |                        |   |   |   |                        |                  |        |           |         |              |                                  |                              |
| LOAD<br>SERVED               | Romex Wire Size (Inch)                                                                                                                                                             | LOAD (VA)                    |         |           | C<br>B | P<br>O<br>L<br>E                | C<br>K<br>T<br>N<br>O. | R | Y | B | C<br>K<br>T<br>N<br>O. | P<br>O<br>L<br>E | C<br>B | LOAD (VA) |         |              | WIRE &<br>CONDUIT<br>SIZE (INCH) | LOAD<br>SERVED               |
|                              |                                                                                                                                                                                    | ØA                           | ØB      | ØC        |        |                                 |                        |   |   |   |                        |                  |        | ØA        | ØB      | ØC           |                                  |                              |
| LIGHTING                     | 2#12, 1#12(G), 3/4"C                                                                                                                                                               | 800.00                       |         |           | 20     | 1P                              | 1                      |   |   |   | 2                      | 1P               | 20     | 800.00    |         |              | 2#12, 1#12(G), 3/4"C             | LIGHTING                     |
| LIGHTING                     | 2#12, 1#12(G), 3/4"C                                                                                                                                                               |                              | 800.00  |           | 20     | 1P                              | 3                      |   |   |   | 4                      | 1P               | 20     |           | 800.00  |              | 2#12, 1#12(G), 3/4"C             | LIGHTING                     |
| LIGHTING                     | 2#12, 1#12(G), 3/4"C                                                                                                                                                               |                              |         | 800.00    | 20     | 1P                              | 5                      |   |   |   | 6                      | 1P               | 15     |           |         | 1200.00      | 2#14, 1#14(G), 3/4"C             | RECEPTACLE - OFFICE          |
| RECEPTACLE - COMMERCIAL AREA | 2#14, 1#14(G), 3/4"C                                                                                                                                                               |                              | 1200.00 |           | 15     | 1P                              | 7                      |   |   |   | 8                      | 1P               | 15     |           | 1200.00 |              | 2#14, 1#14(G), 3/4"C             | RECEPTACLE - COMMERCIAL AREA |
| FACP                         | 2#14, 1#14(G), 3/4"C                                                                                                                                                               | 600.00                       |         |           | 15     | 1P                              | 9                      |   |   |   | 10                     | 1P               | 15     | 1200.00   |         |              | 2#14, 1#14(G), 3/4"C             | BATH - GFCI                  |
| BATH - GFCI                  | 2#14, 1#14(G), 3/4"C                                                                                                                                                               |                              | 1200.00 |           | 15     | 1P                              | 11                     |   |   |   | 12                     | 1P               | 15     |           | 1200.00 |              | 2#14, 1#14(G), 3/4"C             | BATH - GFCI                  |
| BATH - GFCI                  | 2#14, 1#14(G), 3/4"C                                                                                                                                                               |                              |         | 1200.00   | 15     | 1P                              | 13                     |   |   |   | 14                     | 3P               | 60     |           |         | 5000.00      | 2#8, 1#8(G), 3/4"C               | WATER HEATER                 |
| RECEPTACLE(AC)               | 2#6, 1#6(G), 3/4"C                                                                                                                                                                 |                              | 1500.00 |           | 15     | 1P                              | 15                     |   |   |   | 16                     | 1P               | 20     |           |         |              |                                  | SPARE                        |
| SPARE                        |                                                                                                                                                                                    |                              |         |           | 20     | 1P                              | 17                     |   |   |   | 18                     | 1P               | 20     |           |         |              |                                  | SPARE                        |
|                              |                                                                                                                                                                                    |                              |         |           |        |                                 |                        |   |   |   |                        |                  |        |           |         |              |                                  |                              |
|                              |                                                                                                                                                                                    | 1400.00                      | 4700.00 | 2000.00   |        |                                 |                        |   |   |   |                        |                  |        | 2000.00   | 3200.00 | 6200.00      |                                  |                              |
|                              |                                                                                                                                                                                    |                              |         |           |        |                                 |                        |   |   |   |                        |                  |        |           |         |              |                                  |                              |
|                              | <div> <div> <div>ØA =</div> <div>3400.00</div> </div> <div> <div>TOTAL CONNECTED LOAD =</div> <div>19500.00</div> </div> <div> <div>I (AMP) =</div> <div>46.29</div> </div> </div> |                              |         |           |        |                                 |                        |   |   |   |                        |                  |        |           |         |              |                                  |                              |
|                              | <div> <div> <div>ØB =</div> <div>7900.00</div> </div> <div> <div>DEMAND FACTOR =</div> <div>1</div> </div> </div>                                                                  |                              |         |           |        |                                 |                        |   |   |   |                        |                  |        |           |         |              |                                  |                              |
|                              | <div> <div> <div>ØC =</div> <div>8200.00</div> </div> <div> <div>TOTAL CONNECTED LOAD +25% SPARE=</div> <div>24375.00</div> </div> </div>                                          |                              |         |           |        |                                 |                        |   |   |   |                        |                  |        |           |         |              |                                  |                              |

| PANEL TAG:                      |                                                                                                                                                                                                                                                                                                                                                                                                                       | EP2                     |         | MAINS:    |        | 100A/T/250AF, 120/240V, 1P, 60HZ |                        | NEUTRAL: YES |   |   |                        |             |        |           |         |         |                        |                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------|-----------|--------|----------------------------------|------------------------|--------------|---|---|------------------------|-------------|--------|-----------|---------|---------|------------------------|-------------------------------|
| LOCATION:                       |                                                                                                                                                                                                                                                                                                                                                                                                                       | UNIT 1 FIRST FLOOR PLAN |         | MOUNTING: |        | SURFACE MOUNTED                  |                        | GROUND: YES  |   |   |                        |             |        |           |         |         |                        |                               |
| ENCLOSURE:                      |                                                                                                                                                                                                                                                                                                                                                                                                                       | IP42                    |         | VOLTAGE:  |        | 120/240V, 1P, 3W                 |                        |              |   |   |                        |             |        |           |         |         |                        |                               |
| FEEDER ENTRANCE:                |                                                                                                                                                                                                                                                                                                                                                                                                                       | TOP                     |         |           |        |                                  |                        |              |   |   |                        |             |        |           |         |         |                        |                               |
| LOAD<br>SERVED                  | Romex Wire Size (Inch)                                                                                                                                                                                                                                                                                                                                                                                                | LOAD (VA)               |         |           | C<br>B | P<br>O<br>L                      | C<br>K<br>T<br>N<br>O. | R            | Y | B | C<br>K<br>T<br>N<br>O. | P<br>O<br>L | C<br>B | LOAD (VA) |         |         | CONDUIT<br>SIZE (INCH) | LOAD<br>SERVED                |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                       | ØA                      | ØB      | ØC        |        |                                  |                        |              |   |   |                        |             |        | ØA        | ØB      | ØC      |                        |                               |
| LIGHTING                        | 2#12, 1#12(G), 3/4"C                                                                                                                                                                                                                                                                                                                                                                                                  | 800.00                  |         |           | 20     | 1P                               | 1                      |              |   |   | 2                      | 1P          | 15     | 1200.00   |         |         | 2#14, 1#14(G), 3/4"C   | RECEPTACLE - BEDROOM 2 (AFCI) |
| RECEPTACLE - DINING/LIVING ROOM | 2#14, 1#14(G), 3/4"C                                                                                                                                                                                                                                                                                                                                                                                                  |                         | 1200.00 |           | 15     | 1P                               | 3                      |              |   |   | 4                      | 1P          | 15     |           | 1200.00 |         | 2#14, 1#14(G), 3/4"C   | BATH - GFCI                   |
| SMALL APPLIANCE                 | 2#12, 1#12(G), 3/4"C                                                                                                                                                                                                                                                                                                                                                                                                  |                         |         | 1500.00   | 15     | 1P                               | 5                      |              |   |   | 6                      | 1P          | 15     |           |         | 1500.00 | 2#12, 1#12(G), 3/4"C   | SMALL APPLIANCE               |
| REFRIGERATOR (AF/GF)            | 2#12, 1#12(G), 3/4"C                                                                                                                                                                                                                                                                                                                                                                                                  |                         | 1500.00 |           | 20     | 1P                               | 7                      |              |   |   | 8                      | 1P          | 15     |           | 1500.00 |         | 2#14, 1#14(G), 3/4"C   | HOOD                          |
| MICROWAVE (AF/GF)               | 2#12, 1#12(G), 3/4"C                                                                                                                                                                                                                                                                                                                                                                                                  | 2000.00                 |         |           | 20     | 1P                               | 9                      |              |   |   | 10                     | 2P          | 20     | 4500.00   |         |         | 2#10, 1#10(G), 3/4"C   | RANGE                         |
| WATER HEATER                    | 2#8, 1#8(G), 3/4"C                                                                                                                                                                                                                                                                                                                                                                                                    |                         | 5000.00 |           | 60     | 3P                               | 11                     |              |   |   | 12                     | 3P          | 60     |           | 5000.00 |         | 2#6, 1#6(G), 3/4"C     | RECEPTACLE(A/C)               |
| SPARE                           |                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |         |           | 20     | 1P                               | 13                     |              |   |   | 14                     | 1P          | 20     |           |         |         |                        | SPARE                         |
| SPARE                           |                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |         |           | 20     | 1P                               | 15                     |              |   |   | 16                     | 1P          | 20     |           |         |         |                        | SPARE                         |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |         |           |        |                                  |                        |              |   |   |                        |             |        |           |         |         |                        |                               |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                       | 2800.00                 | 7700.00 | 1500.00   |        |                                  |                        |              |   |   |                        |             |        | 5700.00   | 7700.00 | 1500.00 |                        |                               |
|                                 | <div> <div> <div>ØA =</div> <div>8500.00</div> </div> <div> <div>ØB =</div> <div>15400.00</div> </div> <div> <div>ØC =</div> <div>3000.00</div> </div> </div> <div> <div>TOTAL CONNECTED LOAD =</div> <div>26900.00</div> </div> <div> <div>DEMAND FACTOR =</div> <div>1</div> </div> <div> <div>TOTAL CONNECTED LOAD +25% SPARE=</div> <div>33625.00</div> </div> <div> <div>I (AMP) =</div> <div>63.86</div> </div> |                         |         |           |        |                                  |                        |              |   |   |                        |             |        |           |         |         |                        |                               |

| PANEL BOARD SCHEDULE            |                                                                                                                  |                         |         |                                         |        |             |                        |   |   |   |                        |             |        |           |         |              |                        |                               |  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------|---------|-----------------------------------------|--------|-------------|------------------------|---|---|---|------------------------|-------------|--------|-----------|---------|--------------|------------------------|-------------------------------|--|
| PANEL TAG:                      |                                                                                                                  | EP3                     |         | MAINS: 100A/T/250AF, 120/240V, 1P, 60HZ |        |             |                        |   |   |   |                        |             |        |           |         | NEUTRAL: YES |                        |                               |  |
| LOCATION:                       |                                                                                                                  | UNIT 2 FIRST FLOOR PLAN |         | MOUNTING: SURFACE MOUNTED               |        |             |                        |   |   |   |                        |             |        |           |         | GROUND: YES  |                        |                               |  |
| ENCLOSURE:                      |                                                                                                                  | IP42                    |         | VOLTAGE: 120/240V, 1P, 3W               |        |             |                        |   |   |   |                        |             |        |           |         |              |                        |                               |  |
| FEEDER ENTRANCE:                |                                                                                                                  | TOP                     |         |                                         |        |             |                        |   |   |   |                        |             |        |           |         |              |                        |                               |  |
| LOAD<br>SERVED                  | Romex Wire Size (Inch)                                                                                           | LOAD (VA)               |         |                                         | C<br>B | P<br>O<br>L | C<br>K<br>T<br>N<br>O. | R | Y | B | C<br>K<br>T<br>N<br>O. | P<br>O<br>L | C<br>B | LOAD (VA) |         |              | CONDUIT<br>SIZE (INCH) | LOAD<br>SERVED                |  |
|                                 |                                                                                                                  | ØA                      | ØB      | ØC                                      |        |             |                        |   |   |   |                        |             |        | ØA        | ØB      | ØC           |                        |                               |  |
| LIGHTING                        | 2#12, 1#12(G), 3/4"C                                                                                             | 800.00                  |         |                                         | 20     | 1P          | 1                      |   |   |   | 2                      | 1P          | 15     | 1200.00   |         |              | 2#14, 1#14(G), 3/4"C   | RECEPTACLE - BEDROOM 2 (AFCI) |  |
| RECEPTACLE - DINING/LIVING ROOM | 2#14, 1#14(G), 3/4"C                                                                                             |                         | 1200.00 |                                         | 15     | 1P          | 3                      |   |   |   | 4                      | 1P          | 15     |           | 1200.00 |              | 2#14, 1#14(G), 3/4"C   | RECEPTACLE - BEDROOM 2 (AFCI) |  |
| BATH - GFCI                     | 2#14, 1#14(G), 3/4"C                                                                                             |                         |         | 1200.00                                 | 15     | 1P          | 5                      |   |   |   | 6                      | 1P          | 15     |           |         | 1500.00      | 2#12, 1#12(G), 3/4"C   | SMALL APPLIANCE               |  |
| SMALL APPLIANCE                 | 2#12, 1#12(G), 3/4"C                                                                                             |                         | 1500.00 |                                         | 15     | 1P          | 7                      |   |   |   | 8                      | 1P          | 20     |           | 1500.00 |              | 2#12, 1#12(G), 3/4"C   | REFRIGERATOR (AF/GF)          |  |
| HOOD                            | 2#14, 1#14(G), 3/4"C                                                                                             | 1500.00                 |         |                                         | 15     | 1P          | 9                      |   |   |   | 10                     | 1P          | 20     | 2000.00   |         |              | 2#12, 1#12(G), 3/4"C   | MICROWAVE (AF/GF)             |  |
| RANGE 1                         | 2#10, 1#10(G), 3/4"C                                                                                             |                         | 4500.00 |                                         | 20     | 2P          | 11                     |   |   |   | 12                     | 3P          | 60     |           | 5000.00 |              | 2#8, 1#8(G), 3/4"C     | WATER HEATER                  |  |
| RECEPTACLE(AC)                  | 2#6, 1#6(G), 3/4"C                                                                                               |                         |         | 5000.00                                 | 60     | 1P          | 13                     |   |   |   | 14                     | 1P          | 20     |           |         |              |                        | SPARE                         |  |
| SPARE                           |                                                                                                                  |                         |         |                                         | 20     | 1P          | 15                     |   |   |   | 16                     | 1P          | 20     |           |         |              |                        | SPARE                         |  |
|                                 |                                                                                                                  |                         |         |                                         |        |             |                        |   |   |   |                        |             |        |           |         |              |                        |                               |  |
|                                 |                                                                                                                  | 2300.00                 | 7200.00 | 6200.00                                 |        |             |                        |   |   |   |                        |             |        | 3200.00   | 7700.00 | 1500.00      |                        |                               |  |
|                                 |                                                                                                                  |                         |         |                                         |        |             |                        |   |   |   |                        |             |        |           |         |              |                        |                               |  |
|                                 | <div> <div> ØA = 5500.00 </div> <div> TOTAL CONNECTED LOAD = 28100.00 </div> <div> I (AMP) = 66.71 </div> </div> |                         |         |                                         |        |             |                        |   |   |   |                        |             |        |           |         |              |                        |                               |  |
|                                 | <div> <div> ØB = 14900.00 </div> <div> DEMAND FACTOR = 1 </div> </div>                                           |                         |         |                                         |        |             |                        |   |   |   |                        |             |        |           |         |              |                        |                               |  |
|                                 | <div> <div> ØC = 7700.00 </div> <div> TOTAL CONNECTED LOAD +25% SPARE= 35125.00 </div> </div>                    |                         |         |                                         |        |             |                        |   |   |   |                        |             |        |           |         |              |                        |                               |  |

|                         |                           |                               |
|-------------------------|---------------------------|-------------------------------|
| <u>DRAWN BY:</u><br>G.M | <u>CHECKED BY:</u><br>N.B | <u>DATE:</u><br>11/05/2026    |
| <u>JOB NO</u>           | SHT NO<br>JS-EL-04        | <u>SCALE:</u><br>AS SHOWN @A1 |



## PANELBOARD SCHEDULE

| PANEL TAG:                      | EP4                                                         | MAINS: 100AT/250AF,120/240V ,1P,60HZ |         |         |        |             |            |   |   |   |            | NEUTRAL: YES |        |           |         |         |                        |                               |  |
|---------------------------------|-------------------------------------------------------------|--------------------------------------|---------|---------|--------|-------------|------------|---|---|---|------------|--------------|--------|-----------|---------|---------|------------------------|-------------------------------|--|
| LOCATION:                       | UNIT 3 FIRST FLOOR PLAN                                     | MOUNTING: SURFACE MOUNTED            |         |         |        |             |            |   |   |   |            | GROUND: YES  |        |           |         |         |                        |                               |  |
| ENCLOSURE:                      | IP42                                                        | VOLTAGE: 120/240V, 1P,3W             |         |         |        |             |            |   |   |   |            |              |        |           |         |         |                        |                               |  |
| FEEDER ENTRANCE:                | TOP                                                         |                                      |         |         |        |             |            |   |   |   |            |              |        |           |         |         |                        |                               |  |
| LOAD<br>SERVED                  | Romex Wire Size (Inch)                                      | LOAD (VA)                            |         |         | C<br>B | P<br>O<br>L | CKT<br>NO. | R | Y | B | CKT<br>NO. | P<br>O<br>L  | C<br>B | LOAD (VA) |         |         | CONDUIT<br>SIZE (INCH) | LOAD<br>SERVED                |  |
|                                 |                                                             | ØA                                   | ØB      | ØC      |        |             |            |   |   |   |            |              |        | ØA        | ØB      | ØC      |                        |                               |  |
| LIGHTING                        | 2#12, 1#12(G), 3/4"C                                        | 800.00                               |         |         | 20     | 1P          | 1          |   |   |   | 2          | 1P           | 15     | 1200.00   |         |         | 2#14, 1#14(G), 3/4"C   | RECEPTACLE - BEDROOM 2 (AFCI) |  |
| RECEPTACLE - DINING/LIVING ROOM | 2#14, 1#14(G), 3/4"C                                        |                                      | 1200.00 |         | 15     | 1P          | 3          |   |   |   | 4          | 1P           | 15     |           | 1200.00 |         | 2#14, 1#14(G), 3/4"C   | BATH - GFCI                   |  |
| SMALL APPLIANCE                 | 2#12, 1#12(G), 3/4"C                                        |                                      |         | 1500.00 | 15     | 1P          | 5          |   |   |   | 6          | 1P           | 15     |           |         | 1500.00 | 2#12, 1#12(G), 3/4"C   | SMALL APPLIANCE               |  |
| REFRIGERATOR (AF/GF)            | 2#12, 1#12(G), 3/4"C                                        |                                      | 1500.00 |         | 20     | 1P          | 7          |   |   |   | 8          | 1P           | 15     |           | 1500.00 |         | 2#14, 1#14(G), 3/4"C   | HOOD                          |  |
| MICROWAVE (AF/GF)               | 2#12, 1#12(G), 3/4"C                                        | 2000.00                              |         |         | 20     | 1P          | 9          |   |   |   | 10         | 2P           | 20     | 4500.00   |         |         | 2#10, 1#10(G), 3/4"C   | RANGE                         |  |
| WATER HEATER                    | 2#8, 1#8(G), 3/4"C                                          |                                      | 5000.00 |         | 60     | 3P          | 11         |   |   |   | 12         | 3P           | 60     |           | 5000.00 |         | 2#6, 1#6(G), 3/4"C     | RECEPTACLE(AC)                |  |
| SPARE                           |                                                             |                                      |         |         | 20     | 1P          | 13         |   |   |   | 14         | 1P           | 20     |           |         |         |                        | SPARE                         |  |
| SPARE                           |                                                             |                                      |         |         | 20     | 1P          | 15         |   |   |   | 16         | 1P           | 20     |           |         |         |                        | SPARE                         |  |
|                                 |                                                             |                                      |         |         |        |             |            |   |   |   |            |              |        |           |         |         |                        |                               |  |
|                                 |                                                             | 2800.00                              | 7700.00 | 1500.00 |        |             |            |   |   |   |            |              |        | 5700.00   | 7700.00 | 1500.00 |                        |                               |  |
|                                 |                                                             |                                      |         |         |        |             |            |   |   |   |            |              |        |           |         |         |                        |                               |  |
|                                 |                                                             |                                      |         |         |        |             |            |   |   |   |            |              |        |           |         |         |                        |                               |  |
|                                 | ØA = 8500.00 TOTAL CONNECTED LOAD = 26900.00 I(AMP) = 63.86 |                                      |         |         |        |             |            |   |   |   |            |              |        |           |         |         |                        |                               |  |
|                                 | ØB = 15400.00 DEMAND FACTOR = 1                             |                                      |         |         |        |             |            |   |   |   |            |              |        |           |         |         |                        |                               |  |
|                                 | ØC = 3000.00 TOTAL CONNECTED LOAD +25% SPARE= 33625.00      |                                      |         |         |        |             |            |   |   |   |            |              |        |           |         |         |                        |                               |  |

## PANELBOARD SCHEDULE

| PANEL TAG:                      | EP5                                                          | MAINS:    |         |         |        |             |                        |   |   |   |                        | 100A T/250AF,120/240V,1P,60HZ |        |           | NEUTRAL: YES |         |                        |                               |
|---------------------------------|--------------------------------------------------------------|-----------|---------|---------|--------|-------------|------------------------|---|---|---|------------------------|-------------------------------|--------|-----------|--------------|---------|------------------------|-------------------------------|
| LOCATION:                       | UNIT 4 FIRST FLOOR PLAN                                      | MOUNTING: |         |         |        |             |                        |   |   |   |                        | SURFACE MOUNTED               |        |           | GROUND: YES  |         |                        |                               |
| ENCLOSURE:                      | IP42                                                         | VOLTAGE:  |         |         |        |             |                        |   |   |   |                        | 120/240V,1P,3W                |        |           |              |         |                        |                               |
| FEEDER ENTRANCE:                | TOP                                                          |           |         |         |        |             |                        |   |   |   |                        |                               |        |           |              |         |                        |                               |
| LOAD<br>SERVED                  | Romex Wire Size (Inch)                                       | LOAD (VA) |         |         | C<br>B | P<br>O<br>L | C<br>K<br>T<br>N<br>O. | R | Y | B | C<br>K<br>T<br>N<br>O. | P<br>O<br>L                   | C<br>B | LOAD (VA) |              |         | CONDUIT<br>SIZE (INCH) | LOAD<br>SERVED                |
|                                 |                                                              | ØA        | ØB      | ØC      |        |             |                        |   |   |   |                        |                               |        | ØA        | ØB           | ØC      |                        |                               |
| LIGHTING                        | 2#12, 1#12(G), 3/4"C                                         | 800.00    |         |         | 20     | 1P          | 1                      |   |   |   | 2                      | 1P                            | 15     | 1200.00   |              |         | 2#14, 1#14(G), 3/4"C   | RECEPTACLE - BEDROOM 2 (AFCI) |
| RECEPTACLE - DINING/LIVING ROOM | 2#14, 1#14(G), 3/4"C                                         |           | 1200.00 |         | 15     | 1P          | 3                      |   |   |   | 4                      | 1P                            | 15     |           | 1200.00      |         | 2#14, 1#14(G), 3/4"C   | BATH - GFCI                   |
| SMALL APPLIANCE                 | 2#12, 1#12(G), 3/4"C                                         |           |         | 1500.00 | 15     | 1P          | 5                      |   |   |   | 6                      | 1P                            | 15     |           |              | 1500.00 | 2#12, 1#12(G), 3/4"C   | SMALL APPLIANCE               |
| REFRIGERATOR (AF/GF)            | 2#12, 1#12(G), 3/4"C                                         |           | 1500.00 |         | 20     | 1P          | 7                      |   |   |   | 8                      | 1P                            | 15     |           | 1500.00      |         | 2#14, 1#14(G), 3/4"C   | HOOD                          |
| MICROWAVE (AF/GF)               | 2#12, 1#12(G), 3/4"C                                         | 2000.00   |         |         | 20     | 1P          | 9                      |   |   |   | 10                     | 2P                            | 20     | 4500.00   |              |         | 2#10, 1#10(G), 3/4"C   | RANGE                         |
| WATER HEATER                    | 2#8, 1#8(G), 3/4"C                                           |           | 5000.00 |         | 60     | 3P          | 11                     |   |   |   | 12                     | 3P                            | 60     |           | 5000.00      |         | 2#6, 1#6(G), 3/4"C     | RECEPTACLE(AC)                |
| SPARE                           |                                                              |           |         |         | 20     | 1P          | 13                     |   |   |   | 14                     | 1P                            | 20     |           |              |         |                        | SPARE                         |
| SPARE                           |                                                              |           |         |         | 20     | 1P          | 15                     |   |   |   | 16                     | 1P                            | 20     |           |              |         |                        | SPARE                         |
|                                 |                                                              |           |         |         |        |             |                        |   |   |   |                        |                               |        |           |              |         |                        |                               |
|                                 |                                                              | 2800.00   | 7700.00 | 1500.00 |        |             |                        |   |   |   |                        |                               |        | 5700.00   | 7700.00      | 1500.00 |                        |                               |
|                                 |                                                              |           |         |         |        |             |                        |   |   |   |                        |                               |        |           |              |         |                        |                               |
|                                 |                                                              |           |         |         |        |             |                        |   |   |   |                        |                               |        |           |              |         |                        |                               |
|                                 | ØA = 8500.00 TOTAL CONNECTED LOAD = 26900.00 I (AMP) = 63.86 |           |         |         |        |             |                        |   |   |   |                        |                               |        |           |              |         |                        |                               |
|                                 | ØB = 15400.00 DEMAND FACTOR = 1                              |           |         |         |        |             |                        |   |   |   |                        |                               |        |           |              |         |                        |                               |
|                                 | ØC = 3000.00 TOTAL CONNECTED LOAD +25% SPARE= 33625.00       |           |         |         |        |             |                        |   |   |   |                        |                               |        |           |              |         |                        |                               |
|                                 | ØC = 0.00 TOTAL CONNECTED LOAD +25% SPARE= 1.25              |           |         |         |        |             |                        |   |   |   |                        |                               |        |           |              |         |                        |                               |

## PANELBOARD SCHEDULE

[illegible]

OWNER SIGNATURE \_\_\_\_\_

## REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
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| 03  |      |             |          |
| 04  |      |             |          |

**CLIENT/CONSULTANT APPROVAL STATUS**

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

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| DRAWING STATUS |
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DESIGN DRAWING

DISCIPLINE: ELECTRICAL

PROJECT TITLE:

COMMERCIAL MIXED-USE

DRAWING TITLE:

## PANEL SCHEDULES

|                         |                           |                               |
|-------------------------|---------------------------|-------------------------------|
| <u>DRAWN BY:</u><br>G.M | <u>CHECKED BY:</u><br>N.B | <u>DATE:</u><br>11/05/2026    |
| <u>JOB NO</u>           | SHT NO<br>JS-EL-05        | <u>SCALE:</u><br>AS SHOWN @A1 |



| PANELBOARD SCHEDULE             |                        |                                                                                                            |         |         |     |                                          |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|---------------------------------|------------------------|------------------------------------------------------------------------------------------------------------|---------|---------|-----|------------------------------------------|------------|---|---|---|------------|-------|-----|-----------|---------|--------------|----------------------|--------------------------------|--|
| PANEL TAG:                      |                        | EP7                                                                                                        |         |         |     | MAINS: 100A T/250A F, 120/240V, 1P, 60HZ |            |   |   |   |            |       |     |           |         | NEUTRAL: YES |                      |                                |  |
| LOCATION:                       |                        | UNIT 6 SECOND FLOOR PLAN                                                                                   |         |         |     | MOUNTING: SURFACE MOUNTED                |            |   |   |   |            |       |     |           |         | GROUND: YES  |                      |                                |  |
| ENCLOSURE:                      |                        | IP42                                                                                                       |         |         |     | VOLTAGE: 120/240V, 1P, 3W                |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
| FEEDER ENTRANCE                 |                        | TOP                                                                                                        |         |         |     |                                          |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
| LOAD SERVED                     | Romex Wire Size (Inch) | LOAD (VA)                                                                                                  |         |         | C B | P O L                                    | C K T N O. | R | Y | B | C K T N O. | P O L | C B | LOAD (VA) |         |              | CONDUIT SIZE (INCH)  | LOAD SERVED                    |  |
|                                 |                        | ØA                                                                                                         | ØB      | ØC      |     |                                          |            |   |   |   |            |       |     | ØA        | ØB      | ØC           |                      |                                |  |
| LIGHTING                        | 2#12, 1#12(G), 3/4"C   | 800.00                                                                                                     |         |         | 20  | 1P                                       | 1          |   |   |   | 2          | 1P    | 15  | 1200.00   |         |              | 2#14, 1#14(G), 3/4"C | RECEPTACLE - BEDROOM 2 (A FCI) |  |
| RECEPTACLE - DINING/LIVING ROOM | 2#14, 1#14(G), 3/4"C   |                                                                                                            | 1200.00 |         | 15  | 1P                                       | 3          |   |   |   | 4          | 1P    | 15  |           | 1200.00 |              | 2#14, 1#14(G), 3/4"C | BATH - GFCI                    |  |
| SMALL APPLIANCE                 | 2#12, 1#12(G), 3/4"C   |                                                                                                            |         | 1500.00 | 15  | 1P                                       | 5          |   |   |   | 6          | 1P    | 15  |           |         | 1500.00      | 2#12, 1#12(G), 3/4"C | SMALL APPLIANCE                |  |
| REFRIGERATOR (A/F/GF)           | 2#12, 1#12(G), 3/4"C   |                                                                                                            | 1500.00 |         | 20  | 1P                                       | 7          |   |   |   | 8          | 1P    | 15  |           | 1500.00 |              | 2#14, 1#14(G), 3/4"C | HOOD                           |  |
| MICROWAVE (A/F/GF)              | 2#12, 1#12(G), 3/4"C   | 2000.00                                                                                                    |         |         | 20  | 1P                                       | 9          |   |   |   | 10         | 2P    | 20  | 4500.00   |         |              | 2#10, 1#10(G), 3/4"C | RANGE                          |  |
| WATER HEATER                    | 2#8, 1#8(G), 3/4"C     |                                                                                                            | 5000.00 |         | 60  | 3P                                       | 11         |   |   |   | 12         | 3P    | 60  |           | 5000.00 |              | 2#6, 1#6(G), 3/4"C   | RECEPTACLE(A/C)                |  |
| SPARE                           |                        |                                                                                                            |         |         | 20  | 1P                                       | 13         |   |   |   | 14         | 1P    | 20  |           |         |              |                      | SPARE                          |  |
| SPARE                           |                        |                                                                                                            |         |         | 20  | 1P                                       | 15         |   |   |   | 16         | 1P    | 20  |           |         |              |                      | SPARE                          |  |
|                                 |                        |                                                                                                            |         |         |     |                                          |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|                                 |                        | 2800.00                                                                                                    | 7700.00 | 1500.00 |     |                                          |            |   |   |   |            |       |     | 5700.00   | 7700.00 | 1500.00      |                      |                                |  |
|                                 |                        | <div> <div>ØA = 8500.00</div> <div>TOTAL CONNECTED LOAD = 26900.00</div> <div>I (AMP) = 63.86</div> </div> |         |         |     |                                          |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|                                 |                        | <div> <div>ØB = 15400.00</div> <div>DEMAND FACTOR = 1</div> </div>                                         |         |         |     |                                          |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|                                 |                        | <div> <div>ØC = 3000.00</div> <div>TOTAL CONNECTED LOAD +25% SPARE= 33625.00</div> </div>                  |         |         |     |                                          |            |   |   |   |            |       |     |           |         |              |                      |                                |  |

| PANELBOARD SCHEDULE             |                        |                                                                                                                  |         |                                         |     |       |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|---------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------------|-----|-------|------------|---|---|---|------------|-------|-----|-----------|---------|--------------|----------------------|--------------------------------|--|
| PANEL TAG:                      |                        | EP8                                                                                                              |         | MAINS: 100A T/250AF, 120/240V, 1P, 60HZ |     |       |            |   |   |   |            |       |     |           |         | NEUTRAL: YES |                      |                                |  |
| LOCATION:                       |                        | UNIT 7 SECOND FLOOR PLAN                                                                                         |         | MOUNTING: SURFACE MOUNTED               |     |       |            |   |   |   |            |       |     |           |         | GROUND: YES  |                      |                                |  |
| ENCLOSURE:                      |                        | IP42                                                                                                             |         | VOLTAGE: 120/240V, 1P, 3W               |     |       |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
| FEEDER ENTRANCE:                |                        | TOP                                                                                                              |         |                                         |     |       |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
| LOAD SERVED                     | Romex Wire Size (Inch) | LOAD (VA)                                                                                                        |         |                                         | C B | P O L | C K T N O. | R | Y | B | C K T N O. | P O L | C B | LOAD (VA) |         |              | CONDUIT SIZE (INCH)  | LOAD SERVED                    |  |
|                                 |                        | ØA                                                                                                               | ØB      | ØC                                      |     |       |            |   |   |   |            |       |     | ØA        | ØB      | ØC           |                      |                                |  |
| LIGHTING                        | 2#12, 1#12(G), 3/4"C   | 800.00                                                                                                           |         |                                         | 20  | 1P    | 1          |   |   |   | 2          | 1P    | 15  | 1200.00   |         |              | 2#14, 1#14(G), 3/4"C | RECEPTACLE - BEDROOM 2 (A FCI) |  |
| RECEPTACLE - DINING/LIVING ROOM | 2#14, 1#14(G), 3/4"C   |                                                                                                                  | 1200.00 |                                         | 15  | 1P    | 3          |   |   |   | 4          | 1P    | 15  |           | 1200.00 |              | 2#14, 1#14(G), 3/4"C | RECEPTACLE - BEDROOM 2 (A FCI) |  |
| BATH - GFCI                     | 2#14, 1#14(G), 3/4"C   |                                                                                                                  |         | 1200.00                                 | 15  | 1P    | 5          |   |   |   | 6          | 1P    | 15  |           |         | 1500.00      | 2#12, 1#12(G), 3/4"C | SMALL APPLIANCE                |  |
| SMALL APPLIANCE                 | 2#12, 1#12(G), 3/4"C   |                                                                                                                  | 1500.00 |                                         | 15  | 1P    | 7          |   |   |   | 8          | 1P    | 20  |           | 1500.00 |              | 2#12, 1#12(G), 3/4"C | REFRIGERATOR (A F/GF)          |  |
| HOOD                            | 2#14, 1#14(G), 3/4"C   | 1500.00                                                                                                          |         |                                         | 15  | 1P    | 9          |   |   |   | 10         | 1P    | 20  | 2000.00   |         |              | 2#12, 1#12(G), 3/4"C | MICROWAVE (A F/GF)             |  |
| RANGE 1                         | 2#10, 1#10(G), 3/4"C   |                                                                                                                  | 4500.00 |                                         | 20  | 2P    | 11         |   |   |   | 12         | 3P    | 60  |           | 5000.00 |              | 2#8, 1#8(G), 3/4"C   | WATER HEATER                   |  |
| RECEPTACLE(A/C)                 | 2#6, 1#6(G), 3/4"C     |                                                                                                                  |         | 5000.00                                 | 60  | 1P    | 13         |   |   |   | 14         | 1P    | 20  |           |         |              |                      | SPA RE                         |  |
| SPARE                           |                        |                                                                                                                  |         |                                         | 20  | 1P    | 15         |   |   |   | 16         | 1P    | 20  |           |         |              |                      | SPA RE                         |  |
|                                 |                        |                                                                                                                  |         |                                         |     |       |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|                                 |                        | 2300.00                                                                                                          | 7200.00 | 6200.00                                 |     |       |            |   |   |   |            |       |     | 3200.00   | 7700.00 | 1500.00      |                      |                                |  |
|                                 |                        |                                                                                                                  |         |                                         |     |       |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|                                 |                        | <div> <div> ØA = 5500.00 </div> <div> TOTAL CONNECTED LOAD = 28100.00 </div> <div> I (AMP) = 66.71 </div> </div> |         |                                         |     |       |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|                                 |                        | <div> <div> ØB = 14900.00 </div> <div> DEMAND FACTOR = 1 </div> </div>                                           |         |                                         |     |       |            |   |   |   |            |       |     |           |         |              |                      |                                |  |
|                                 |                        | <div> <div> ØC = 7700.00 </div> <div> TOTAL CONNECTED LOAD +25% SPARE= 35125.00 </div> </div>                    |         |                                         |     |       |            |   |   |   |            |       |     |           |         |              |                      |                                |  |

| PANELBOARD SCHEDULE             |                        |                                                                                                                                                                                                                                                                                    |         |                                       |        |         |            |   |   |   |            |         |        |              |         |         |                        |                                |  |
|---------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------------|--------|---------|------------|---|---|---|------------|---------|--------|--------------|---------|---------|------------------------|--------------------------------|--|
| PANEL TAG:                      |                        | EP9                                                                                                                                                                                                                                                                                |         | MAINS: 100A/T/250AF,120/240V ,1P,60HZ |        |         |            |   |   |   |            |         |        | NEUTRAL: YES |         |         |                        |                                |  |
| LOCATION:                       |                        | UNIT 8 SECOND FLOOR PLAN                                                                                                                                                                                                                                                           |         | MOUNTING: SURFACE MOUNTED             |        |         |            |   |   |   |            |         |        | GROUND: YES  |         |         |                        |                                |  |
| ENCLOSURE:                      |                        | IP42                                                                                                                                                                                                                                                                               |         | VOLTAGE: 120/240V,1P,3W               |        |         |            |   |   |   |            |         |        |              |         |         |                        |                                |  |
| FEEDER ENTRANCE                 |                        | TOP                                                                                                                                                                                                                                                                                |         |                                       |        |         |            |   |   |   |            |         |        |              |         |         |                        |                                |  |
| LOAD<br>SERVED                  | Romex Wire Size (Inch) | LOAD (VA)                                                                                                                                                                                                                                                                          |         |                                       | C<br>B | P<br>OL | CKT<br>NO. | R | Y | B | CKT<br>NO. | P<br>OL | C<br>B | LOAD (VA)    |         |         | CONDUIT<br>SIZE (INCH) | LOAD<br>SERVED                 |  |
|                                 |                        | ØA                                                                                                                                                                                                                                                                                 | ØB      | ØC                                    |        |         |            |   |   |   |            |         |        | ØA           | ØB      | ØC      |                        |                                |  |
| LIGHTING                        | 2#12, 1#12(G), 3/4"C   | 800.00                                                                                                                                                                                                                                                                             |         |                                       | 20     | 1P      | 1          |   |   |   | 2          | 1P      | 15     | 1200.00      |         |         | 2#14, 1#14(G), 3/4"C   | RECEPTACLE - BEDROOM 2 (A FCI) |  |
| RECEPTACLE - DINING/LIVING ROOM | 2#14, 1#14(G), 3/4"C   |                                                                                                                                                                                                                                                                                    | 1200.00 |                                       | 15     | 1P      | 3          |   |   |   | 4          | 1P      | 15     |              | 1200.00 |         | 2#14, 1#14(G), 3/4"C   | RECEPTACLE - BEDROOM 2 (A FCI) |  |
| BATH - GFCI                     | 2#14, 1#14(G), 3/4"C   |                                                                                                                                                                                                                                                                                    |         | 1200.00                               | 15     | 1P      | 5          |   |   |   | 6          | 1P      | 15     |              |         | 1500.00 | 2#12, 1#12(G), 3/4"C   | SMALL APPLIANCE                |  |
| SMALL APPLIANCE                 | 2#12, 1#12(G), 3/4"C   |                                                                                                                                                                                                                                                                                    | 1500.00 |                                       | 15     | 1P      | 7          |   |   |   | 8          | 1P      | 20     |              | 1500.00 |         | 2#12, 1#12(G), 3/4"C   | REFRIGERATOR (A F/GF)          |  |
| HOOD                            | 2#14, 1#14(G), 3/4"C   | 1500.00                                                                                                                                                                                                                                                                            |         |                                       | 15     | 1P      | 9          |   |   |   | 10         | 1P      | 20     | 2000.00      |         |         | 2#12, 1#12(G), 3/4"C   | MICROWAVE (A F/GF)             |  |
| RANGE 1                         | 2#10, 1#10(G), 3/4"C   |                                                                                                                                                                                                                                                                                    | 4500.00 |                                       | 20     | 2P      | 11         |   |   |   | 12         | 3P      | 60     |              | 5000.00 |         | 2#8, 1#8(G), 3/4"C     | WATER HEATER                   |  |
| RECEPTACLE(A/C)                 | 2#6, 1#6(G), 3/4"C     |                                                                                                                                                                                                                                                                                    |         | 5000.00                               | 60     | 1P      | 13         |   |   |   | 14         | 1P      | 20     |              |         |         |                        | SPARE                          |  |
| SPARE                           |                        |                                                                                                                                                                                                                                                                                    |         |                                       | 20     | 1P      | 15         |   |   |   | 16         | 1P      | 20     |              |         |         |                        | SPARE                          |  |
|                                 |                        |                                                                                                                                                                                                                                                                                    |         |                                       |        |         |            |   |   |   |            |         |        |              |         |         |                        |                                |  |
|                                 |                        | 2300.00                                                                                                                                                                                                                                                                            | 7200.00 | 6200.00                               |        |         |            |   |   |   |            |         |        | 3200.00      | 7700.00 | 1500.00 |                        |                                |  |
|                                 |                        | <div style="display: flex; justify-content: space-between;"> <div> ØA = 5500.00<br/> ØB = 14900.00<br/> ØC = 7700.00 </div> <div> TOTAL CONNECTED LOAD = 28100.00<br/> DEMAND FACTOR = 1<br/> TOTAL CONNECTED LOAD +25% SPARE= 35125.00 </div> <div> I (AMP) = 66.71 </div> </div> |         |                                       |        |         |            |   |   |   |            |         |        |              |         |         |                        |                                |  |

OWNER SIGNATURE

## REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
| 01  |      |             |          |
| 02  |      |             |          |
| 03  |      |             |          |
| 04  |      |             |          |

**CLIENT/CONSULTANT APPROVAL STATUS**

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

**DRAWING STATUS:**

## DESIGN DRAWING

DISCIPLINE: ELECTRICAL

PROJECT TITLE:

## COMMERCIAL MIXED-USE

DRAWING TITLE:

## PANEL SCHEDULES

|                         |                           |                               |
|-------------------------|---------------------------|-------------------------------|
| <u>DRAWN BY:</u><br>G.M | <u>CHECKED BY:</u><br>N.B | <u>DATE:</u><br>11/05/2026    |
| <u>JOB NO</u>           | SHT NO<br>JS-EL-06        | <u>SCALE:</u><br>AS SHOWN @A1 |







GENERAL PLUMBING NOTES:

1. LOCATION AND SIZES OF ALL EXISTING SERVICES ARE APPROXIMATE. VERIFY ALL UTILITIES PRIOR TO COMMENCING WORK.

2. FOR EXACT LOCATION OF PLUMBING FIXTURES, DRAINS AND EQUIPMENT SEE ARCHITECTURAL DRAWINGS.

3. ALL DOMESTIC WATER AND VENT PIPING SHOWN TO BE RUN ABOVE CEILING UNLESS OTHERWISE NOTED.

4. ALL SANITARY PIPING SHOWN TO BE RUN BELOW FLOOR UNLESS OTHERWISE NOTED.

5. ENTIRE INSTALLATION INCLUDING MATERIALS, EQUIPMENT AND WORKMANSHIP SHALL CONFIRM WITH ALL APPLICABLE LAWS, CODES AND REGULATIONS OF MUNICIPAL, STATE AND FEDERAL AUTHORITIES, ALSO THE PLUMBING CODE 2018 IPC, AND OTHER REGULATORY BODIES HAVING JURISDICTION OVER THE CLASS OF WORK.

6. THE CONTRACTOR SHALL MAKE TESTS FOR ACCEPTANCE AND APPROVAL AS REQUIRED BY THE CODE. REQUIRED TESTS SHALL BE PERFORMED IN THE PRESENCE OF OWNER’S REPRESENTATIVE UNLESS WAIVED IN WRITING.

7. THE CONTRACTOR SHALL FIELD COORDINATE THE WORK WITH THAT OF ALL OTHER TRADES.

8. ALL DOMESTIC WATER PIPING SHALL BE INSULATED AS REQUIRED.

9. DOMESTIC WATER PIPING MATERIAL: ASTM B88 TYPE L OR M COPPER TUBE WITH WROUGHT OR CAST COPPER FITTINGS AND LEAD–FREE SOLDERED JOINTS. INSTALL DOMESTIC WATER PIPING LEVEL WITHOUT PITCH AND PLUMB.

THE PEXA PIPING SHALL CONFORM TO THE REQUIREMENTS OF ASTM F876–XX FOR CROSSLINKED POLYETHYLENE (PEXA) TUBING, WITH A MINIMUM PRESSURE RATING OF 200 PSI AT 73.4°F (23°C). DRAIN, WASTE, STORM, AND VENT PIPING MATERIAL: FOR BELOW GROUND SOIL, WASTE AND BENT PIPING SHOULD BE SERVICE WEIGHT HUB AND SPIGOT CAST IRON CONFORMING TO ASTM A74, WITH LEAD AND OAKUM JOINTS AS ALLOWED BY PLUMBING CODE; FOR ABOVE GROUND SOIL, WASTE AND VENT PIPING SHOULD BE HUBLESS CAST–IRON.

PVC DMV PIPING MAY ALSO BE USED FOR DRAINING APPLICATIONS. PVC DWV PIPE SHALL BE MADE FROM TYPE 1, GRADE 1, PVC COMPOUND AS SPECIFIED IN ASTM D1784. THE PVC DWV PIPE SHALL HAVE A NOMINAL INSIDE DIAMETER (ID) OF 4 INCHES (101.6 MM). THE PIPE SHALL CONFORM TO THE STANDARD DIMENSIONS FOR 4–INCH PVC DWV PIPE AS SPECIFIED IN ASTM D2665. EACH LENGTH OF PVC DWV PIPE SHALL BE CLEARLY MARKED WITH THE MANUFACTURER’S NAME OR TRADEMARK, PIPE SIZE, MATERIAL DESIGNATION (E.G., PVC), AND THE ASTM STANDARD DESIGNATION (E.G., ASTM D2665).

10. ANY ITEM OF MATERIALS NOT INDICATED ON THE DRAWINGS OR NOT SPECIFIED AND WHICH ARE REQUIRED FOR THE COMPLETE INSTALLATION OF ANY PART OF THE WORK DESCRIBED, SHALL BE FURNISHED AS CODE REQUIRED.

11. INSTALL CLEANOUT AT OR NEAR THE BOTTOM OF SANITARY/WASTE STACK.

12. PROVIDE PROPER ACCESS PANELS AS SPECIFIED IN WALLS AND CEILINGS FOR ANY EQUIPMENT, DEVICES, VALVES, SHOCK ABSORBERS, TRAP PRIMERS, CLEANOUTS ETC. REQUIRING ACCESS.

| PLUMBING FIXTURE AND FAUCETS REFERENCED STANDARDS |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARK                                              | DESCRIPTION                  | STANDARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| WC–1                                              | WATER CLOSET                 | WATER CLOSETS SHALL CONFIRM TO THE WATER CONSUMPTION REQUIREMENTS OF SECTION 604.4 AND SHALL CONFIRM TO ANSI Z124.4. ASME A112.19.2M, CSA B45.1, CSA B45.4 OR CSA B45.5. WATER CLOSETS SHALL CONFIRM TO THE HYDRAULIC PERFORMANCE REQUIREMENTS OF ASME A112.19.19.9M, CSA B45.1, CSA B45.4 OR CSA B45.5. ELECTRO HYDRAULIC WATER CLOSETS SHALL COMPLY WITH ASME A112.19.13.                                                                                                                                                                                                                                                                                                             |
| L–1                                               | LAVATORY                     | LAVATORIES SHALL CONFIRM TO ANSI Z124.3. ASME A112.19.1M, ASME A112.19.2M, ASME A112.19.3M, ASME A112.19.4M, ASME A112.19.9M, CSA B45.2, CSA B45.3, CSA B45.4, ROUP WASH UP EQUIPMENT SHALL CONFIRM TO THE REQUIREMENTS OF SECTION 402. EVERY 20 INCHES (608MM) OF RIM SPACE SHALL BE CONSIDERED AS ONE LAVATORY.                                                                                                                                                                                                                                                                                                                                                                       |
| KS–1                                              | KITCHEN SINK                 | SINKS SHALL CONFIRM TO ANSI Z124.6, ASME A112.19.1M, ASME A112.19.2M, ASME A112.19.3M, ASME A112.19.4M, ASME A112.19.9M, CSA B45.1, CSA B45.2, CSA B45.3, OR CSA B45.4,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                   | FAUCETS AND FIXTURE FITTINGS | FAUCETS AND FIXTURE FITTINGS SHALL CONFORM TO ASME A112.18.1 OR CSA B125. FAUCETS AND FIXTURE FITTINGS THAT SUPPLY DRINKING WATER FOR HUMAN INGESTION SHALL CONFORM TO THE REQUIREMENTS OF NSF 61, SECTION 9. FLEXIBLE WATER CONNECTORS EXPOSED TO CONTINUOUS PRESSURE SHALL CONFORM TO THE REQUIREMENTS OF SECTION 605.6                                                                                                                                                                                                                                                                                                                                                               |
|                                                   |                              | HAND HELD SHOWERS SHALL CONFORM TO ASSE 1014 OR CSA B125.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                   |                              | SHOWER AND TUB SHOWER COMBINATION VALVES SHALL BE BALANCED PRESSURE, THERMOSTATIC OR COMBINATION BALANCED PRESSURE/THERMOSTATIC VALVES THAT CONFORM TO THE REQUIREMENTS OF ASSE 1016 OR CSA B125, MULTIPLE (GANG) SHOWERS SUPPLIED WITH A SINGLE TEMPERED WATER SUPPLY PIPE SHALL HAVE THE WATER SUPPLY FOR SUCH SHOWERS CONTROLLED BY A MASTER THERMOSTATIC MIXING VALVE COMPLYING WITH ASSE 1017. SHOWER AND TUB SHOWER COMBINATION VALVES AND MASTER THERMOSTATIC MIXING VALVES REQUIRED BY THIS SECTION SHALL BE EQUIPPED WITH A MEANS TO LIMIT THE MAXIMUM SETTING OF THE VALVE TO 120°F (49°C), WHICH SHALL BE FIELD ADJUSTED IN ACCORDANCE WITH THE MANUFACTURER’S INSTRUCTIONS. |
|                                                   |                              | FAUCETS AND FIXTURE FITTINGS WITH HOSE CONNECTED OUTLETS SHALL CONFORM TO ASME A112.18.3M.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                   |                              | TEMPERATURE ACTUATED, FLOW REDUCTION DEVICES. WHERE INSTALLED FOR INDIVIDUAL FIXTURE FITTINGS, SHALL CONFORM TO ASSE 1062. SUCH VALVES SHALL NOT BE USED ALONE AS A SUBSTITUTE FOR THE BALANCED PRESSURE. THERMOSTATIC OR COMBINATION SHOWER VALVES REQUIRED IN SECTION 424.3                                                                                                                                                                                                                                                                                                                                                                                                           |

PLUMBING NOTES:

1. ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE NOTED.

2. ALL PLUMBING INSTALLATIONS SHALL CONFIRM WITH THE REQUIREMENTS OF MANUFACTURER RECOMMENDATION AND OR 2018 InPC (2006 IPC W/AMENDMENTS).

3. ALL PLUMBING MATERIALS SHALL BE AS SPECIFIED IN THE CONTRACT SPECS AND BOQ.

4. ALL PIPE SIZES SHOWN ARE NOMINAL DIAMETER IN INCHES. UNLESS OTHERWISE MENTIONED.

5. INSTALLER SHOULD NOT SCALE THE DRAWING.

6. ALL PIPE PENETRATIONS IN WALL & SLABS SHALL BE THROUGH PIPE SLEEVES.

7. IF ANY ERROR OR OMISSION IS FOUND IN DRAWING THIS OFFICE MAY BE CONSULTED FOR RECTIFICATION.

| TABLE 604.4<br>MAXIMUM FLOW RATES AND CONSUMPTION FOR PLUMBING FIXTURES AND FIXTURE FITTINGS                                                                                                                        |                                    |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--|
| PLUMBING FIXTURE<br>R FIXTURE FITTING                                                                                                                                                                               | MAXIMUM FLOW RATE<br>OR QUANTITY b |  |
| LAVATORY, PRIVATE                                                                                                                                                                                                   | 1.5 gpm at 60 psi                  |  |
| SINK FAUCET                                                                                                                                                                                                         | 1.5 gpm at 60 psi                  |  |
| WATER CLOSET                                                                                                                                                                                                        | 1.28 GALLONS PER FLUSHING CYCLE    |  |
| FOR SI:                                                                                                                                                                                                             |                                    |  |
| 1 GALLON = 3.785 L, 1 GALLON PER MINUTE = 3.785 L/M,<br>1 POUND PER SQUARE INCH = 6.895 KPA.                                                                                                                        |                                    |  |
| A. A HAND HELD SHOWER SPRAY IS A SHOWER HEAD.                                                                                                                                                                       |                                    |  |
| B. CONSUMPTION TOLERANCES SHALL BE DETERMINED FROM REFERENCED STANDARDS.                                                                                                                                            |                                    |  |
| C. WATER FACTOR IN GALLONS PER CYCLE CUBIC FOOT                                                                                                                                                                     |                                    |  |
| ALL PLUMBING FIXTURES SHALL CONTRIBUTE TO A LEEDS SOLUTION AND SHALL MEET EPA CRITERIA AND BEEN CONSIDERED AS WATER EFFICIENT PRODUCTS AS STIPULATED BY THE LEEDS TA 2009 REFERENCE GUIDE (20% REDUCTION REQUIRED). |                                    |  |

13. SANITARY: SCHEDULE 40 PVC OR CAST IRON.

SUPPLY: COPPER OR PEX (IF ALLOWED LOCALLY).

VENT: EXTEND MIN. 6” ABOVE ROOF.

| PLUMBING FIXTURES |              |                   |                 |                     |                    |          |         |             |        |      |      |                                                                                                                                          |
|-------------------|--------------|-------------------|-----------------|---------------------|--------------------|----------|---------|-------------|--------|------|------|------------------------------------------------------------------------------------------------------------------------------------------|
| MARK              | ITEM         | MAKE              | NAME            | MODEL               | TRIM               | TRAP     | STOPS   | CONNECTIONS |        |      |      | REMARKS                                                                                                                                  |
|                   |              |                   |                 |                     |                    |          |         | SAN         | VENT   | HW   | CW   |                                                                                                                                          |
| WC                | WATER CLOSET | AMERICAN STANDARD | CADET 3 1.6 GPF | 2386.012            | WITH UNIT          | INTEGRAL | NOTE #4 | 4"          | 2"     | -    | 1/2" | FLOOR MOUNTED, TANK TYPE WITH FLUSH RIGHT SYSTEM, FLUSH HANDLE ON WIDE SIDE OF STALL. SEAT: CHURCH NO. 9500C WHITE WITH CONCEALED CHECK. |
| LAV               | LAVATORY     | AMERICAN STANDARD | LUCERNE         | 0355.012 4" CENTERS | 2385.003 (0.5 GPM) | NOTE #3  | NOTE #4 | 1 1/2"      | 1 1/2" | 1/2" | 1/2" | WITH FLOOR MOUNTED CONCEALED ARM WALL CARRIER, GRID DRAIN & SINGLE HANDLE FAUCET                                                         |

OCCUPANT LOAD SHALL BE USED TO DETERMINE ALL FIXTURE COUNTS PER THE 2018 InPC (2006 IPC W/AMENDMENTS MINIMUM NUMBER OF REQUIRED PLUMBING FIXTURES ASSEMBLY – AUDITORIUMS WITHOUT PERMANENT SEATING, ART GALLERIES, EXHIBITION HALLS, MUSEUMS, LECTURE HALLS, LIBRARIES, ARCADES AND GYMNASIUMS (NOTE D)

607.2 HOT OR TEMPERED WATER SUPPLY TO FIXTURES. THE DEVELOPED LENGTH OF HOT OR TEMPERED WATER PIPING, FROM THE SOURCE OF HOT WATER TO THE FIXTURES THAT REQUIRE HOT OR TEMPERED WATER, SHALL NOT EXCEED 50 FEET (15 240 MM). RECIRCULATING SYSTEM PIPING AND HEAT–TRACED PIPING SHALL BE CONSIDERED TO BE SOURCES OF HOT OR TEMPERED WATER.

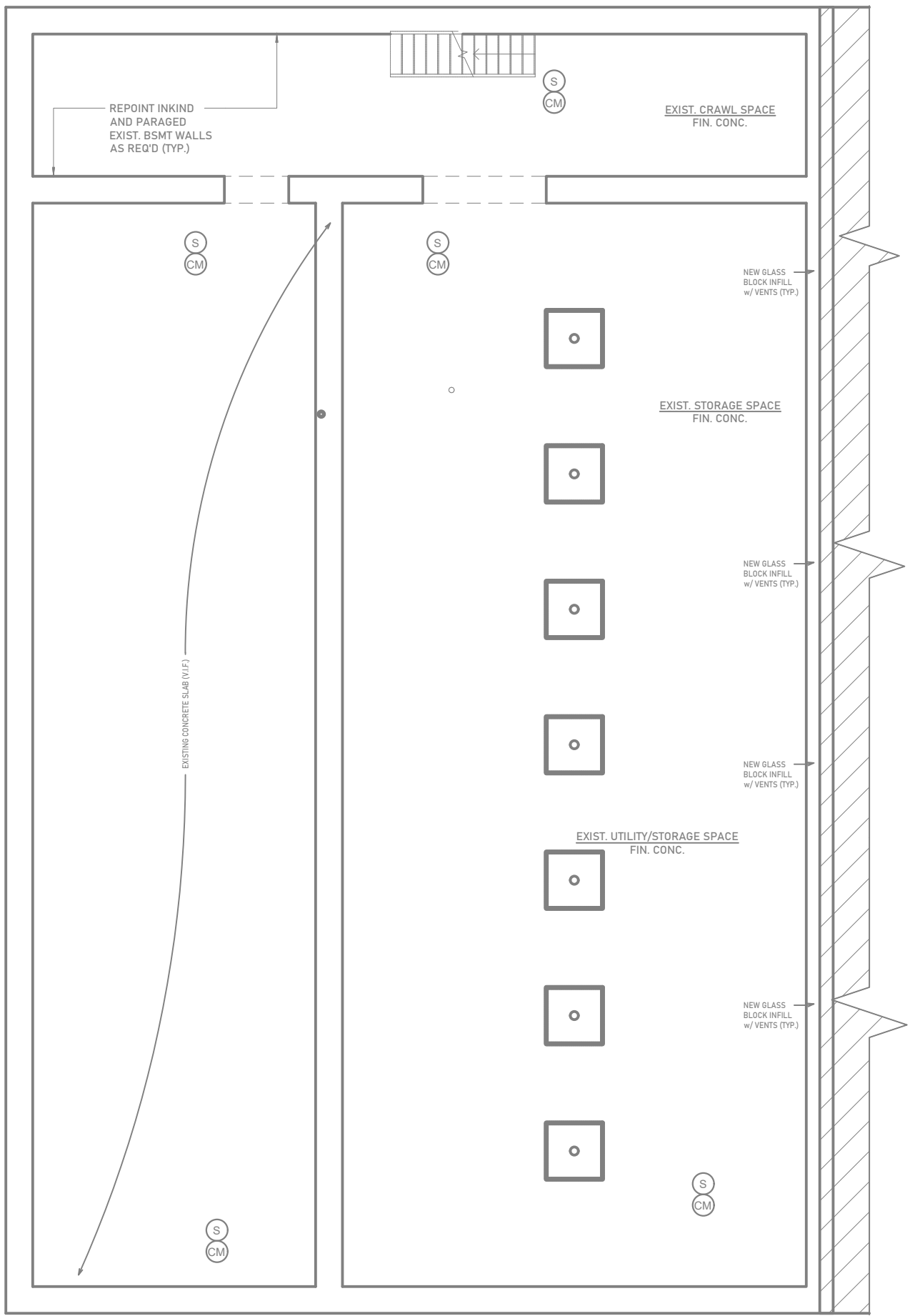
607.1.1 TEMPERATURE LIMITING MEANS. A THERMOSTAT CONTROL FOR A WATER HEATER SHALL NOT SERVE AS THE TEMPERATURE LIMITING MEANS FOR THE PURPOSES OF COMPLYING WITH THE REQUIREMENTS OF THIS CODE FOR MAXIMUM ALLOWABLE HOT OR TEMPERED WATER DELIVERY TEMPERATURE AT FIXTURES.

607.1.2 TEMPERED WATER TEMPERATURE CONTROL. TEMPERED WATER SHALL BE SUPPLIED THROUGH A WATER TEMPERATURE LIMITING DEVICE THAT CONFORMS TO ASSE 1070/ASME A112.1070/CSA B125.70 AND SHALL LIMIT THE TEMPERED WATER TO NOT GREATER THAN 110°F (43°C). THIS PROVISION SHALL NOT SUPERSEDE THE REQUIREMENT FOR PROTECTIVE SHOWER VALVES IN ACCORDANCE WITH SECTION 412.3.

607.3 THERMAL EXPANSION CONTROL. WHERE A STORAGE WATER HEATER IS SUPPLIED WITH COLD WATER THAT PASSES THROUGH A CHECK VALVE, PRESSURE REDUCING VALVE OR BACKFLOW PREVENTER, A THERMAL EXPANSION CONTROL DEVICE SHALL BE CONNECTED TO THE WATER HEATER COLD WATER SUPPLY PIPE AT A POINT THAT IS DOWNSTREAM OF ALL CHECK VALVES, PRESSURE REDUCING VALVES AND BACKFLOW PREVENTERS. THERMAL EXPANSION TANKS SHALL BE SIZED IN ACCORDANCE WITH THE TANK MANUFACTURER’S INSTRUCTIONS AND SHALL BE SIZED SUCH THAT THE PRESSURE IN THE WATER DISTRIBUTION SYSTEMSHALL NOT EXCEED THAT REQUIRED BY SECTION 604.8.

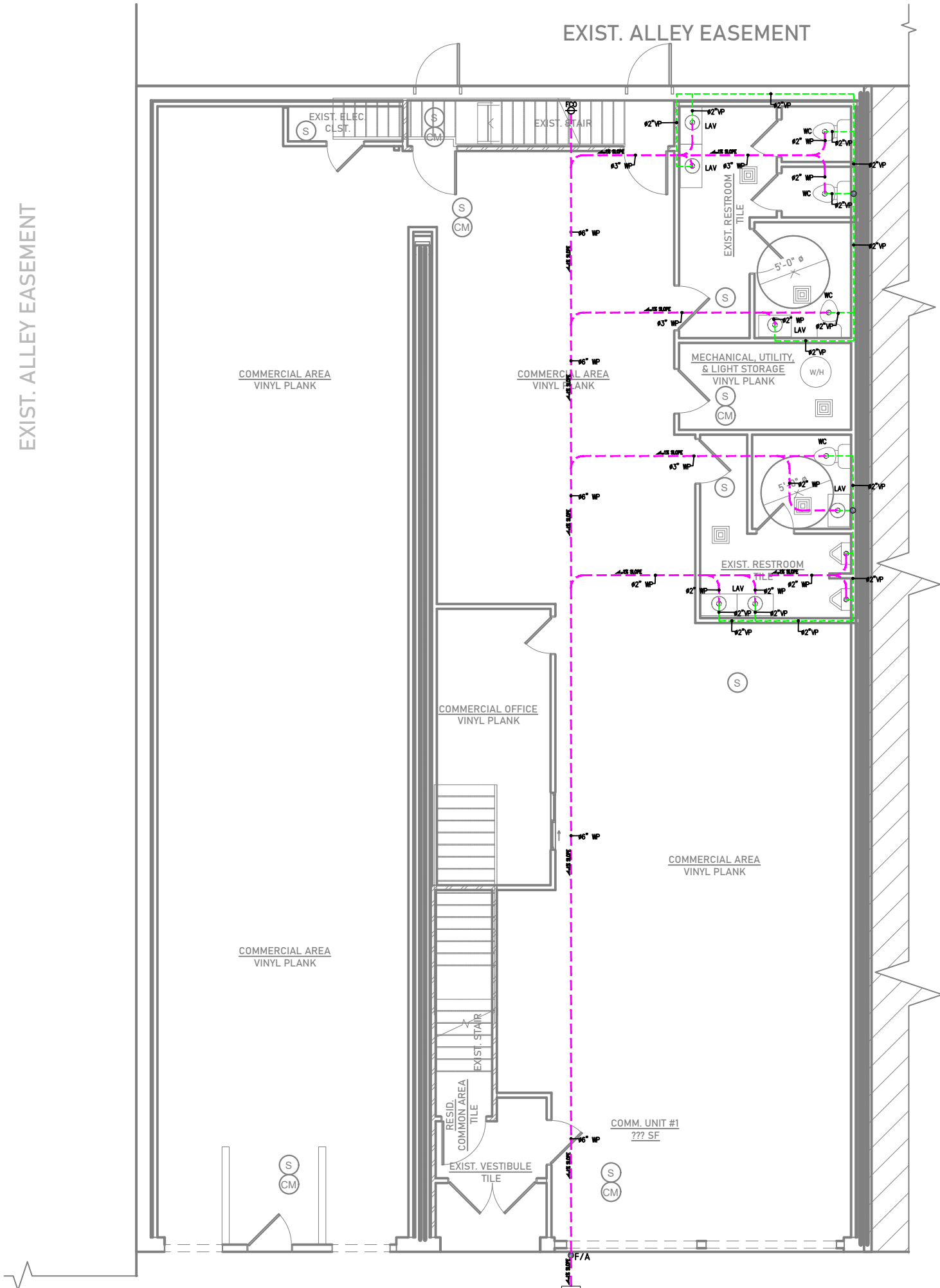
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|                                                                                                                                            |      |             |                    |                     |
|                                                                                                                                            |      |             |                    |                     |
| OWNER SIGNATURE                                                                                                                            |      |             |                    |                     |
|                                                                                                                                            |      |             |                    |                     |
| REVISION STATUS                                                                                                                            |      |             |                    |                     |
| Rev                                                                                                                                        | DATE | DESCRIPTION | DRAWN BY           |                     |
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| 01                                                                                                                                         |      |             |                    |                     |
| 02                                                                                                                                         |      |             |                    |                     |
| 03                                                                                                                                         |      |             |                    |                     |
| 04                                                                                                                                         |      |             |                    |                     |
| CLIENT/CONSULTANT APPROVAL STATUS                                                                                                          |      |             |                    |                     |
| APPROVED                                                                                                                                   | (A)  |             |                    |                     |
| APPROVED WITH NOTES                                                                                                                        | (B)  |             |                    |                     |
| RESUBMIT                                                                                                                                   | (C)  |             |                    |                     |
| DISAPPROVED                                                                                                                                | (D)  |             |                    |                     |
| OWNER:                                                                                                                                     |      |             |                    |                     |
|                                                                                                                                            |      |             |                    |                     |
| ADDRESS:                                                                                                                                   |      |             |                    |                     |
| 508 MAIN STREET, JOHNSTOWN, PA 15901                                                                                                       |      |             |                    |                     |
| DESIGN CONSULTANT:<br>TSABA CORP<br>NADIF BRACEY<br>254 CHAPMAN RD, STE 208 #13130<br>NEWARK, DELAWARE 19702<br>EMAIL:NADIF@TS9DESIGNS.COM |      |             |                    |                     |
| DRAWING STATUS:                                                                                                                            |      |             |                    |                     |
| DESIGN DRAWING                                                                                                                             |      |             |                    |                     |
| DISCIPLINE: PLUMBING                                                                                                                       |      |             |                    |                     |
| PROJECT TITLE:                                                                                                                             |      |             |                    |                     |
| COMMERCIAL MIXED-USE                                                                                                                       |      |             |                    |                     |
| DRAWING TITLE:                                                                                                                             |      |             |                    |                     |
| NOTES                                                                                                                                      |      |             |                    |                     |
| DRAWN BY:<br>G.M                                                                                                                           |      |             | CHECKED BY:<br>N.B | DATE:<br>11/05/2026 |
| JOB NO                                                                                                                                     |      |             | SHT NO<br>JS-PL-01 | SCALE:<br>NTS @A1   |



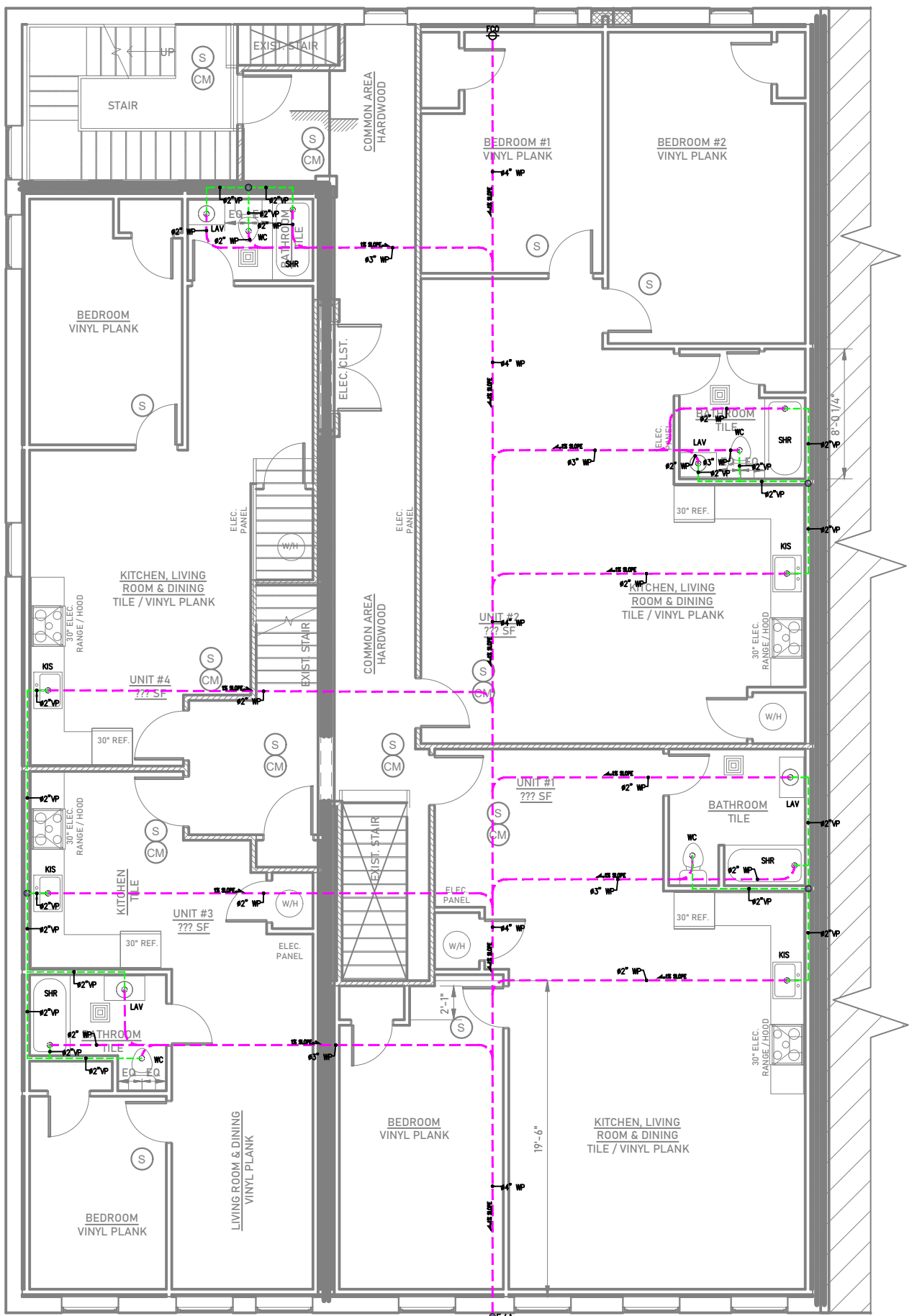


PL 02 SCALE BASEMENT FLOOR DRAINAGE PLAN 3" = 1'-0"

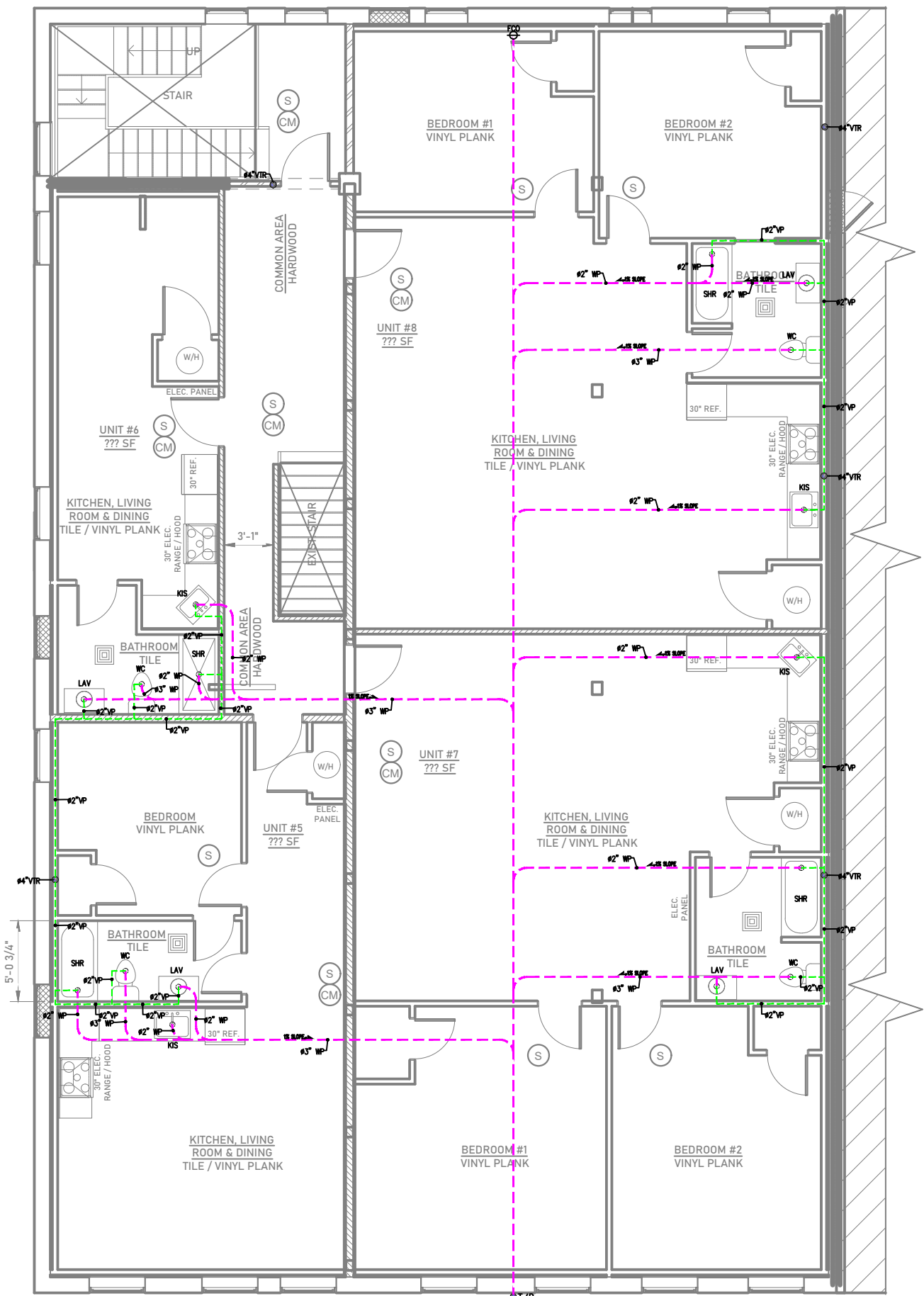
EXIST. ALLEY EASEMENT



PL 03 SCALE FIRST FLOOR DRAINAGE PLAN 3" = 1'-0"



PL 04 SCALE SECOND FLOOR DRAINAGE PLAN 3" = 1'-0"



PL 05 SCALE THIRD FLOOR DRAINAGE PLAN 3" = 1'-0"

| PLUMBING LEGEND: |                         |
|------------------|-------------------------|
| SYMBOL           | DESCRIPTION             |
| ---              | HWS - HOT WATER SUPPLY  |
| ---              | CWS - COLD WATER SUPPLY |
| ---              | SP-SOIL PIPE            |
| ---              | WP-WASTE PIPE           |
| ---              | SHUT OFF VALVE          |
| ---              | GATE VALVE              |
| WH               | WATER HEATER            |
| GT               | GREASE TRAP             |
| WC               | WATER CLOSET            |
| KS               | KITCHEN SINK            |
| AS               | ABLUTION SPRAY          |
| SHR              | SHOWER                  |
| BT               | BATH TUB                |
| FCD              | FLOOR CLEANOUT          |
| FD               | FLOOR DRAIN             |
| LAV              | LAVATORY                |
| FA               | FROM ABOVE              |

OWNER SIGNATURE

| REVISION STATUS |      |             |          |
|-----------------|------|-------------|----------|
| Rev             | DATE | DESCRIPTION | DRAWN BY |
| 00              |      |             |          |
| 01              |      |             |          |
| 02              |      |             |          |
| 03              |      |             |          |
| 04              |      |             |          |

| CLIENT/CONSULTANT APPROVAL STATUS |     |
|-----------------------------------|-----|
| APPROVED                          | (A) |
| APPROVED WITH NOTES               | (B) |
| RESUBMIT                          | (C) |
| DISAPPROVED                       | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE: PLUMBING

PROJECT TITLE:

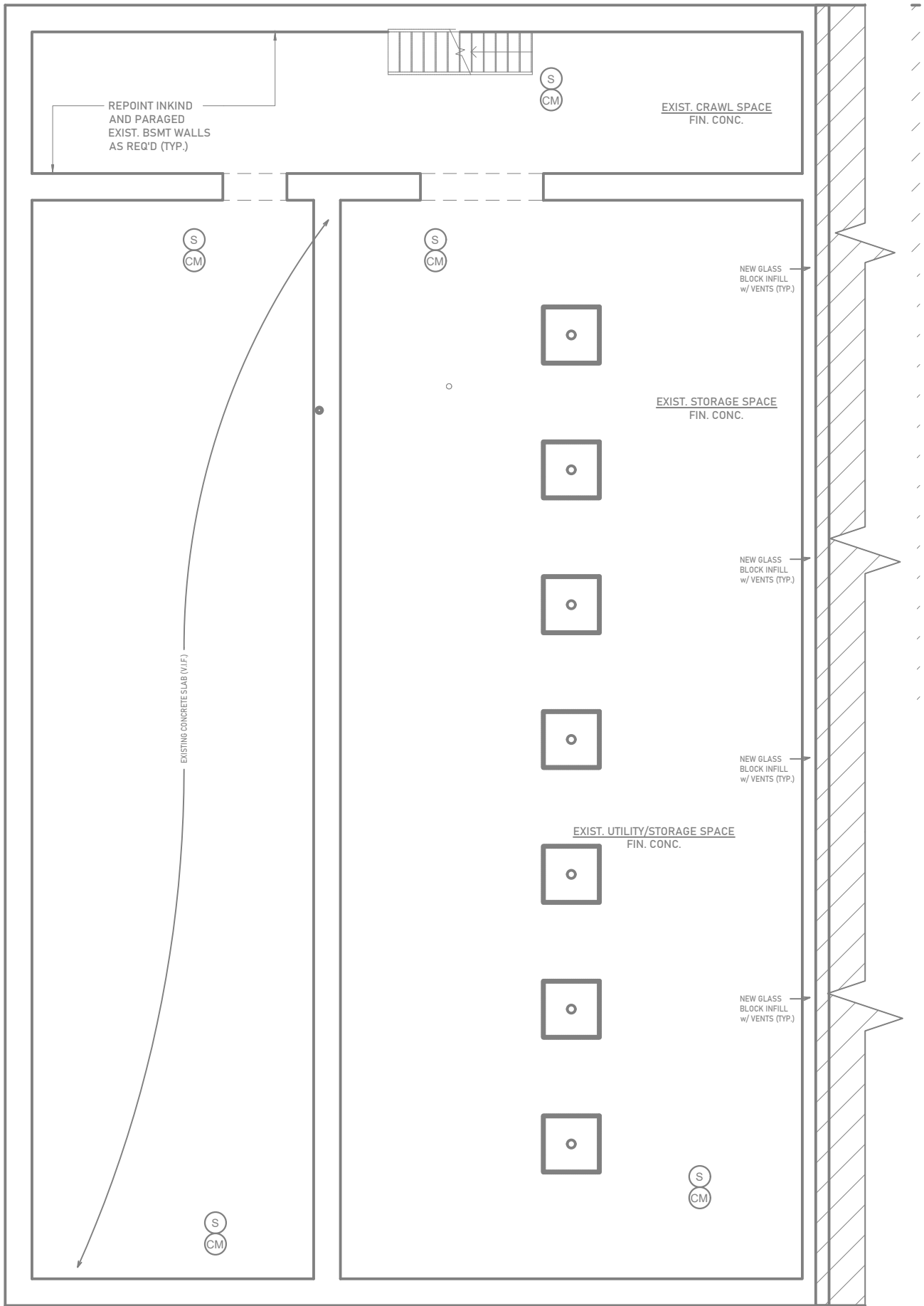
COMMERCIAL MIXED-USE

DRAWING TITLE:

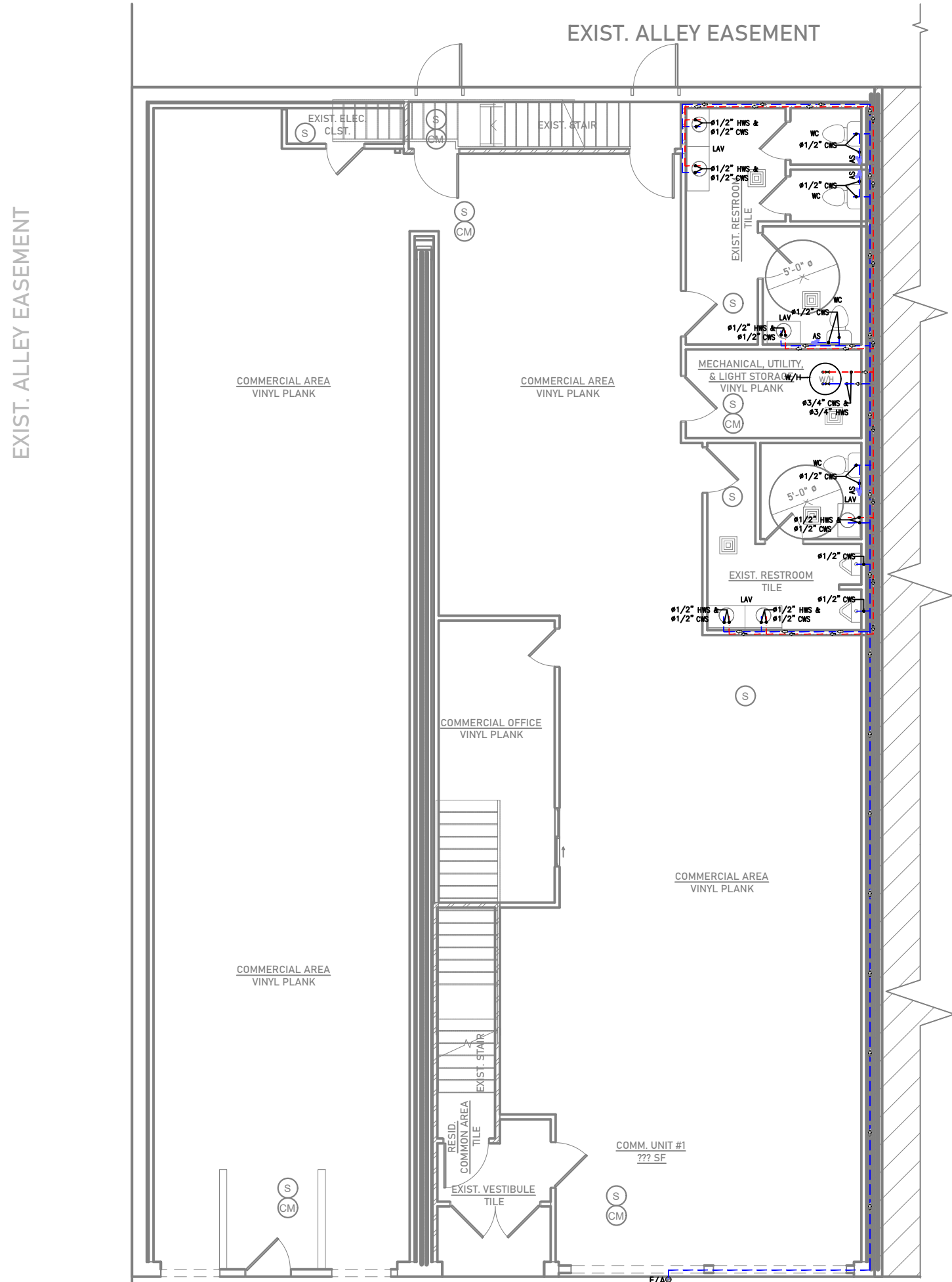
BASEMENT, FIRST, SECOND &  
THIRD FLOOR DRAINAGE PLANS

|                  |                    |                        |
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| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026    |
| JOB NO           | SHT NO<br>JS-PL-02 | SCALE:<br>AS SHOWN @A1 |

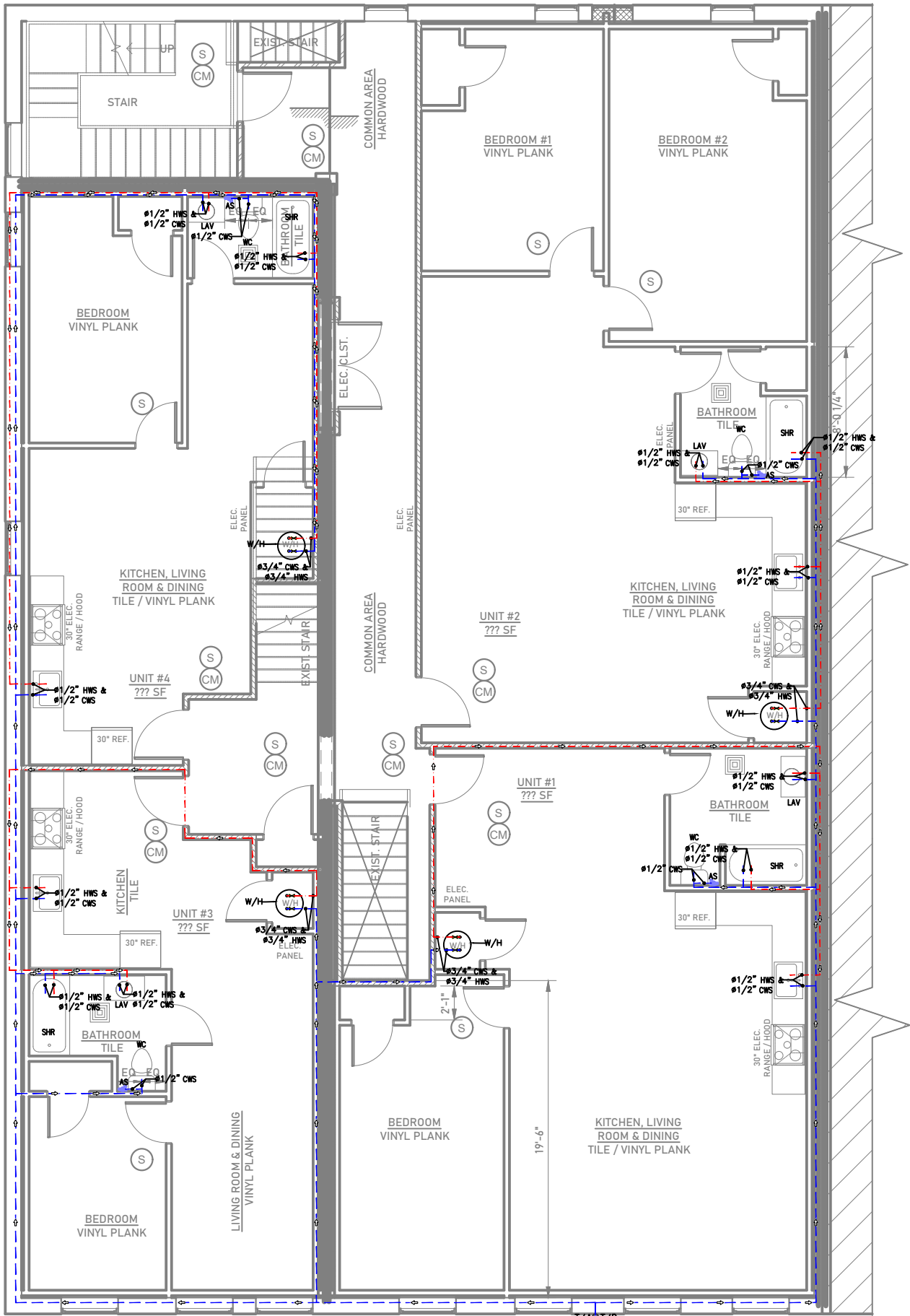




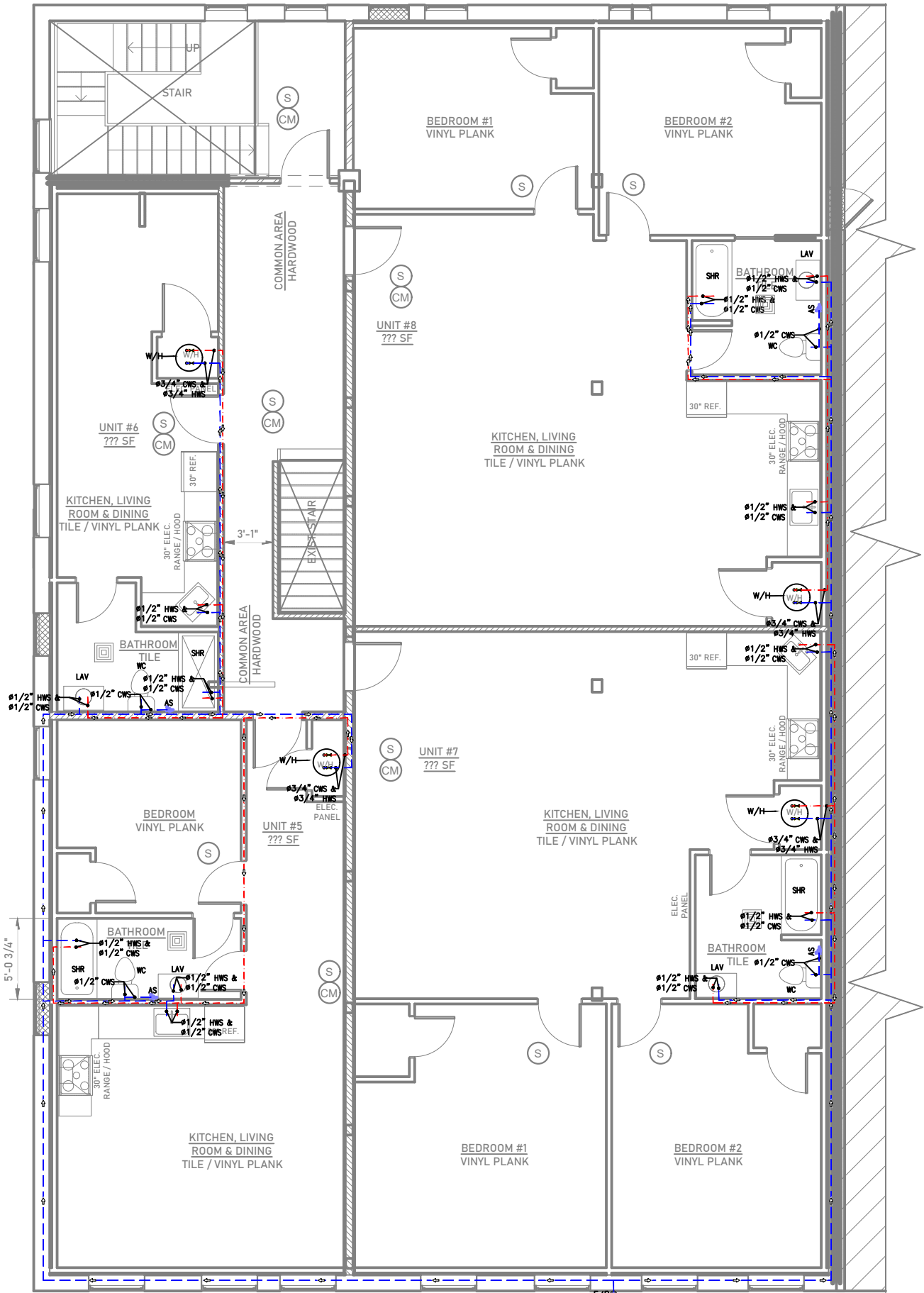
**PL 03** BASEMENT FLOOR WATER SUPPLY PLAN  
SCALE 3" = 1'-0"



**PL 03** FIRST FLOOR WATER SUPPLY PLAN  
SCALE 3" = 1'-0"



**PL 03** SECOND FLOOR WATER SUPPLY PLAN  
SCALE 3" = 1'-0"



**PL 03** THIRD FLOOR WATER SUPPLY PLAN  
SCALE 3" = 1'-0"

| PLUMBING LEGEND: |                         |
|------------------|-------------------------|
| SYMBOL           | DESCRIPTION             |
| ---              | HWS - HOT WATER SUPPLY  |
| ---              | CWS - COLD WATER SUPPLY |
| ---              | SP-SOIL PIPE            |
| ---              | WP-WASTE PIPE           |
| ---              | SHUT OFF VALVE          |
| ---              | GATE VALVE              |
| WH               | WATER HEATER            |
| GT               | GREASE TRAP             |
| WC               | WATER CLOSET            |
| KS               | KITCHEN SINK            |
| AS               | ABLUTION SPRAY          |
| SHR              | SHOWER                  |
| BT               | BATH TUB                |
| FCD              | FLOOR CLEANOUT          |
| FD               | FLOOR DRAIN             |
| LAV              | LAVATORY                |
| FA               | FROM ABOVE              |

OWNER SIGNATURE

| REVISION STATUS |      |             |          |
|-----------------|------|-------------|----------|
| Rev             | DATE | DESCRIPTION | DRAWN BY |
| 00              |      |             |          |
| 01              |      |             |          |
| 02              |      |             |          |
| 03              |      |             |          |
| 04              |      |             |          |

| CLIENT/CONSULTANT APPROVAL STATUS |     |
|-----------------------------------|-----|
| APPROVED                          | (A) |
| APPROVED WITH NOTES               | (B) |
| RESUBMIT                          | (C) |
| DISAPPROVED                       | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE: PLUMBING

PROJECT TITLE:

COMMERCIAL MIXED-USE

DRAWING TITLE:

BASEMENT, FIRST, SECOND & THIRD  
FLOOR WATER SUPPLY PLANS

|                  |                    |                        |
|------------------|--------------------|------------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026    |
| JOB NO           | SHT NO<br>JS-PL-03 | SCALE:<br>AS SHOWN @A1 |





|                                                                                                                       |      |                            |            |
|-----------------------------------------------------------------------------------------------------------------------|------|----------------------------|------------|
| <b>OWNER SIGNATURE</b>                                                                                                |      |                            |            |
|                                                                                                                       |      |                            |            |
| <b>REVISION STATUS</b>                                                                                                |      |                            |            |
| Rev                                                                                                                   | DATE | DESCRIPTION                | DRAWN BY   |
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| 01                                                                                                                    |      |                            |            |
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| 04                                                                                                                    |      |                            |            |
| <b>CLIENT/CONSULTANT APPROVAL STATUS</b>                                                                              |      |                            |            |
| <b>APPROVED</b>                                                                                                       |      |                            | <b>(A)</b> |
| <b>APPROVED WITH NOTES</b>                                                                                            |      |                            | <b>(B)</b> |
| <b>RESUBMIT</b>                                                                                                       |      |                            | <b>(C)</b> |
| <b>DISAPPROVED</b>                                                                                                    |      |                            | <b>(D)</b> |
| <b>OWNER:</b>                                                                                                         |      |                            |            |
|                                                                                                                       |      |                            |            |
| <b>ADDRESS:</b>                                                                                                       |      |                            |            |
|                                                                                                                       |      |                            |            |
| 508 MAIN STREET, JOHNSTOWN, PA 15901                                                                                  |      |                            |            |
| <b>DESIGN CONSULTANT:</b>                                                                                             |      |                            |            |
| TSABA CORP<br>NADIF BRACEY<br>254 CHAPMAN RD, STE 208 #13130<br>NEWARK, DELAWARE 19702<br>EMAIL: NADIF@TS9DESIGNS.COM |      |                            |            |
| <b>DRAWING STATUS:</b>                                                                                                |      |                            |            |
| DESIGN DRAWING                                                                                                        |      |                            |            |
| <b>DISCIPLINE:</b>                                                                                                    |      |                            |            |
| PLUMBING                                                                                                              |      |                            |            |
| <b>PROJECT TITLE:</b>                                                                                                 |      |                            |            |
| COMMERCIAL MIXED-USE                                                                                                  |      |                            |            |
| <b>DRAWING TITLE:</b>                                                                                                 |      |                            |            |
| DRAINAGE SYSTEM RISER DIAGRAM                                                                                         |      |                            |            |
| <b><u>DRAWN BY:</u></b>                                                                                               |      | <b><u>CHECKED BY:</u></b>  |            |
| G.M                                                                                                                   |      | N.B                        |            |
| <b><u>JOB NO</u></b>                                                                                                  |      | <b><u>SCALE:</u></b>       |            |
| SHT NO<br>JS-PL-04                                                                                                    |      | 11/05/2026<br>AS SHOWN @A1 |            |

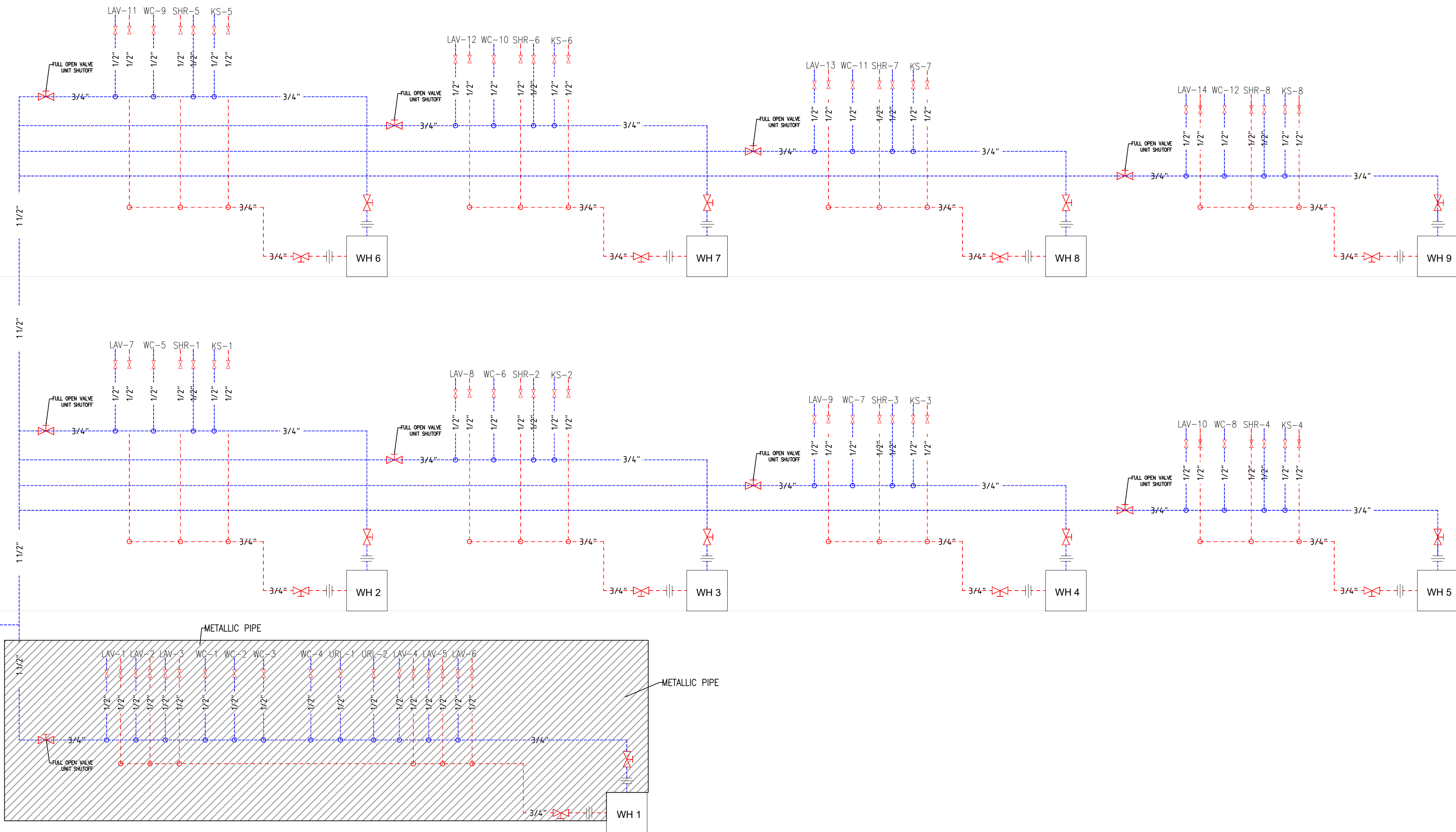


ROOF

THIRD FLOOR

SECOND FLOOR

FIRST FLOOR



WATER SUPPLY SYSTEM RISER DIAGRAM  
NTS  
SCALE

CALC C — WATER SUPPLY FIXTURE UNITS (WSFU) | 508 MAIN STREET — JOHNSTOWN, PA 15901 | Commercial Mixed-Use

Code: 2018 IPC Table 610.3 (WSFU) & Table 604.3 (Pipe Sizing @ 60 psi min.) | Date: May 24, 2026

BUILDING PARAMETERS

|                                  |                                                                    |
|----------------------------------|--------------------------------------------------------------------|
| Residential dwelling units:      | 8 8 apartments — 9 water heaters total (1 per unit + 1 commercial) |
| Lavatories per residential unit: | 2 2 per unit assumed — verify with architectural drawings          |

WSFU SCHEDULE — 2018 IPC Table 610.3 | 9 Water Heaters Total (1 per residential unit + 1 commercial)

| Fixture Type                                       | Qty | WSFU Each (IPC 610.3) | Total WSFU | Hot? | Cold? | IPC 610.3 Reference                                 |
|----------------------------------------------------|-----|-----------------------|------------|------|-------|-----------------------------------------------------|
| Water closet, private (1.6 gpf) — residential      | 8   | 2.50                  | 20.00      | No   | Yes   | Table 610.3 — WC private: 2.5 WSFU (cold only)      |
| Water closet, public — commercial                  | 2   | 2.50                  | 5.00       | No   | Yes   | Table 610.3 — WC public 1.6 gpf: 2.5 WSFU           |
| Lavatory, private — residential                    | 16  | 1.00                  | 16.00      | Yes  | Yes   | Table 610.3 — Lavatory private: 1.0 WSFU            |
| Lavatory, public — commercial                      | 2   | 1.50                  | 3.00       | Yes  | Yes   | Table 610.3 — Lavatory public: 1.5 WSFU             |
| Bathtub / shower combo — residential               | 8   | 1.50                  | 12.00      | Yes  | Yes   | Table 610.3 — Tub/shower combo: 1.5 WSFU            |
| Kitchen sink — residential                         | 8   | 1.50                  | 12.00      | Yes  | Yes   | Table 610.3 — Kitchen sink: 1.5 WSFU                |
| Service / mop sink — commercial                    | 1   | 3.00                  | 3.00       | Yes  | Yes   | Table 610.3 — Service sink: 3.0 WSFU                |
| Drinking fountain — commercial                     | 1   | 0.75                  | 0.75       | No   | Yes   | Table 610.3 — Drinking fountain: 0.75 WSFU          |
| Electric water heater 50-gal (per unit + comm.)    | 9   | 0.00                  | 0.00       | N/A  | N/A   | WH per unit — each independently fed per IPC §606.1 |
| TOTAL WATER SUPPLY FIXTURE UNITS (WSFU) — BUILDING |     |                       | 71.75      |      |       |                                                     |

PIPE SIZING — 2018 IPC Table 604.3 @ 60 psi minimum service pressure

| Supply Segment                              | WSFU  | Min. Pipe Size — IPC 604.3 @60 psi | Max Dev. Length | Specified Size   | Material      | Notes                                                         |
|---------------------------------------------|-------|------------------------------------|-----------------|------------------|---------------|---------------------------------------------------------------|
| Building cold water service (from meter)    | 71.75 | 1 1/2" min — >30 WSFU @60 psi      | 200 ft          | 1 1/2" Cu Type L | Copper Type L | Confirm meter size with Johnstown Water Authority             |
| Residential cold branch — per unit          | ~20   | 3/4" min                           | 80 ft           | 3/4" Cu Type L   | Copper Type L | Riser to unit water heater and fixtures                       |
| Residential hot branch — per unit (from WH) | ~12   | 1/2"–3/4" min                      | 50 ft           | 3/4" Cu Type L   | Copper Type L | Individual WH to unit fixtures — IPC §503.3: 4ft copper at WH |
| Commercial branch (1st floor)               | ~10   | 3/4" min                           | 40 ft           | 3/4" Cu Type L   | Copper Type L | Restrooms, service sink, drinking fountain                    |
| PRV at service entry                        | —     | Req'd if > 80 psi — §604.8         | —               | Bronze PRV       | Bronze/brass  | Verify Johnstown Water Authority pressure at meter            |
| Thermal expansion tank                      | —     | Required — IPC §607.3              | —               | Per mfr.         | Bladder type  | Required where backflow preventer is installed                |

PIPING MATERIALS: Above-slab supply — Copper Type L (ASTMB88) with lead-free solder, or PEX-A (ASTM F876/F877, SDR-9) with ASTM F1960 expansion fittings for residential. Commercial: metallic (copper) per IPC. Each water heater independently fed — shutoff of one unit does not affect others (IPC §606.1). T&P relief valve discharges to outside wall per IPC §504.6. FLOOD ZONE: All water heaters elevated to BFE + 2 ft min. (IBC §1612). Provide thermal expansion tank per IPC §607.3. Confirm available pressure with Johnstown Water Authority (614) 533-2800.

PLUMBING LEGEND:

| SYMBOL                                   | DESCRIPTION             |
|------------------------------------------|-------------------------|
| <span style="color: red;">---</span>     | HWS — HOT WATER SUPPLY  |
| <span style="color: blue;">---</span>    | CWS — COLD WATER SUPPLY |
| <span style="color: magenta;">---</span> | SP—SOIL PIPE            |
| <span style="color: magenta;">---</span> | WP—WASTE PIPE           |
| <span style="color: blue;">—</span>      | SHUT OFF VALVE          |
| <span style="color: blue;">—</span>      | GATE VALVE              |
| WH                                       | WATER HEATER            |
| GT                                       | GREASE TRAP             |
| WC                                       | WATER CLOSET            |
| KS                                       | KITCHEN SINK            |
| AS                                       | ABLUTION SPRAY          |
| SHR                                      | SHOWER                  |
| BT                                       | BATH TUB                |
| FCD                                      | FLOOR CLEANOUT          |
| FD                                       | FLOOR DRAIN             |
| LAV                                      | LAVATORY                |
| FA                                       | FROM ABOVE              |

OWNER SIGNATURE

REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
| 01  |      |             |          |
| 02  |      |             |          |
| 03  |      |             |          |
| 04  |      |             |          |

CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE:

PLUMBING

PROJECT TITLE:

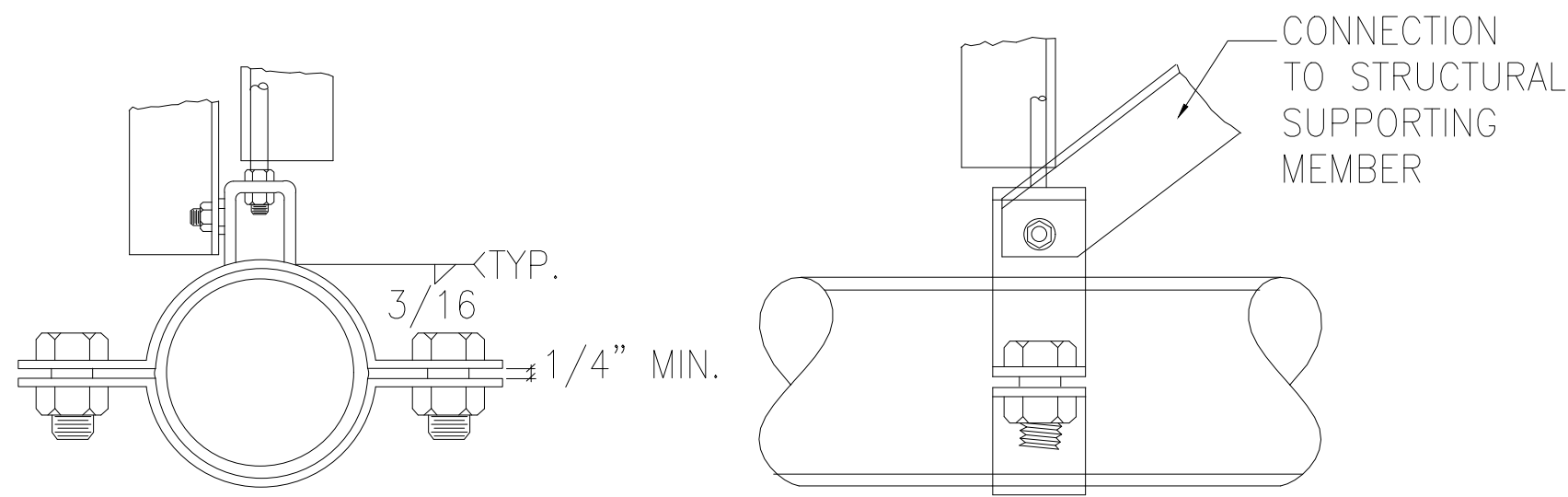
COMMERCIAL MIXED-USE

DRAWING TITLE:

WATER SUPPLY SYSTEM RISER  
DIAGRAM

|                  |                    |                        |
|------------------|--------------------|------------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026    |
| JOB NO           | SHT NO<br>JS-PL-05 | SCALE:<br>AS SHOWN @A1 |

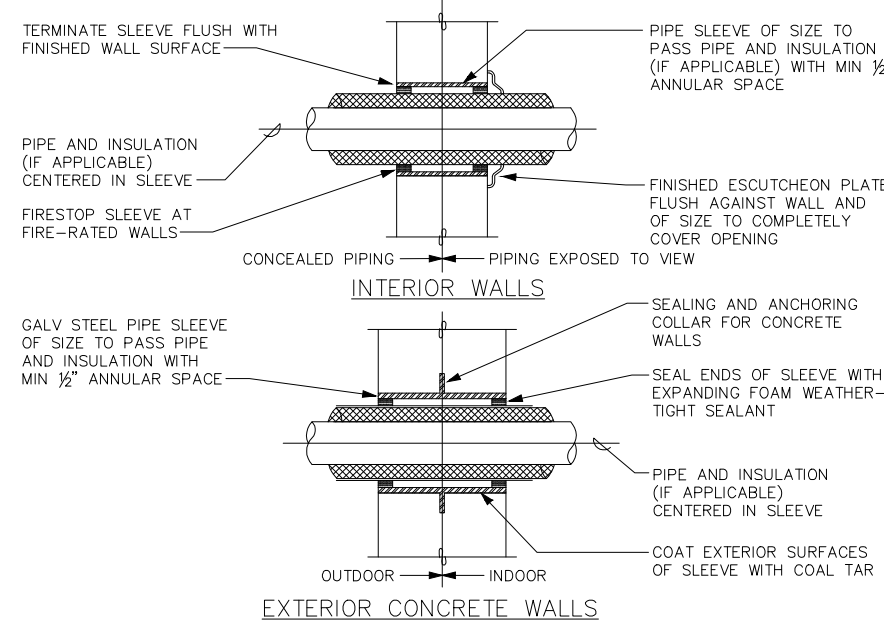




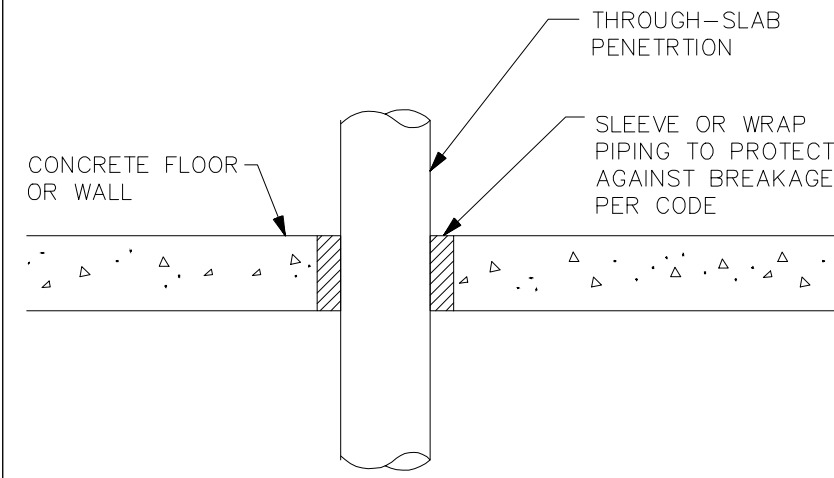
1. TYPICAL LONGITUDINAL PIPE BRACING  
SCALE: NTS

1. THIS BRACING DETAIL APPLIES ONLY FOR COLD WATER PIPE AND GAS PIPE WHERE MOVEMENT OF THE PIPE DUE TO TEMPERATURE DIFFERENTIAL IS NEGLIGIBLE
2. IT IS THE RESPONSIBILITY OF THE USER OF THIS GUIDELINE TO ASCERTAIN THAT AN ADEQUATE BRACING AND ANCHORAGE DEVICE BE DESIGNED FOR PIPE WHENEVER THE MOVEMENT DUE TO THERMAL DIFFERENTIAL EXISTS. PROVIDE ONE ANCHOR POINT IN ONE PIPE RUN. ALLOW FOR LONGITUDINAL PIPE MOVEMENT AT THE OPPOSITE END OF THE ANCHOR POINT.

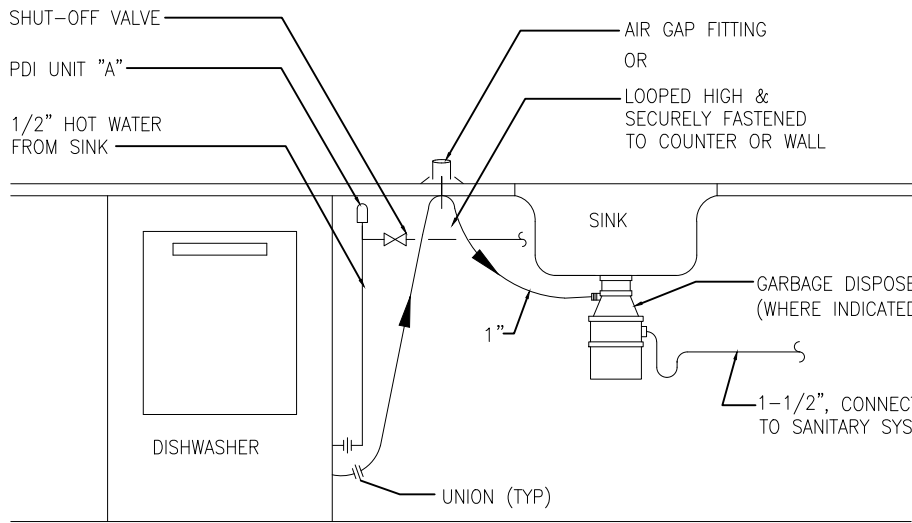
1. LONGITUDINAL PIPE BRACING  
SCALE: NTS



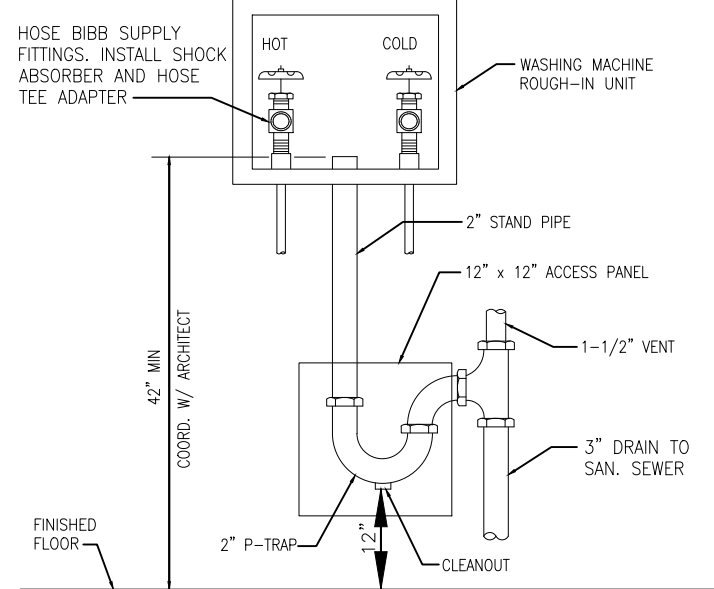
1. PIPE SLEEVES THROUGH WALLS  
SCALE: NTS



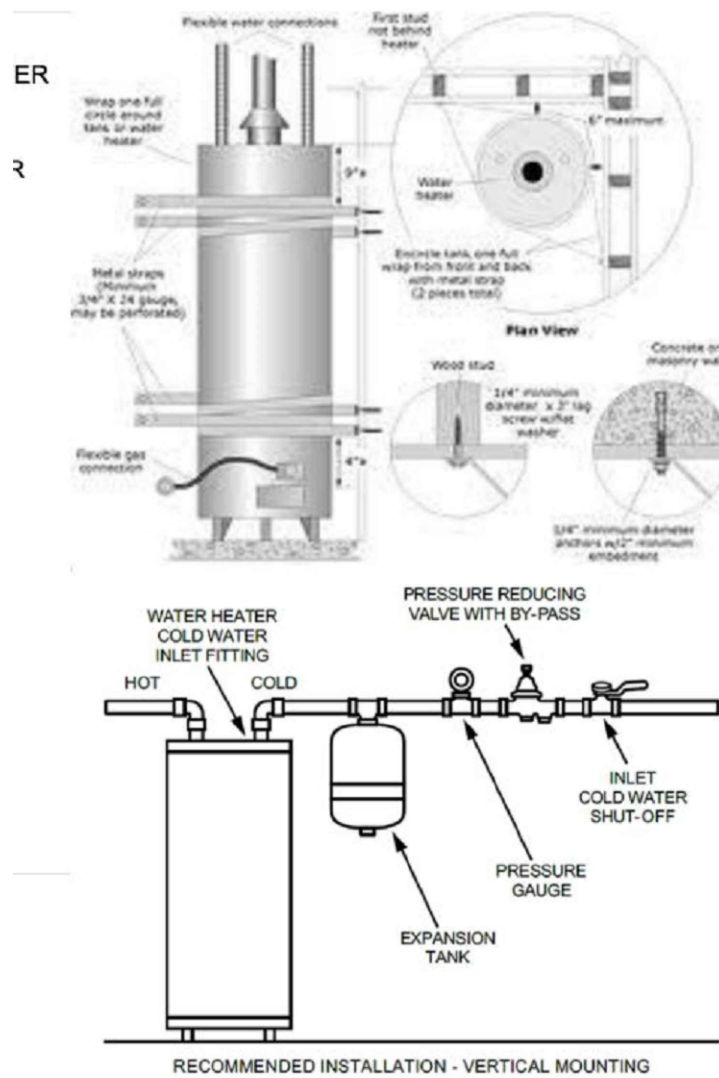
1. PIPE SLAB PENETRATION  
SCALE: NTS



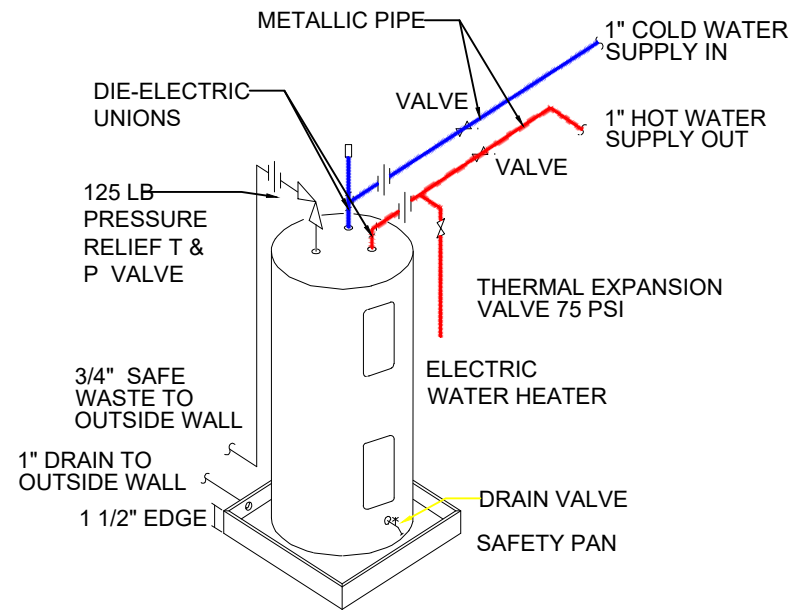
1. RESIDENTIAL DISHWASHER PIPING DETAIL  
SCALE: NTS



1. RESIDENTIAL WASHING MACHINE CONNECTION DETAIL  
SCALE: NTS

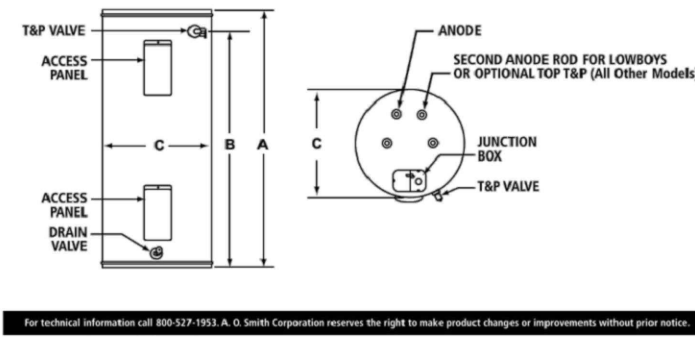


1. WATER HEATER INSTALLATION DETAILS  
SCALE: NTS



1. WATER HEATER INSTALLATION DETAILS  
SCALE: NTS

| AC Smith. COMMERCIAL-GRADE RESIDENTIAL ELECTRIC WATER HEATERS |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |
|---------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Model Number                                                  | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element |
| EW100                                                         | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      |
| EW150                                                         | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      |
| EW200                                                         | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      |
| EW250                                                         | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      |
| EW300                                                         | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      |
| EW350                                                         | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      |
| EW400                                                         | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      |
| EW450                                                         | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      |
| EW500                                                         | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      |
| EW550                                                         | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      |
| EW600                                                         | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      |
| EW650                                                         | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      |
| EW700                                                         | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      |
| EW750                                                         | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      |
| EW800                                                         | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      |
| EW850                                                         | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      |
| EW900                                                         | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      |
| EW950                                                         | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      |
| EW1000                                                        | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     |



1. WATER HEATER INSTALLATION DETAILS  
SCALE: NTS

| AC Smith. COMMERCIAL-GRADE RESIDENTIAL ELECTRIC WATER HEATERS |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |
|---------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Model Number                                                  | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element | Standard Heating Element |
| EW100                                                         | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      | 100                      |
| EW150                                                         | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      | 150                      |
| EW200                                                         | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      | 200                      |
| EW250                                                         | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      | 250                      |
| EW300                                                         | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      | 300                      |
| EW350                                                         | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      | 350                      |
| EW400                                                         | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      | 400                      |
| EW450                                                         | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      | 450                      |
| EW500                                                         | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      | 500                      |
| EW550                                                         | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      | 550                      |
| EW600                                                         | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      | 600                      |
| EW650                                                         | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      | 650                      |
| EW700                                                         | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      | 700                      |
| EW750                                                         | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      | 750                      |
| EW800                                                         | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      | 800                      |
| EW850                                                         | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      | 850                      |
| EW900                                                         | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      | 900                      |
| EW950                                                         | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      | 950                      |
| EW1000                                                        | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     | 1000                     |

1. WATER HEATER INSTALLATION DETAILS  
SCALE: NTS

| MATERIAL                                                                       | PIPE OR TUBE SIZE (inches) | MAXIMUM VELOCITY (feet per second) |
|--------------------------------------------------------------------------------|----------------------------|------------------------------------|
| All pipe and tube material                                                     | 1/2 and smaller            | 5                                  |
| Brass pipe                                                                     | 3/4 and larger             | 8                                  |
| Copper or copper-alloy pipe and tubing (hot water systems)                     | 1/2 and larger             | 5                                  |
| Copper or copper-alloy pipe and tubing (cold water systems)                    | 3/4 and larger             | 8                                  |
| Chlorinated polyvinyl (CPVC) plastic pipe                                      | 1/2 - 1                    | 8                                  |
| Cross-linked polyethylene (PEX) plastic tubing                                 | 1/2 and larger             | 8                                  |
| Cross-linked polyethylene/aluminum/cross-linked polyethylene (PEX-AL-PEX) pipe | 3/4 and larger             | 8                                  |
| Galvanized steel pipe                                                          | 1/2 - 4                    | 10                                 |
| PE-AL-PE                                                                       | 4 and larger               | 12                                 |
| PE-AL-PE                                                                       | 1/2 and larger             | 8                                  |

1. WATER DISTRIBUTION PIPE SIZING AND WATER SERVICE PIPE SIZING  
SCALE: NTS

OWNER SIGNATURE

REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
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| 03  |      |             |          |
| 04  |      |             |          |

CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE: PLUMBING

PROJECT TITLE:

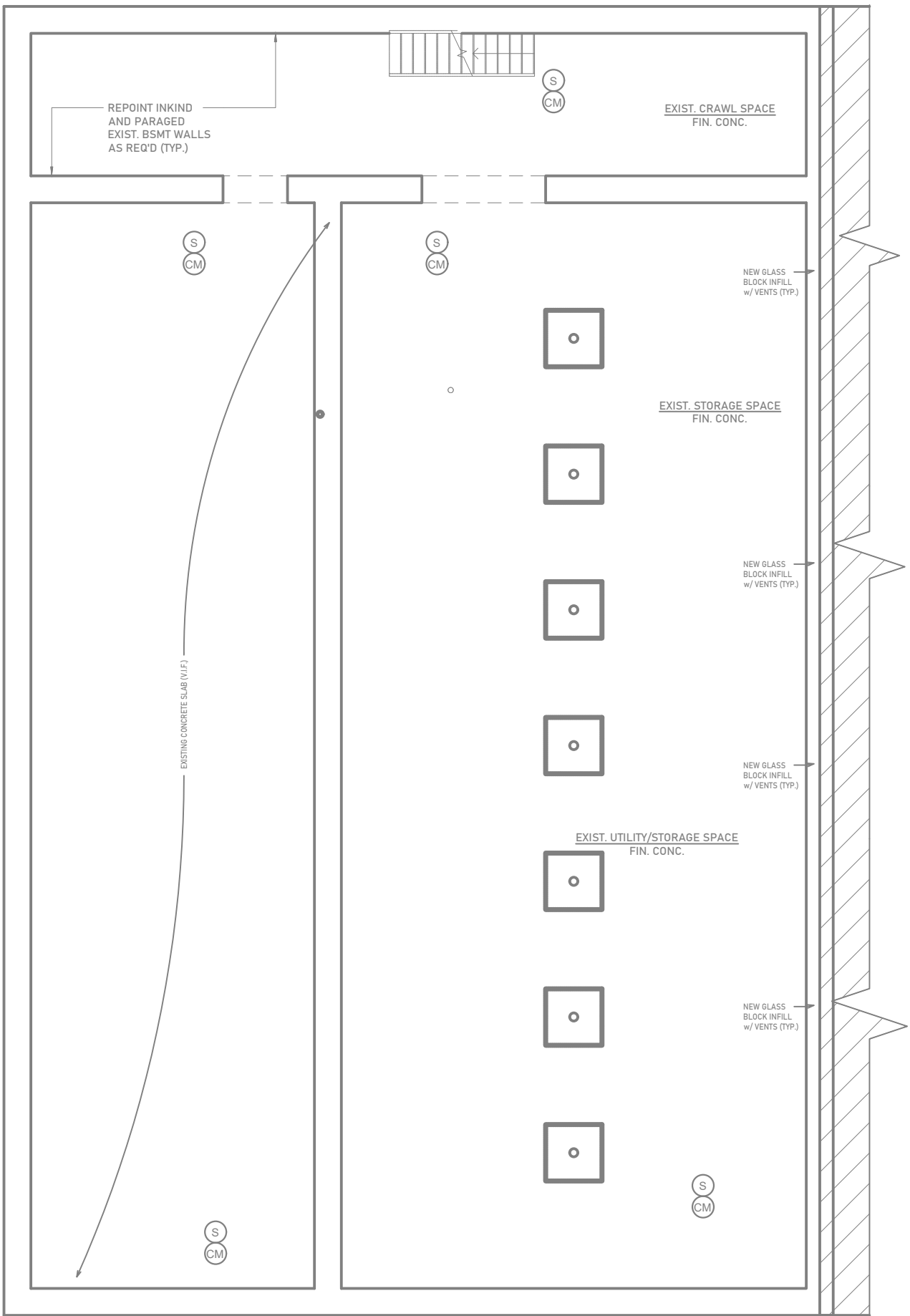
COMMERCIAL MIXED-USE

DRAWING TITLE:

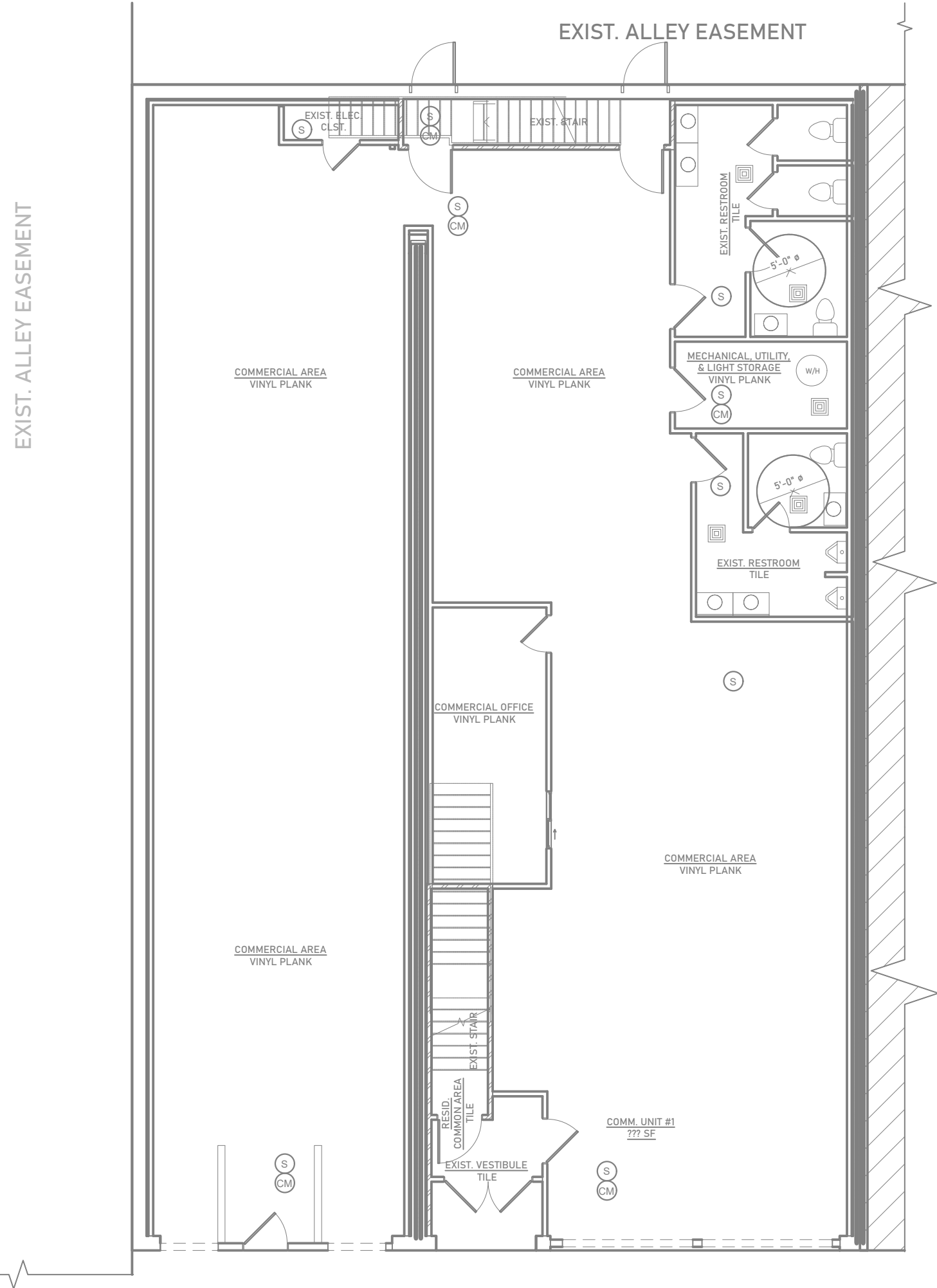
DETAILS

|                  |                    |                     |
|------------------|--------------------|---------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026 |
| JOB NO           | SHT NO<br>JS-PL-06 | SCALE:<br>NTS @A1   |

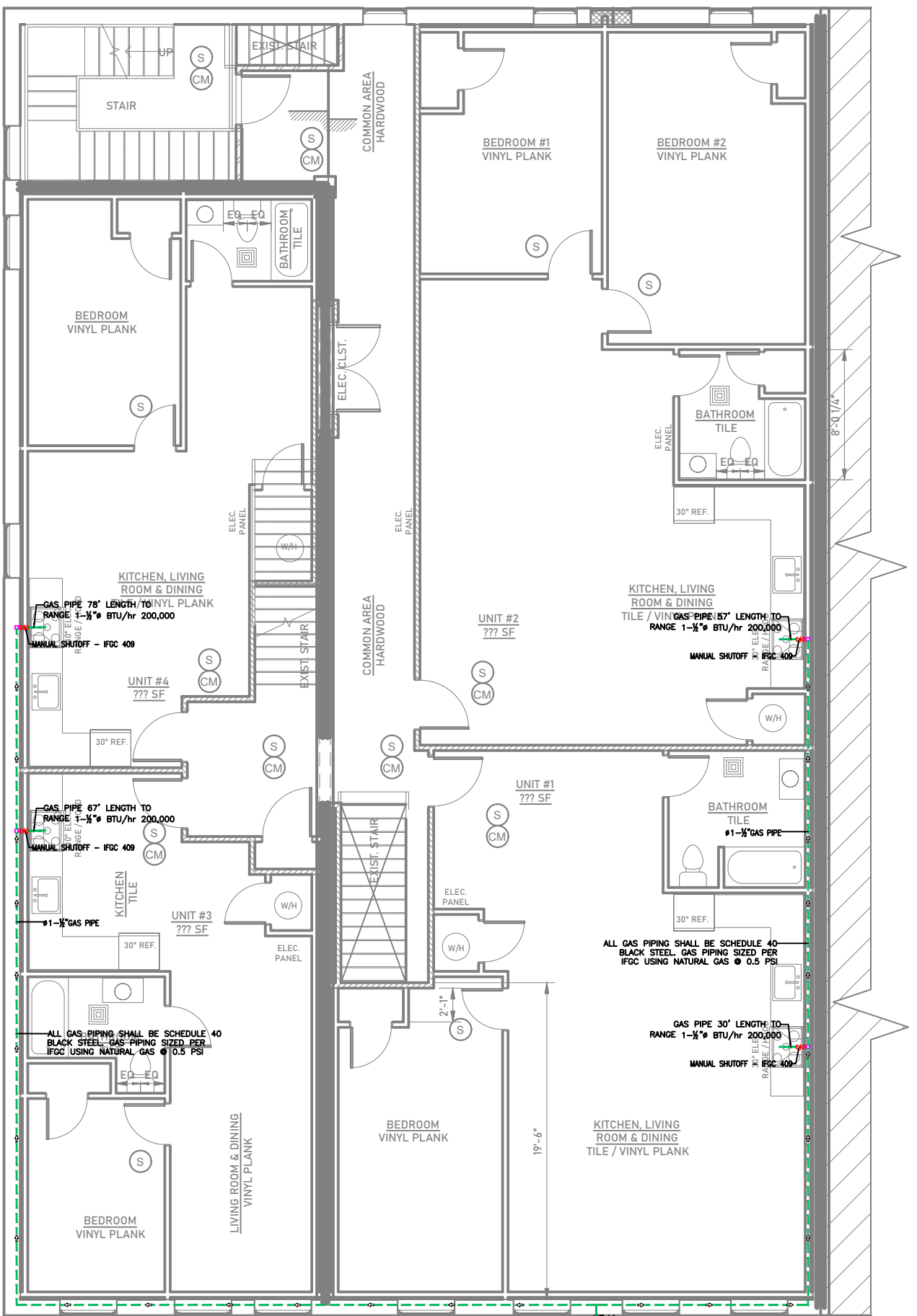




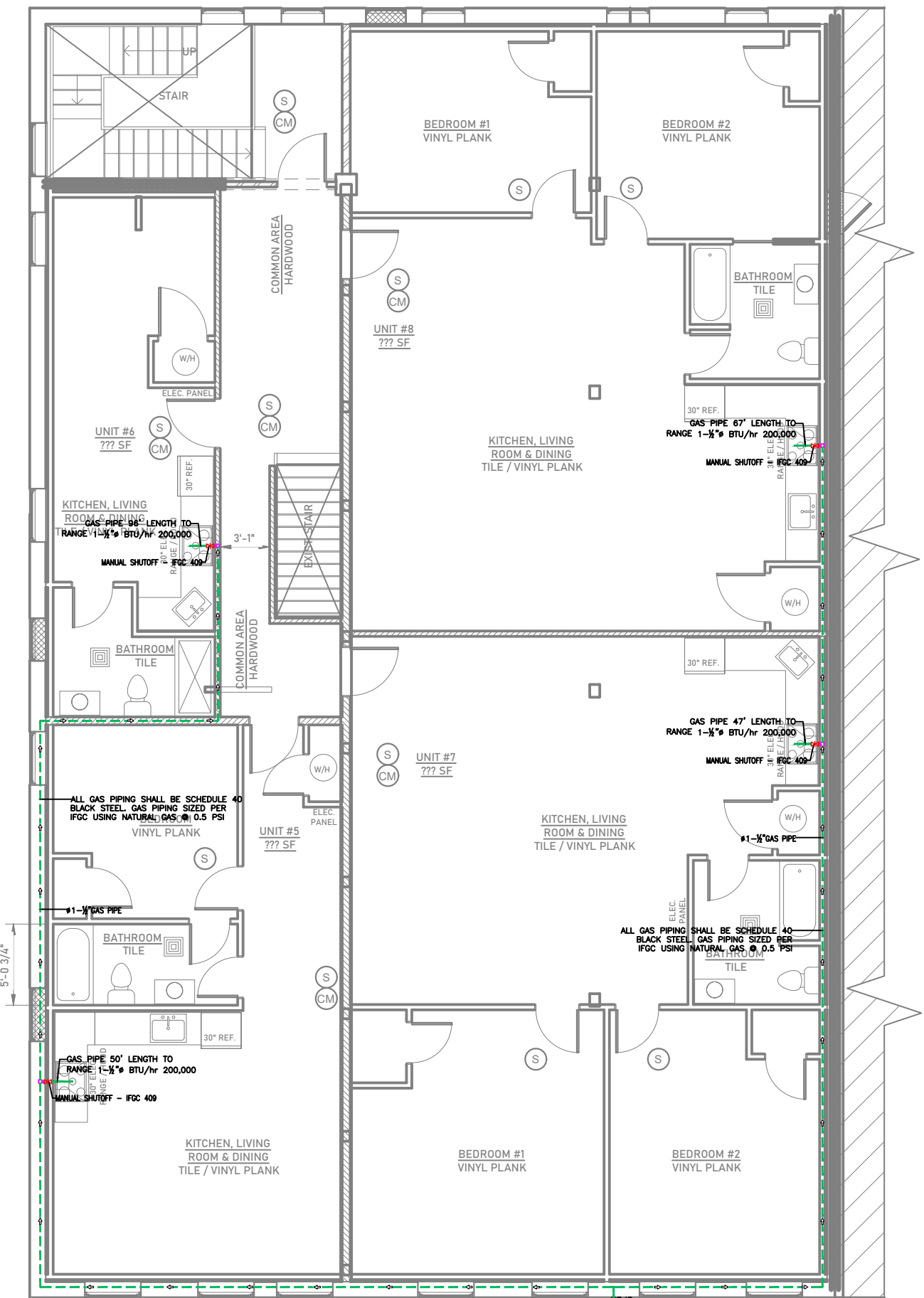
PL 07 SCALE  
BASEMENT FLOOR GAS SUPPLY PLAN  
3" = 1'-0"



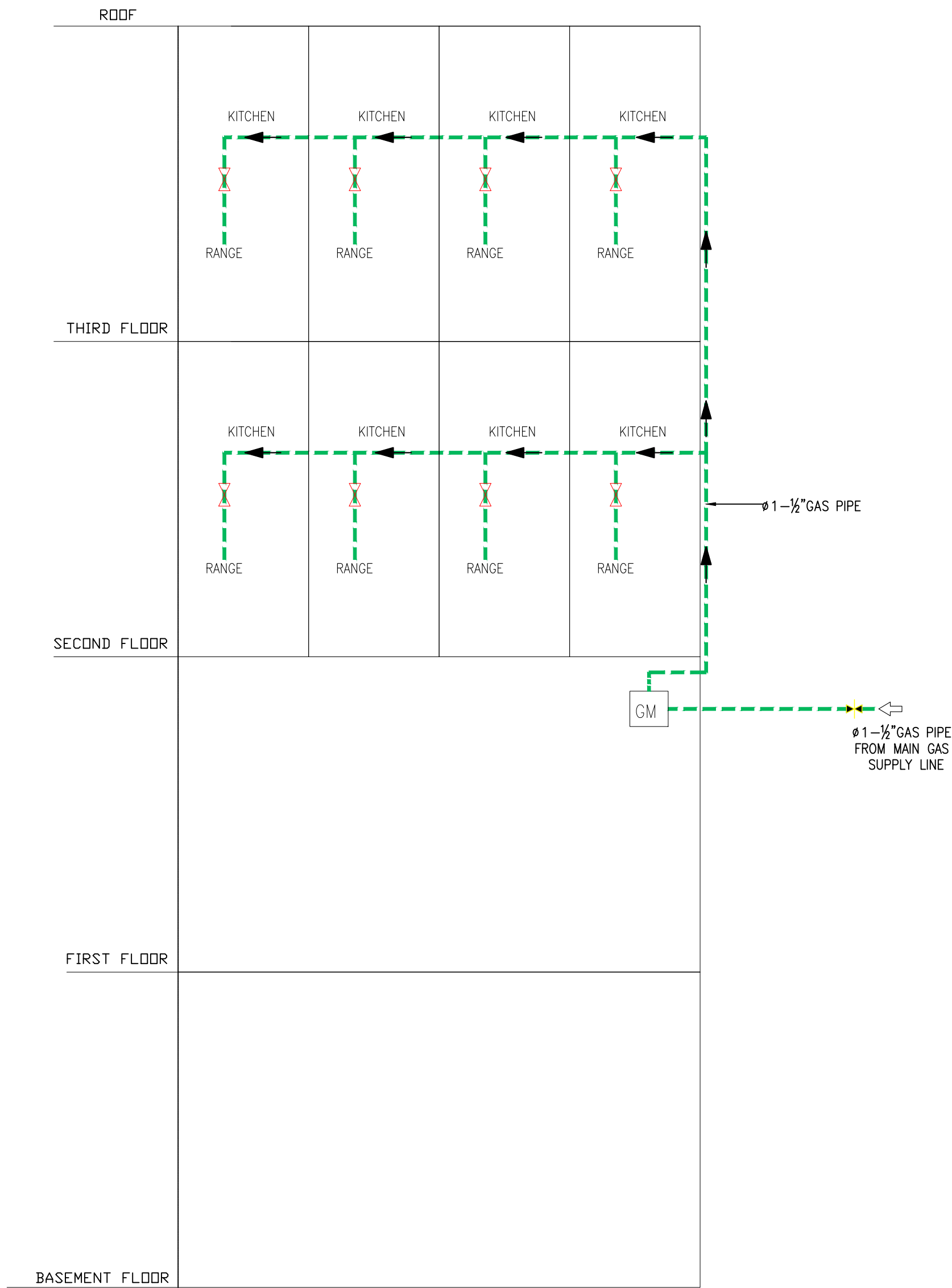
PL 07 SCALE  
FIRST FLOOR GAS SUPPLY PLAN  
3" = 1'-0"



PL 07 SCALE  
SECOND FLOOR GAS SUPPLY PLAN  
3" = 1'-0"



PL 07 SCALE  
THIRD FLOOR GAS SUPPLY PLAN  
3" = 1'-0"



PL 07 SCALE  
GAS SUPPLY SYSTEM RISER DIAGRAM  
NTS

NOTES:

FFLOOD-GAS-01: GAS METER, PRESSURE REGULATOR, SHUTOFF VALVES, AND GAS-FIRED EQUIPMENT SHALL BE LOCATED AT OR ABOVE BFE/DPE/RFE WHERE PRACTICAL. GAS SYSTEM SHALL BE ANCHORED AND INSTALLED TO PREVENT FLOATATION, BREAKAWAY, FLOODWATER INFILTRATION, AND DISCHARGE OF GAS INTO FLOODWATERS.

FLOOD-GAS-02: GAS PIPING BELOW BFE/DPE/RFE SHALL BE CORROSION-RESISTANT, ADEQUATELY SUPPORTED, AND PROTECTED FROM HYDROSTATIC/HYDRODYNAMIC FLOOD LOADS. PROVIDE SHUTOFF VALVES ABOVE FLOOD ELEVATION WHERE PRACTICAL.

| GAS LEGEND: |                  |
|-------------|------------------|
| SYMBOL      | DESCRIPTION      |
|             | GAS SUPPLY PIPE  |
|             | SHUT OFF VALVE   |
|             | GATE VALVE       |
|             | TEE WITH REDUCER |
|             | GAS CAP          |

OWNER SIGNATURE

REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
| 01  |      |             |          |
| 02  |      |             |          |
| 03  |      |             |          |
| 04  |      |             |          |

CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

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DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE:

PLUMBING

PROJECT TITLE:

COMMERCIAL MIXED-USE

DRAWING TITLE:

BASEMENT, FIRST, SECOND, THIRD  
FLOOR GAS SUPPLY PLANS, RISER  
DIAGRAM

|                  |                    |                        |
|------------------|--------------------|------------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026    |
| JOB NO           | SHT NO<br>JS-PL-07 | SCALE:<br>AS SHOWN @A1 |



CALC G — GAS LOAD & PIPE SIZING | 508 MAIN STREET — JOHNSTOWN, PA 15901 | Commercial Mixed-Use

Code: 2018 IFGC / NFPA 54 | 8 Residential Units — Natural Gas Ranges | Date: May 24, 2026

| GAS LOAD TABLE — 8 Residential Units Each with 1 Six-Burner Range |                      |                              |                              |                         |                                                                  |
|-------------------------------------------------------------------|----------------------|------------------------------|------------------------------|-------------------------|------------------------------------------------------------------|
| Equipment                                                         | BTU/hr per Appliance | Run Length to Appliance (ft) | Pipe Size (IFGC Table 402.4) | Location                | Note                                                             |
| 6-Burner Range                                                    | 200,000              | 67                           | 1½"                          | Apt. Unit 1 — 2nd Floor | Per IFGC Table 402.4(2) — 1½" @ max 96 ft run for 200,000 BTU/hr |
| 6-Burner Range                                                    | 200,000              | 78                           | 1½"                          | Apt. Unit 2 — 2nd Floor | Per IFGC Table 402.4(2) — 1½" @ max 96 ft run for 200,000 BTU/hr |
| 6-Burner Range                                                    | 200,000              | 30                           | 1½"                          | Apt. Unit 3 — 2nd Floor | Per IFGC Table 402.4(2) — 1½" @ max 96 ft run for 200,000 BTU/hr |
| 6-Burner Range                                                    | 200,000              | 57                           | 1½"                          | Apt. Unit 4 — 2nd Floor | Per IFGC Table 402.4(2) — 1½" @ max 96 ft run for 200,000 BTU/hr |
| 6-Burner Range                                                    | 200,000              | 50                           | 1½"                          | Apt. Unit 5 — 3rd Floor | Per IFGC Table 402.4(2) — 1½" @ max 96 ft run for 200,000 BTU/hr |
| 6-Burner Range                                                    | 200,000              | 96                           | 1½"                          | Apt. Unit 6 — 3rd Floor | Per IFGC Table 402.4(2) — 1½" @ max 96 ft run for 200,000 BTU/hr |
| 6-Burner Range                                                    | 200,000              | 47                           | 1½"                          | Apt. Unit 7 — 3rd Floor | Per IFGC Table 402.4(2) — 1½" @ max 96 ft run for 200,000 BTU/hr |
| 6-Burner Range                                                    | 200,000              | 67                           | 1½"                          | Apt. Unit 8 — 3rd Floor | Per IFGC Table 402.4(2) — 1½" @ max 96 ft run for 200,000 BTU/hr |
| TOTAL GAS LOAD (BTU/hr)                                           |                      | 1,600,000                    |                              |                         |                                                                  |

| PIPE SIZING — 2018 IFGC / NFPA 54   Max. Run 96 ft (longest branch) per gas load table |                |                |                                                               |           |      |
|----------------------------------------------------------------------------------------|----------------|----------------|---------------------------------------------------------------|-----------|------|
| Segment                                                                                | Load           | IFGC Pipe Size | Basis                                                         | Specified | Note |
| Building gas service entry (from utility main)                                         | 1,600,000      | 1½" steel min. | IFGC Tbl 402.4(2), maxrun includes longest 96 ft branch       | 1½"       |      |
| Vertical riser — all floors                                                            | 1,600,000      | 1½" steel      | Full building load through riser                              | 1½"       |      |
| Branch to each residential unit (per unit)                                             | 200,000 BTU/hr | 1" steel       | Single appliance per unit per IFGC Table 402.4(2)             | 1"        |      |
| Flexible connector at range (per unit)                                                 | 200,000 BTU/hr | ¾" CSST flex   | IFGC §403.5 — flexible connector at appliance connection only | ¾" flex   |      |

8 residential 6-burner ranges at 200,000 BTU/hr each — total gas load calculated by formula above. No commercial gas equipment — commercial units are all-electric. Confirm meter sizing, inlet pressure, and service capacity with Peoples Natural Gas (800) 764-0111 prior to rough-in. FLOOD ZONE: Gas meter, regulator, and main shutoff elevated to BFE + 2 ft min. (IBC §1612). Install manual shutoff valve at building entry operable from above BFE. Coordinate elevated meter installation with Peoples Natural Gas.

| GAS LEGEND: |                  |
|-------------|------------------|
| SYMBOL      | DESCRIPTION      |
|             | GAS SUPPLY PIPE  |
|             | SHUT OFF VALVE   |
|             | GATE VALVE       |
|             | TEE WITH REDUCER |
|             | GAS CAP          |

| Natural Gas Pipe Sizing Chart |      |                        |      |     |        |        |     |        |      |
|-------------------------------|------|------------------------|------|-----|--------|--------|-----|--------|------|
| Length of Pipe in Feet        |      | Size of Pipe in Inches |      |     |        |        |     |        |      |
| Pipe In                       | Feet | 1/2"                   | 3/4" | 1"  | 1-1/4" | 1-1/2" | 2"  | 2-1/2" | 4"   |
| 10                            | 100  | 230                    | 307  | 383 | 512    | 608    | 960 | 1414   |      |
| 20                            | 175  | 160                    | 205  | 259 | 371    | 435    | 675 | 1000   | 1525 |
| 30                            | 210  | 125                    | 160  | 205 | 281    | 330    | 500 | 750    | 1125 |
| 40                            | 240  | 100                    | 125  | 160 | 210    | 250    | 375 | 562    | 844  |
| 50                            | 265  | 85                     | 105  | 135 | 180    | 215    | 320 | 480    | 720  |
| 60                            | 285  | 75                     | 95   | 120 | 160    | 190    | 285 | 425    | 638  |
| 70                            | 300  | 68                     | 88   | 110 | 145    | 175    | 265 | 395    | 593  |
| 80                            | 310  | 62                     | 82   | 105 | 135    | 165    | 245 | 365    | 545  |
| 90                            | 315  | 58                     | 78   | 100 | 130    | 160    | 230 | 345    | 515  |
| 100                           | 320  | 55                     | 75   | 95  | 125    | 155    | 220 | 330    | 495  |
| 125                           | 335  | 48                     | 68   | 85  | 110    | 135    | 200 | 300    | 450  |
| 150                           | 345  | 42                     | 60   | 75  | 100    | 120    | 180 | 270    | 405  |
| 175                           | 355  | 38                     | 55   | 70  | 95     | 115    | 165 | 245    | 365  |
| 200                           | 360  | 35                     | 50   | 65  | 90     | 110    | 155 | 230    | 345  |
| 250                           | 370  | 30                     | 45   | 60  | 85     | 105    | 140 | 210    | 315  |
| 300                           | 375  | 27                     | 40   | 55  | 80     | 100    | 130 | 195    | 290  |

Natural Gas flow is given in thousands of BTU/hr. — One cubic foot of L.P. gas = 1000 BTU.  
Nominal pressure at the burner for Natural Gas is 3.5" of water column. (Typical machine supply 9"-12")  
Pipe lengths must include additional length for all fittings. Add approximately 6 feet of pipe per fitting.  
Natural Gas Example: A machine with a burner that requires 400,000 BTU would need a 1-1/4" pipe for a 30' long run.

OWNER SIGNATURE

REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
| 01  |      |             |          |
| 02  |      |             |          |
| 03  |      |             |          |
| 04  |      |             |          |

CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE:

PLUMBING

PROJECT TITLE:

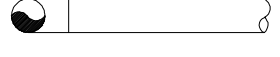
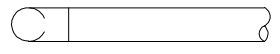
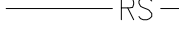
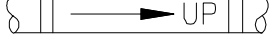
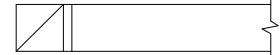
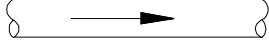
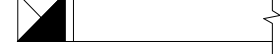
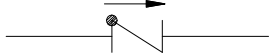
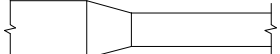
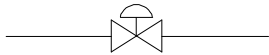
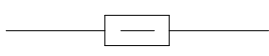
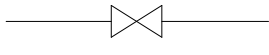
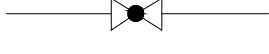
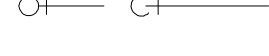
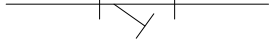
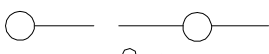
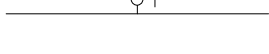
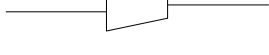
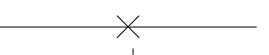
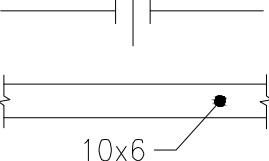
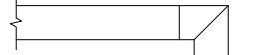
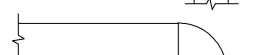
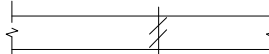
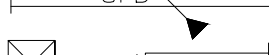
COMMERCIAL MIXED-USE

DRAWING TITLE:

GAS LOAD TABLE

|                  |                    |                        |
|------------------|--------------------|------------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026    |
| JOB NO           | SHT NO<br>JS-PL-08 | SCALE:<br>AS SHOWN @A1 |



| MECHANICAL LEGEND                                                                   |         |                                                                   |                                                                                      |         |                                                                                 |
|-------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------|
| SYMBOL                                                                              | ABBREV. | DESCRIPTION                                                       | SYMBOL                                                                               | ABBREV. | DESCRIPTION                                                                     |
|    |         | COORDINATE WITH ELECTRICAL                                        |    |         | LINED DUCTWORK (OR PLENUM)                                                      |
|    | CD      | CONDENSATE DRAIN (A.C)                                            |    |         | DUCT RISE IN DIRECTION OF FLOW                                                  |
|    | D       | DRAIN                                                             |    |         | DUCT DROP IN DIRECTION OF FLOW                                                  |
|    | RD      | REFRIGERANT DISCHARGE                                             |    |         | ROUND DUCT UP                                                                   |
|    | RL      | REFRIGERANT LIQUID                                                |    |         | ROUND DUCT DOWN                                                                 |
|    | RS      | REFRIGERANT SUCTION                                               |    |         | SUPPLY DUCT UP                                                                  |
|    |         | PIPE DOWN                                                         |    |         | SUPPLY DUCT DOWN                                                                |
|    |         | PIPE UP                                                           |    | RA/OA   | RETURN AIR DUCT/OUTSIDE AIR DUCT UP                                             |
|    |         | PIPE RISE (OR DN. FOR DROP)                                       |    |         | RETURN AIR DUCT/OUTSIDE AIR DUCT DOWN                                           |
|    |         | DIRECTION OF FLOW IN PIPE                                         |    |         | EXHAUST AIR DUCT UP                                                             |
|    | AV      | AIR VENT (VALVE)                                                  |    |         | EXHAUST AIR DUCT DOWN                                                           |
|    | CHV     | CHECK VALVE                                                       |    |         | DUCT TRANSITION                                                                 |
|    | CV (2W) | CONTROL VALVE (2-WAY)                                             |    | CD      | CEILING DIFFUSER                                                                |
|    | CV (3W) | CONTROL VALVE (3-WAY)                                             |    | RR      | RETURN REGISTER                                                                 |
|    | FCD     | AUTOMATIC FLOW CONTROL DEVICE                                     |    | ER      | EXHAUST REGISTER                                                                |
|    | SOV     | SHUT OFF VALVE                                                    |    | T'STAT  | THERMOSTAT OR TEMPERATURE SENSOR<br>(NUMBER INDICATES EQUIPMENT OR ZONE SERVED) |
|    |         | GLOBE/BALL/BUTTERFLY VALVE                                        |    | H'STAT  | HUMIDISTAT                                                                      |
|    | BV      | COMBINATION BALANCING & SHUT-OFF VALVE                            |    |         | CARBON DIOXIDE SENSOR                                                           |
|    | FEV     | FLOW ELEMENT VENTURI                                              |   | CFM     | CUBIC FEET PER MINUTE                                                           |
|    |         | VALVE ON RISE OR DROP                                             |  |         | SYMBOL, SEE EQUIPMENT SCHEDULE                                                  |
|    | STR.    | STRAINER                                                          |                                                                                      |         |                                                                                 |
|   | CL      | CAPPED LINE                                                       |                                                                                      |         |                                                                                 |
|  | DN.     | DOWN OR DROP                                                      |                                                                                      |         |                                                                                 |
|  | UP      | RISE OR RISER                                                     |                                                                                      |         |                                                                                 |
|  | RV      | PRESSURE RELIEF VALVE                                             |                                                                                      |         |                                                                                 |
|  | PG      | PRESSURE GAUGE WITH BALL VALVE                                    |                                                                                      |         |                                                                                 |
|  | R.      | ECCENTRIC REDUCER                                                 |                                                                                      |         |                                                                                 |
|  | R.      | CONCENTRIC REDUCER                                                |                                                                                      |         |                                                                                 |
|  | FC      | FLEXIBLE CONNECTION (PIPE)                                        |                                                                                      |         |                                                                                 |
|  | PA      | PIPE ANCHOR                                                       |                                                                                      |         |                                                                                 |
|  | U       | UNION                                                             |                                                                                      |         |                                                                                 |
|  |         | DUCTWORK (1ST NUMBER INDICATES WIDTH SHOWN), NET INSIDE DIMENSION |                                                                                      |         |                                                                                 |
|  | TV      | SQUARE ELBOW WITH TURNING VANES                                   |                                                                                      |         |                                                                                 |
|  |         | RADIUS ELBOW                                                      |                                                                                      |         |                                                                                 |
|  | MVD     | MANUAL VOLUME DAMPER                                              |                                                                                      |         |                                                                                 |
|  | MOD     | MOTOR OPERATED DAMPER                                             |                                                                                      |         |                                                                                 |
|  | BDD     | BACKDRAFT DAMPER                                                  |                                                                                      |         |                                                                                 |
|  | FD      | FIRE DAMPER                                                       |                                                                                      |         |                                                                                 |
|  | SD      | DUCT MOUNTED SMOKE DETECTOR                                       |                                                                                      |         |                                                                                 |
|  | SFD     | AUTOMATIC SMOKE AND FIRE DAMPER                                   |                                                                                      |         |                                                                                 |
|  | FLEX    | FLEXIBLE CONNECTION (DUCTWORK)                                    |                                                                                      |         |                                                                                 |
|  | FLEX    | FLEXIBLE CONNECTION OR SEISMIC JOINT                              |                                                                                      |         |                                                                                 |

| HVAC ABBREVIATIONS |                               |        |                              |
|--------------------|-------------------------------|--------|------------------------------|
| A                  | AMPERES                       | HZ     | FREQUENCY                    |
| AC                 | AIR CONDITIONING              | IN     | INCH OR INCHES               |
| AD                 | ACCESS DOOR                   | KW     | KILOWATT                     |
| AFF                | ABOVE FINISHED FLOOR          | LG     | LENGTH                       |
| AL                 | ACOUSTICAL LINING             | LAT    | LEAVING AIR TEMPERATURE      |
| BHP                | BRAKE HORSEPOWER              | LBS    | POUNDS                       |
| BTU                | BRITISH THERMAL UNIT          | LDB    | LEAVING DRY BULB TEMPERATURE |
| BTUH               | BTU PER HOUR                  | LIN FT | LINEAR FEET                  |
| CD                 | CEILING DIFFUSER              | LWB    | LEAVING WET BULB TEMPERATURE |
| CFM                | CUBIC FEET PER MINUTE         | MAX    | MAXIMUM                      |
| CG                 | CEILING GRILLE                | MBH    | THOUSAND BTU PER HOUR        |
| CLG                | CEILING                       | MHP    | MOTOR HORSEPOWER             |
| COMPR              | COMPRESSOR                    | MIN    | MINIMUM                      |
| CR                 | CEILING REGISTER              | NIC    | NOT IN CONTRACT              |
| DB                 | DRY BULB                      | NO.    | NUMBER                       |
| DIAM               | DIAMETER                      | NTS    | NOT TO SCALE                 |
| DN                 | DOWN                          | RA     | RETURN AIR                   |
| DWG                | DRAWING                       | RM     | ROOM                         |
| DX                 | DIRECT EXPANSION              | RPM    | REVOLUTIONS PER MINUTE       |
| EAT                | ENTERING AIR TEMPERATURE      | SP     | STATIC PRESSURE              |
| EDB                | ENTERING DRY BULB TEMPERATURE | SPEC   | SPECIFICATION                |
| EF                 | EXHAUST FAN                   | TEMP   | TEMPERATURE                  |
| EWB                | ENTERING WET BULB             | TG     | TOP GRILLE                   |
| EWT                | ENTERING WATER TEMPERATURE    | TV     | TURNING VANES                |
| °F                 | DEGREES FAHRENHEIT            | TYP    | TYPICAL                      |
| FC                 | FLEXIBLE CONNECTION           | W      | WIDTH                        |
| FD                 | FIRE DAMPER                   | W/     | WITH                         |
| FIN FL             | FINISHED FLOOR                | W/O    | WITHOUT                      |
| FLA                | FULL LOAD AMPERES             | WB     | WET BULB                     |
| FPM                | FEET PER MINUTE               | WMS    | WIRE MESH SCREEN             |
| FT                 | FEET                          | SG     | SUPPLY GRILLE                |
| HD                 | HEAD                          | RG     | RETURN GRILLE                |
| HR                 | HOUR                          | SP     | SMOKE PURGE                  |
| MAU                | MAKE UP AIR UNIT              |        |                              |

OWNER SIGNATURE

REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
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| 03  |      |             |          |
| 04  |      |             |          |

CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE: MECHANICAL

PROJECT TITLE:

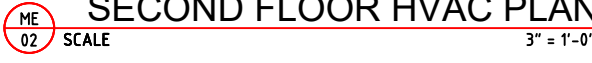
COMMERCIAL MIXED-USE

DRAWING TITLE:

NOTES

|                  |                    |                     |
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| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026 |
| JOB NO           | SHT NO<br>JS-ME-01 | SCALE:<br>NTS @A1   |





|                                                                  |      |              |          |
|------------------------------------------------------------------|------|--------------|----------|
| OWNER SIGNATURE                                                  |      |              |          |
| <b>REVISION STATUS</b>                                           |      |              |          |
| REV                                                              | DATE | DESCRIPTION  | DRAWN BY |
| 00                                                               |      |              |          |
| 01                                                               |      |              |          |
| 02                                                               |      |              |          |
| 03                                                               |      |              |          |
| 04                                                               |      |              |          |
| <b>CLIENT/CONSULTANT APPROVAL STATUS</b>                         |      |              |          |
| APPROVED                                                         |      |              | (A)      |
| APPROVED WITH NOTES                                              |      |              | (B)      |
| RESUBMIT                                                         |      |              | (C)      |
| DISAPPROVED                                                      |      |              | (D)      |
| OWNER:                                                           |      |              |          |
|                                                                  |      |              |          |
| ADDRESS:                                                         |      |              |          |
|                                                                  |      |              |          |
| 608 MAIN STREET, JOHNSTOWN, PA 15901                             |      |              |          |
| DESIGN CONSULTANT:                                               |      |              |          |
| TSABA CORP                                                       |      |              |          |
| NADIF BRACEY                                                     |      |              |          |
| 254 CHAPMAN RD, STE 208 #13130                                   |      |              |          |
| NEWARK, DELAWARE 19702                                           |      |              |          |
| EMAIL:NADIF@TS9DESIGNS.COM                                       |      |              |          |
| DRAWING STATUS:                                                  |      |              |          |
| <b>DESIGN DRAWING</b>                                            |      |              |          |
| DISCIPLINE:                                                      |      |              |          |
| <b>MECHANICAL</b>                                                |      |              |          |
| PROJECT TITLE:                                                   |      |              |          |
| <b>COMMERCIAL MIXED-USE</b>                                      |      |              |          |
| DRAWING TITLE:                                                   |      |              |          |
| <b>FIRST, SECOND, THIRD FLOORS<br/>&amp; ROOF TOP HVAC PLANS</b> |      |              |          |
| DRAWN BY:                                                        |      | CHECKED BY:  |          |
| G.M                                                              |      | N.B          |          |
| DATE:                                                            |      | DATE:        |          |
| 11/05/2026                                                       |      | 11/05/2026   |          |
| JOB NO                                                           |      | SCALE:       |          |
| SHT NO                                                           |      | AS SHOWN @A1 |          |
| JS-ME-02                                                         |      |              |          |



CALC I — HVAC / MINI-SPLIT SCHEDULE | 508 MAIN STREET — JOHNSTOWN, PA 15901 | Commercial Mixed-Use

Source: Mechanical Equipment Schedule | 4 Condensing Units + Exhaust Fans + Roof Ventilators | Date: May 24, 2026

| CONDENSING UNIT SCHEDULE |                                   |      |        |          |         |         |          |       |             |                                                                             |
|--------------------------|-----------------------------------|------|--------|----------|---------|---------|----------|-------|-------------|-----------------------------------------------------------------------------|
| Tag                      | Location / Zones Served           | Tons | BTU/hr | V/Ph/Hz  | Refrig. | MCA (A) | MOCP (A) | Panel | Notes       |                                                                             |
| CU-4T                    | Commercial 1st floor + basement   | 4.0  | 48,000 | 208/1/60 | R-410A  |         | 30       | 40    | Comm. panel | Roof-mounted; curb by structural; serves commercial retail + basement mech. |
| CU-3T                    | Residential 2nd floor — Units 1–4 | 3.0  | 36,000 | 208/1/60 | R-410A  |         | 30       | 40    | Res. panel  | Roof-mounted; multi-split indoor units; curb by structural                  |
| CU-2T                    | Residential 3rd floor — Units 5–8 | 2.0  | 24,000 | 208/1/60 | R-410A  |         | 30       | 40    | Res. panel  | Roof-mounted; multi-split indoor units; curb by structural                  |
| CU-1T                    | Basement / utility / common areas | 1.0  | 12,000 | 208/1/60 | R-410A  |         | 30       | 40    | Comm. panel | Roof-mounted; serves basement mechanical and common areas                   |

| EXHAUST FAN SCHEDULE — Ø4" 150 CFM exhaust fans (all bathrooms and commercial restrooms) |                                                |     |          |           |               |                                  |             |     |                                                             |
|------------------------------------------------------------------------------------------|------------------------------------------------|-----|----------|-----------|---------------|----------------------------------|-------------|-----|-------------------------------------------------------------|
| Tag                                                                                      | Location                                       | CFM | V/Ph/Hz  | Duct Dia. | Mounting      | Control                          | Sound (max) | Qty | Notes                                                       |
| EF-1 thru 8                                                                              | Residential bathrooms — all 8 units            | 150 | 120/1/60 | Ø4"       | Ceiling-mount | Switch-controlled — 0.25 in.w.g. | ≤1.0 sone   | 8   | Back-draft damper; 1 per residential bathroom; IMC §403.4.2 |
| EF-C1, C2                                                                                | Commercial restrooms — 1st floor (2 restrooms) | 150 | 120/1/60 | Ø4"       | Ceiling-mount | Switch-controlled                | ≤1.0 sone   | 2   | 1 per commercial restroom; IMC §403.4.2 — 50 CFM min.       |

| ROOF VENTILATORS — Aura Ventilator Fans (per mechanical plans) |        |           |                    |                       |                     |                                            |
|----------------------------------------------------------------|--------|-----------|--------------------|-----------------------|---------------------|--------------------------------------------|
| Tag                                                            | Zone   | Roof Area | Required Free Area | Calculation: SF × 150 | Specified Unit      | Code Reference                             |
| RV-1                                                           | Zone A | 1,406     | 9.37               | 9.37                  | Aura ventilator fan | IMC §1203 — 1 SF free area per 150 SF roof |
| RV-2                                                           | Zone B | 1,406     | 9.37               | 9.37                  | Aura ventilator fan | IMC §1203                                  |
| RV-3                                                           | Zone C | 976       | 6.51               | 6.51                  | Aura ventilator fan | IMC §1203                                  |

CALC D — OUTDOOR AIR (VENTILATION) | 508 MAIN STREET — JOHNSTOWN, PA 15901 | Commercial Mixed-Use

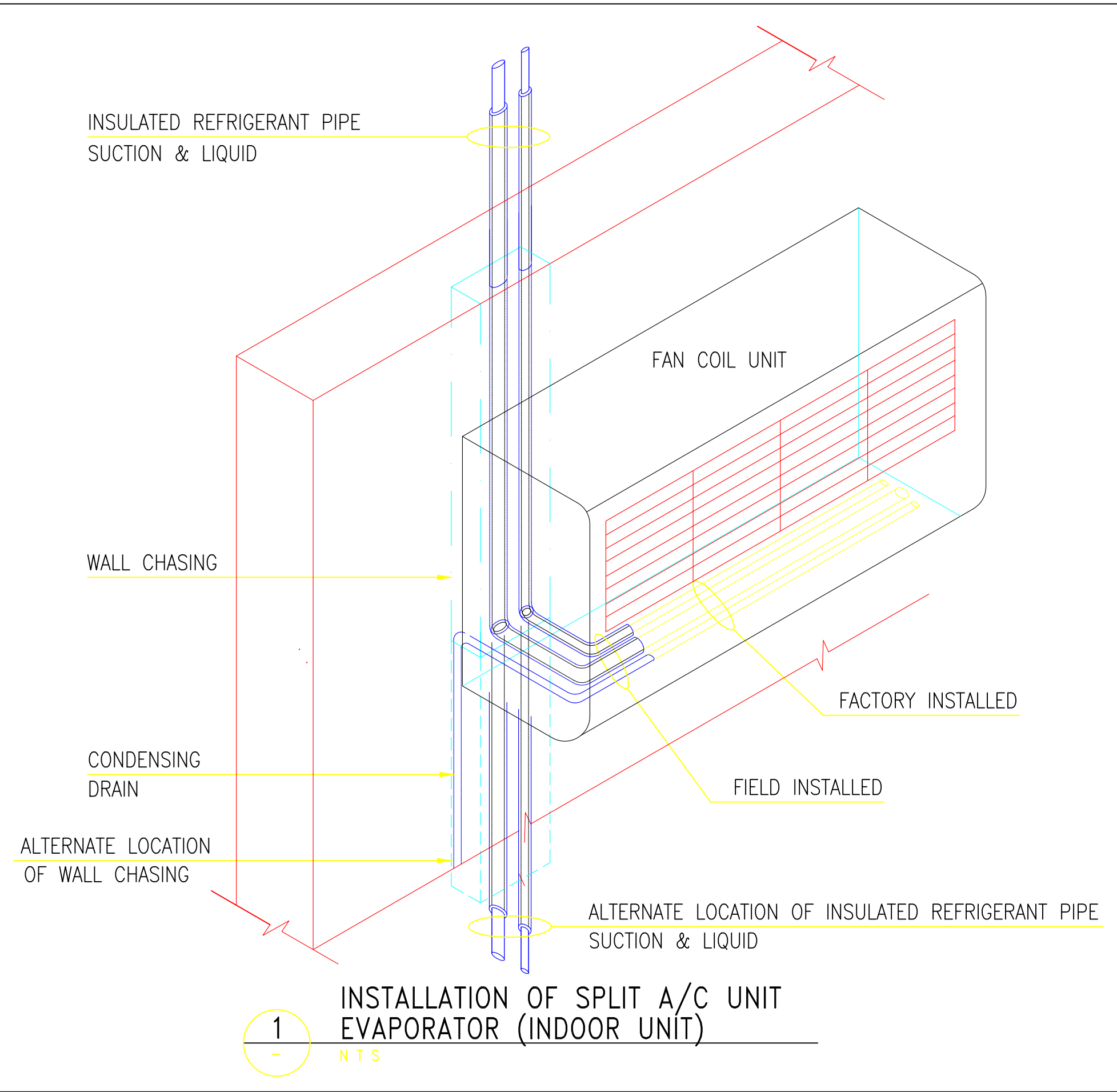
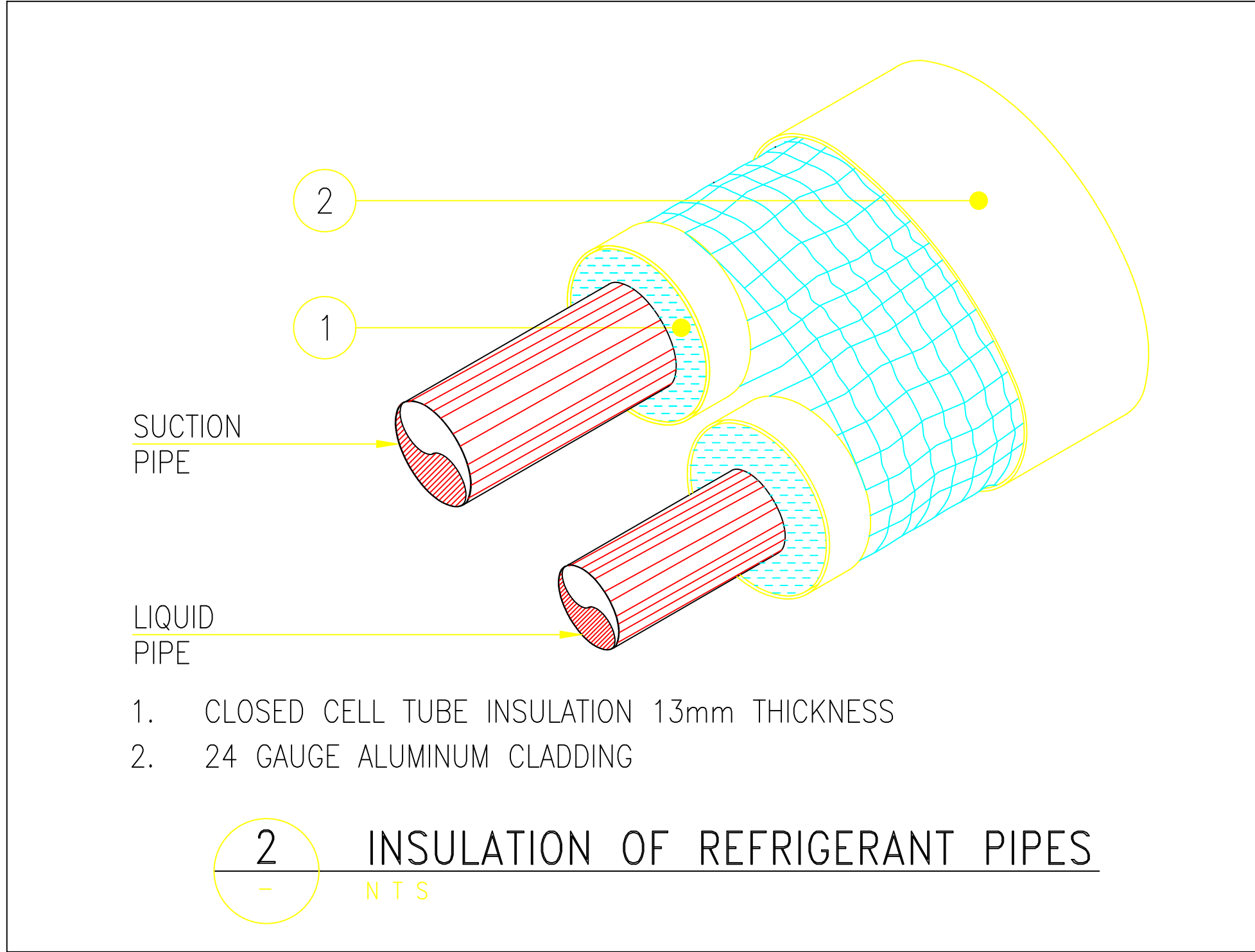
Code: 2018 IMC Table 403.3.1 / ASHRAE 62.1-2016 | Ductless Mini-Split System | Date: May 24, 2026

| DESIGN BASIS        |                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| Standard:           | 2018 IMC Table 403.3.1 — OA = (Area × Ra) + (Occupants × Rp)                               |
| HVAC system:        | Ductless mini-split — roof condensing units + wall-mount indoor units per mechanical plans |
| Res. area rate Ra:  | 0.06 CFM/SF (IMC Table 403.3.1 — R-2 residential occupancy)                                |
| Comm. area rate Ra: | 0.12 CFM/SF (IMC Table 403.3.1 — Retail mercantile occupancy)                              |
| People rate Rp:     | 5.0 CFM/person residential   7.5 CFM/person commercial retail (IMC Table 403.3.1)          |
| Bath exhaust min:   | 50 CFM intermittent per bathroom — IMC §403.4.2                                            |

| BUILDING AREA AND OCCUPANCY INPUTS               |     |                                                                             |
|--------------------------------------------------|-----|-----------------------------------------------------------------------------|
| Commercial unit area per unit (SF, estimated):   | 650 | 2 units × 650 SF = 1,300 SF commercial — verify with architectural drawings |
| Commercial occupant load per unit (1 per 60 SF): | 11  | IBC Table 1004.5 — mercantile: 1 per 60 SF                                  |
| 1-BR unit floor area (SF, estimated — 6 units):  | 380 | Estimated from plan dimensions — verify with architectural drawings         |
| 2-BR unit floor area (SF, estimated — 2 units):  | 420 | Estimated from plan dimensions — verify with architectural drawings         |

| UNIT-BY-UNIT OA CALCULATION — Q = (Area × Ra) + (Occupants × Rp) — 2018 IMC Table 403.3.1 |           |                                            |              |            |         |         |                 |                                                       |  |
|-------------------------------------------------------------------------------------------|-----------|--------------------------------------------|--------------|------------|---------|---------|-----------------|-------------------------------------------------------|--|
| Unit / Space                                                                              | SF (est.) | Occ.                                       | Ra<br>CFM/SF | Rp CFM/occ | Area OA | Occ. OA | Total OA<br>CFM | Notes                                                 |  |
| Commercial Unit A (mercantile, 1st Fl.)                                                   | 650       | IBC Table 1004.5 — mercantile: 1 per 60 SF | 0.12         | 7.5        | 78.0    | #VALUE! | #VALUE!         | CU-4T — commercial mini-split                         |  |
| Commercial Unit B (mercantile, 1st Fl.)                                                   | 650       | IBC Table 1004.5 — mercantile: 1 per 60 SF | 0.12         | 7.5        | 78.0    | #VALUE! | #VALUE!         | CU-4T — commercial mini-split                         |  |
| Residential Common / Lobby                                                                | 200       |                                            | 5            | 0.06       | 5.0     | 12.0    | 25.0            | 37.0 Common area estimate                             |  |
| Apt. Unit 1 (1-BR, 2nd Fl.)                                                               | 380       |                                            | 3            | 0.06       | 5.0     | 22.8    | 15.0            | 37.8 CU-3T — 2nd floor                                |  |
| Apt. Unit 2 (1-BR, 2nd Fl.)                                                               | 380       |                                            | 3            | 0.06       | 5.0     | 22.8    | 15.0            | 37.8 CU-3T — 2nd floor                                |  |
| Apt. Unit 3 (1-BR, 2nd Fl.)                                                               | 380       |                                            | 3            | 0.06       | 5.0     | 22.8    | 15.0            | 37.8 CU-3T — 2nd floor                                |  |
| Apt. Unit 4 (1-BR, 2nd Fl.)                                                               | 380       |                                            | 3            | 0.06       | 5.0     | 22.8    | 15.0            | 37.8 CU-3T — 2nd floor                                |  |
| Apt. Unit 5 (1-BR, 3rd Fl.)                                                               | 380       |                                            | 3            | 0.06       | 5.0     | 22.8    | 15.0            | 37.8 CU-2T — 3rd floor                                |  |
| Apt. Unit 6 (1-BR, 3rd Fl.)                                                               | 380       |                                            | 3            | 0.06       | 5.0     | 22.8    | 15.0            | 37.8 CU-2T — 3rd floor                                |  |
| Apt. Unit 7 (2-BR, 3rd Fl.)                                                               | 420       |                                            | 4            | 0.06       | 5.0     | 25.2    | 20.0            | 45.2 CU-2T — 3rd floor                                |  |
| Apt. Unit 8 (2-BR, 3rd Fl.)                                                               | 420       |                                            | 4            | 0.06       | 5.0     | 25.2    | 20.0            | 45.2 CU-2T — 3rd floor                                |  |
| Bathroom exhaust — all units (10 total)                                                   | 0         |                                            | 0            | 0.00       | 0.0     | 0.0     | 0.0             | 500.0 10 × 50 CFM per IMC §403.4.2 (8 res. + 2 comm.) |  |
| Basement / mechanical / storage                                                           | 1,200     |                                            | 2            | 0.06       | 5.0     | 72.0    | 10.0            | 82.0 CU-1T — basement utility                         |  |
| TOTAL REQUIRED OUTDOOR AIR (CFM)                                                          |           |                                            |              |            |         |         | #VALUE!         |                                                       |  |

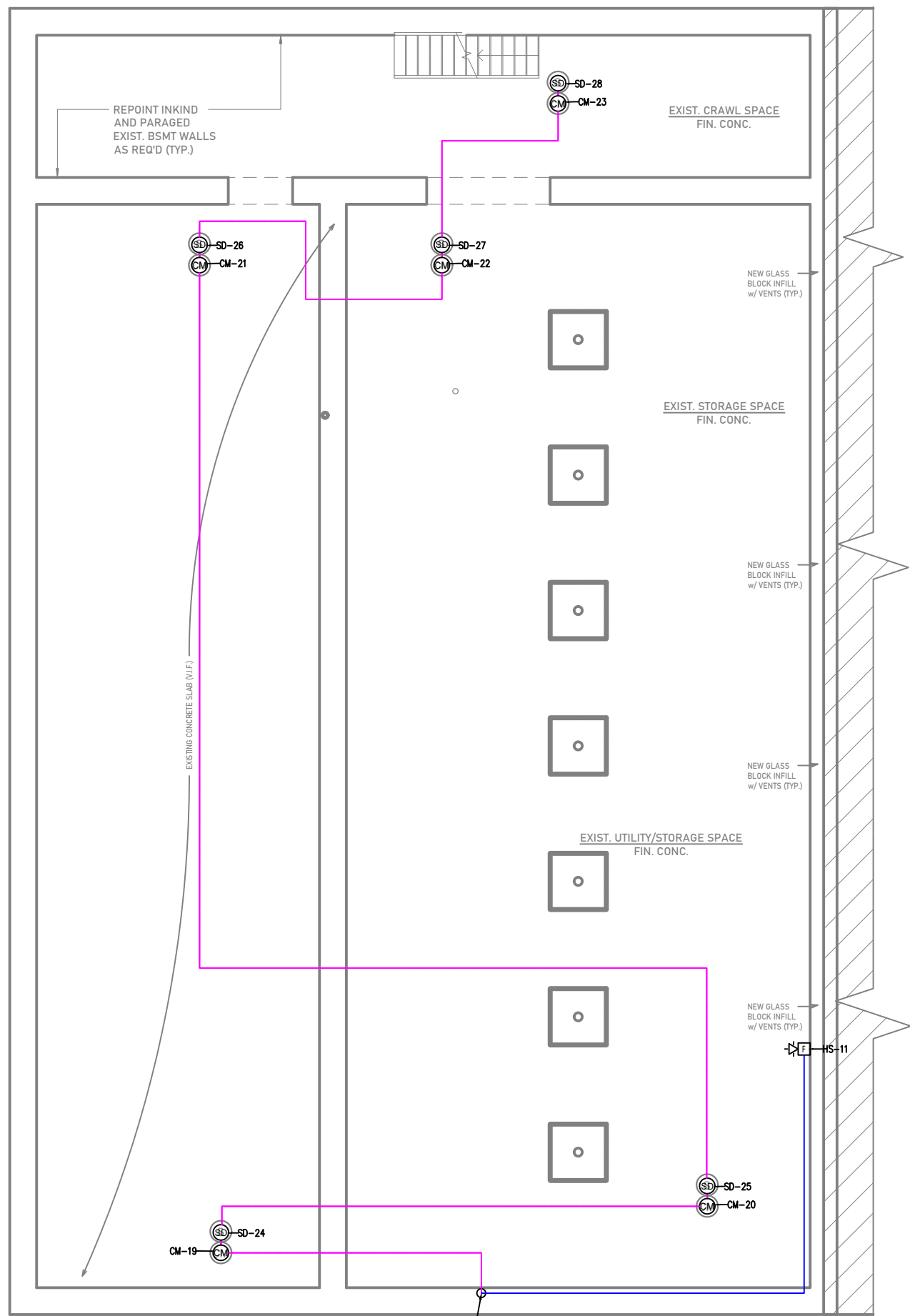
| VENTILATION SUMMARY                   |                            |                |                                                               |        |          |
|---------------------------------------|----------------------------|----------------|---------------------------------------------------------------|--------|----------|
| Parameter                             | Requirement                | Code Reference | Provided / Design                                             | Status |          |
| Bath exhaust — residential (8 units)  | 50 CFM min. per bathroom   | IMC §403.4.2   | 8 × Ø4" @ 150 CFM per mechanical plans                        |        | COMPLIES |
| Restroom exhaust — commercial (2)     | 50 CFM min. per WC room    | IMC §403.4.2   | 2 × Ø4" @ 150 CFM per mechanical plans                        |        | COMPLIES |
| Roof ventilation (3 Aura ventilators) | 1 SF free area per 150 SF  | IMC §1203      | Zones A/B: 9.37 SF ea.   Zone C: 6.51 SF per mechanical plans |        | COMPLIES |
| OA intakes — FLOOD ZONE               | Intakes at BFE + 2 ft min. | IBC §1612      | All condensing units roof-mounted — satisfies elevation req.  |        | COMPLIES |



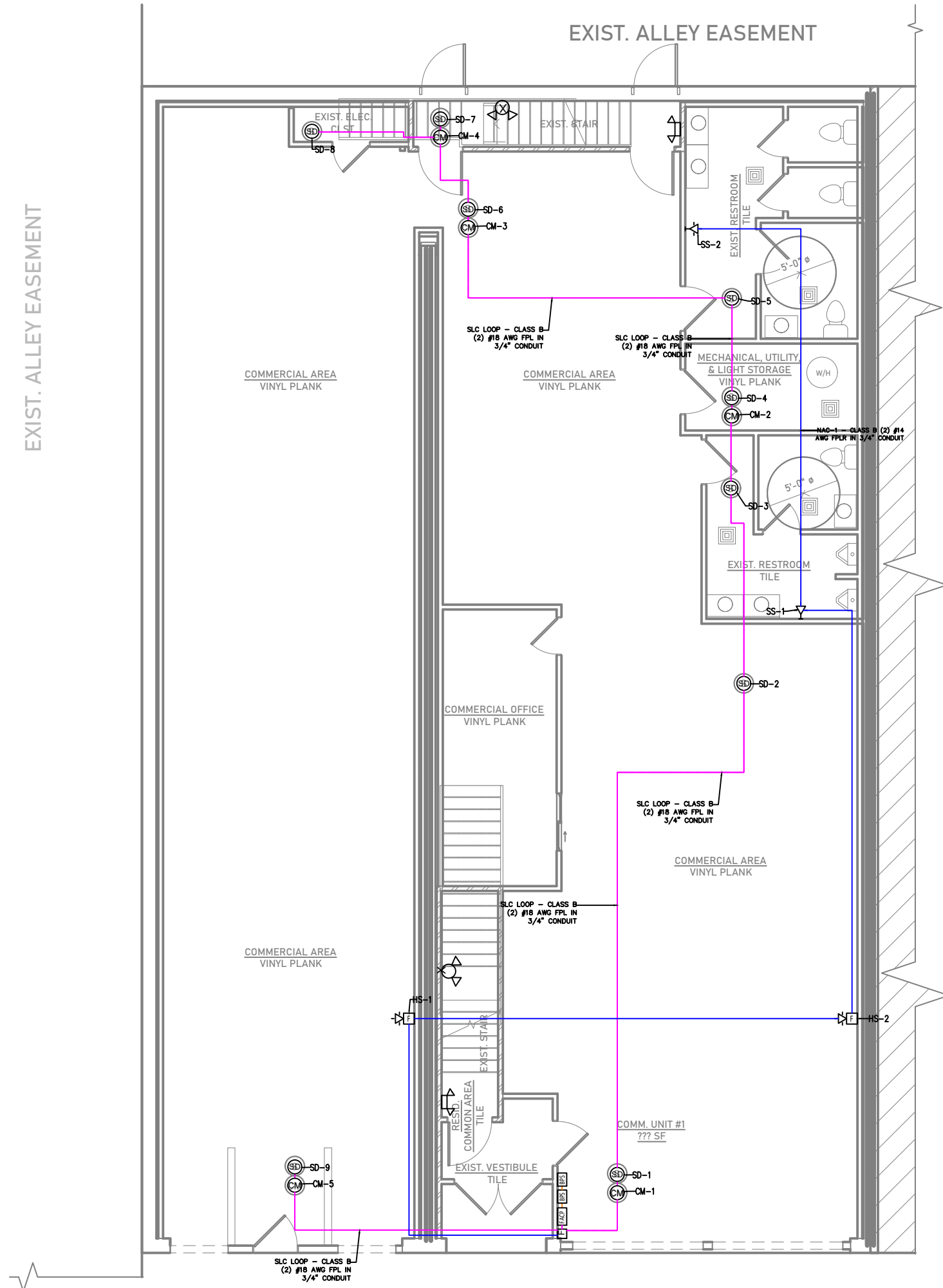
OWNER SIGNATURE

| REVISION STATUS                                                                                                                            |                    |                     |          |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|----------|
| Rev                                                                                                                                        | DATE               | DESCRIPTION         | DRAWN BY |
| 00                                                                                                                                         |                    |                     |          |
| 01                                                                                                                                         |                    |                     |          |
| 02                                                                                                                                         |                    |                     |          |
| 03                                                                                                                                         |                    |                     |          |
| 04                                                                                                                                         |                    |                     |          |
| CLIENT/CONSULTANT APPROVAL STATUS                                                                                                          |                    |                     |          |
| APPROVED                                                                                                                                   |                    |                     | (A)      |
| APPROVED WITH NOTES                                                                                                                        |                    |                     | (B)      |
| RESUBMIT                                                                                                                                   |                    |                     | (C)      |
| DISAPPROVED                                                                                                                                |                    |                     | (D)      |
| OWNER:                                                                                                                                     |                    |                     |          |
| ADDRESS:                                                                                                                                   |                    |                     |          |
| 508 MAIN STREET, JOHNSTOWN, PA 15901                                                                                                       |                    |                     |          |
| DESIGN CONSULTANT:<br>TSABA CORP<br>NADIF BRACEY<br>254 CHAPMAN RD, STE 208 #13130<br>NEWARK, DELAWARE 19702<br>EMAIL:NADIF@TS9DESIGNS.COM |                    |                     |          |
| DRAWING STATUS:<br>DESIGN DRAWING                                                                                                          |                    |                     |          |
| DISCIPLINE: MECHANICAL                                                                                                                     |                    |                     |          |
| PROJECT TITLE:<br>COMMERCIAL MIXED-USE                                                                                                     |                    |                     |          |
| DRAWING TITLE:<br>HVAC DETAILS & MECHANICAL SCHEDULE                                                                                       |                    |                     |          |
| DRAWN BY:<br>G.M                                                                                                                           | CHECKED BY:<br>N.B | DATE:<br>11/05/2026 |          |
| JOB NO                                                                                                                                     | SHT NO<br>JS-ME-03 | SCALE:<br>NTS @A1   |          |

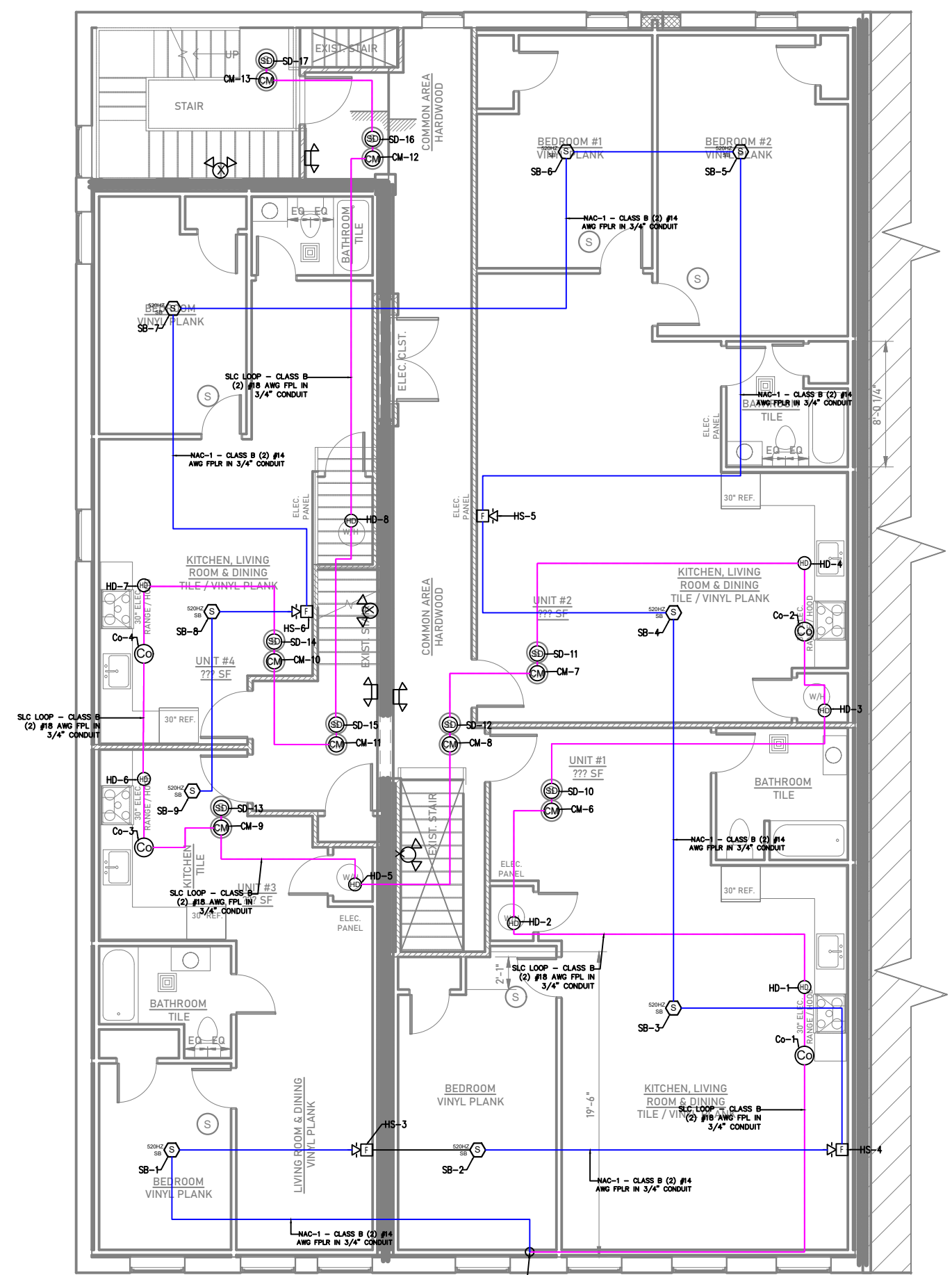




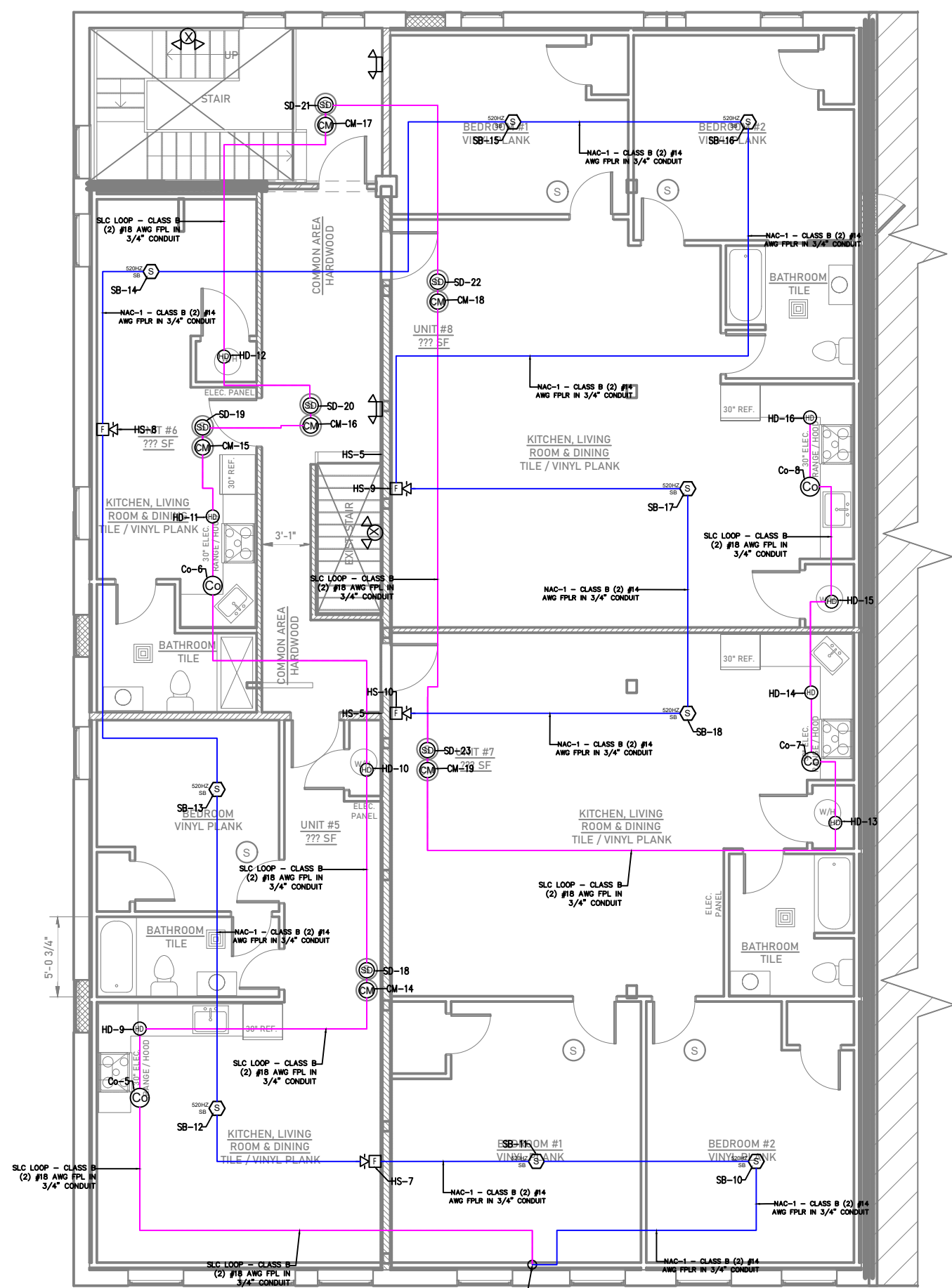
FA 01 BASEMENT FLOOR FIRE ALARM PLAN  
3" = 1'-0"



FA 01 FIRST FLOOR FIRE ALARM PLAN  
3" = 1'-0"



FA 01 SECOND FLOOR FIRE ALARM PLAN  
3" = 1'-0"



FA 01 THIRD FLOOR FIRE ALARM PLAN  
3" = 1'-0"

#### GENERAL FIRE ALARM INSTALLATION NOTES

**VERIFY EXISTING CONDITIONS:** The fire alarm contractor shall verify all existing site conditions prior to beginning work and make any necessary adjustments to the installation, as required by field conditions.

**WIRE SIZE & TYPE:** Minimum wire size for fire alarm circuits shall be 18 AWG, unless otherwise specified by the manufacturer's requirements or the fire alarm control panel specifications.

**CODE COMPLIANCE:** All work shall comply with the latest edition of the National Fire Alarm and Signaling Code (NFPA 72), National Electrical Code (NEC), and all other applicable local and state codes and regulations.

**CONDUIT & WIRING:** All fire alarm wiring shall be installed in conduit unless otherwise specified or permitted by local code. Wire must be fire-rated and approved for fire alarm system use.

**DEVICE MOUNTING LOCATIONS:** All fire alarm devices (smoke detectors, heat detectors, pull stations, notification appliances) must be installed in locations per the approved fire alarm plans and in accordance with NFPA 72 spacing requirements.

**ALARM TESTING:** Contractor to conduct full system testing upon completion of installation, including testing of all notification appliances and initiating devices, in compliance with NFPA 72. Testing shall be documented, and results provided to the owner.

**SYSTEM LABELING & DOCUMENTATION:** All fire alarm circuits, devices, and wiring shall be properly labeled in accordance with code and as per the fire alarm system design documents. As-built drawings and manuals shall be provided to the owner upon completion.

**COORDINATION WITH OTHER TRADES:** The fire alarm contractor shall coordinate with other trades, including electrical, mechanical, and general contractors, to avoid conflicts and ensure proper installation.

**BATTERY BACKUP:** Ensure that the fire alarm system's battery backup complies with the system requirements for standby power, as outlined in NFPA 72.

**MONITORING:** Contractor to ensure fire alarm system is properly connected to the required central monitoring station or fire department as per project specifications and local jurisdiction requirements.

#### NOTES:

**FLOOD FIRE ALARM MATERIALS NOTE:**  
FACP, NAC BOOSTER POWER SUPPLIES, BATTERIES, SURGE PROTECTIVE DEVICES, COMMUNICATORS, AND FIRE ALARM CONTROL EQUIPMENT SHALL BE LOCATED ABOVE BFE/DFE/RFE. FIRE ALARM RACEWAYS BELOW BFE/DFE/RFE SHALL BE RMC, IMC, OR SCH. 80 PVC WITH WET-LOCATION-RATED FITTINGS. FIRE ALARM CONDUCTORS BELOW BFE/DFE/RFE SHALL BE LISTED FOR WET LOCATIONS WHERE INSTALLED IN RACEWAYS SUBJECT TO FLOODING.

FIRE ALARM JUNCTION BOXES, TERMINAL CABINETS, MODULE BOXES, AND DEVICE BACKBOXES BELOW BFE/DFE/RFE SHALL BE CORROSION-RESISTANT AND GASKETED WHERE EXPOSED TO MOISTURE. DEVICES EXPOSED TO FLOODWATER SHALL BE REPLACED AFTER FLOOD EVENT UNLESS MANUFACTURER AND AHJ APPROVE REUSE.

#### FIRE ALARM SEQUENCE OF OPERATIONS

##### Overview

The fire alarm system is designed to ensure prompt detection and notification of a fire event, enabling safe evacuation and response. The system includes sounder basins in each bedroom, horn strobes throughout the building, fire alarm pull stations, and heat and duct detectors.

##### Sequence of Operations

##### Detection Phase

**Heat Detector Activation:** The heat detector is strategically placed in the duct to monitor for abnormal temperature increases indicative of a fire. When the heat detector senses a temperature rise above its threshold, it triggers the fire alarm system.

**Duct Detector Activation:** The duct detector monitors air flow and temperature in the HVAC system. If smoke is detected in the ductwork, the duct detector sends a signal to the fire alarm panel.

**Alarm Activation**

**Signal Processing:** Upon receiving signals from the heat detector or duct detector, the fire alarm panel processes these inputs and determines the appropriate response.

**Alarm Notification:** The fire alarm panel activates the alarm system to alert occupants. This includes:

**Sounder Basins:** The sounder basins in each bedroom emit a 540 Hz tone. This sound is specifically designed to be easily heard and recognized, even during sleep.

**Horn Strobes:** Horn strobes are activated throughout the building. The horn emits a loud alarm sound while the strobe light provides a visual alert, ensuring that both hearing and visually impaired occupants are notified.

**Fire Alarm Pull Stations:** If any occupant manually activates a fire alarm pull station, the system immediately processes this input as a fire emergency and initiates the same alarm notification procedures as described above.

##### Notification and Evacuation

**Alarm Signal Transmission:** Once the alarm is triggered, the system sends an alarm signal to the central monitoring station (if applicable) and activates all alert devices.

**Evacuation Instructions:** The horn strobes and sounder basins continue to operate until the fire alarm panel is manually reset or until the fire is deemed to be under control. Occupants are instructed to evacuate the building immediately upon hearing the alarm.

**System Reset**

**Alarm Panel Reset:** After the alarm event has been addressed and the cause of the alarm has been resolved, the fire alarm panel can be reset to return the system to its normal monitoring state. The system must be inspected to ensure it is fully operational before it can be reset.

| ELECTRICAL SYMBOL LEGEND                                                                                                                                                                                                  |                                    |            |                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------|--------------------------------------------|
| <b>NOTES:</b><br>1. ALL SYMBOLS MAY NOT APPLY.<br>2. MOUNTING HEIGHTS AS SHOWN UNLESS NOTED OTHERWISE ON THE DRAWING.<br>3. MOUNTED HEIGHTS ARE FROM FINISHED FLOOR TO CENTER OF DEVICE OR FROM FINL GRADE AS APPLICABLE. |                                    |            |                                            |
| [FACP]                                                                                                                                                                                                                    | FIRE ALARM ANNUNCIATOR             | [SD]       | SMOKE DETECTOR                             |
| [FACP]                                                                                                                                                                                                                    | FIRE ALARM CONTROL PANEL           | [HD]       | HEAT DETECTOR                              |
| [PST]                                                                                                                                                                                                                     | PULL STATION 4W2" AFF              | [CND]      | CARBON MONOXIDE DETECTOR                   |
| [SPK]                                                                                                                                                                                                                     | SPEAKER STROBE                     | [DD]       | DUCT DETECTOR                              |
| [HNS]                                                                                                                                                                                                                     | HORN-STROBE                        | [RTS]      | REMOTE TEST STATION                        |
| [SD]                                                                                                                                                                                                                      | STROBE - CEILING MOUNTED           | [SDS]      | SMOKE DETECTOR WITH LF SOUNDER             |
| [HNS]                                                                                                                                                                                                                     | HORN-STROBE - CEILING MOUNTED      | [TS]       | TAMPER SWITCH                              |
| [BPS]                                                                                                                                                                                                                     | BOOSTER POWER SUPPLY               | [PS]       | PRESSURE SWITCH                            |
| [ED]                                                                                                                                                                                                                      | ELECTRONIC DOOR HOLD-OPEN DEVICE   | [RM]       | RELAY MODULE                               |
| [ES]                                                                                                                                                                                                                      | EXIT SIGN                          | [FASL]     | FIRE ALARM SIREN LIGHT                     |
| [ECLC]                                                                                                                                                                                                                    | EXIT SIGN EMERGENCY LIGHTING COMBO | [CND-BATT] | CARBON-MONOXIDE DETECTOR W/ BATTERY (XS15) |

#### OWNER SIGNATURE

#### REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
| 01  |      |             |          |
| 02  |      |             |          |
| 03  |      |             |          |
| 04  |      |             |          |

#### CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

OWNER:

ADDRESS:

508 MAIN STREET, JOHNSTOWN, PA 15901

DESIGN CONSULTANT:  
TSABA CORP  
NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE:

FIRE ALARM

PROJECT TITLE:

COMMERCIAL MIXED-USE

DRAWING TITLE:

BASEMENT, FIRST, SECOND, THIRD  
FLOOR FIRE ALARM PLANS

|                  |                    |                        |
|------------------|--------------------|------------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026    |
| JOB NO           | SHT NO<br>JS-FA-01 | SCALE:<br>AS SHOWN @A1 |





|                         |                           |                            |
|-------------------------|---------------------------|----------------------------|
| <u>DRAWN BY:</u><br>G.M | <u>CHECKED BY:</u><br>N.B | <u>DATE:</u><br>11/05/2026 |
| <u>JOB NO</u>           | SHT NO<br>JS-FA-02        | <u>SCALE:</u><br>NTS @A1   |



CALC E — FIRE ALARM BATTERY BACKUP | 508 MAIN STREET — JOHNSTOWN, PA 15901 | Commercial Mixed-Use

Code: NFPA 72 (2019) §10.6.7 | AHJ: City of Johnstown / Johnstown Fire Bureau | Date: May 24, 2026

Device quantities are engineering estimates based on building program (2 commercial + 8 residential units). Device currents are from manufacturer data per NFPA 72 device specifications. FA contractor to verify all quantities and currents against submitted FACP and final device data sheets prior to installation.

| DEVICE CURRENT SCHEDULE — Fire Alarm System   Basis: Building program (2 commercial units + 8 apartments) + NFPA 72 spacing |     |                  |                    |                |                  |                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------|-----|------------------|--------------------|----------------|------------------|-----------------------------------------------------------------------|
| Device Description                                                                                                          | Qty | Standby (mA ea.) | Total Standby (mA) | Alarm (mA ea.) | Total Alarm (mA) | Basis / NFPA 72 Reference                                             |
| FACP — addressable, main electrical room                                                                                    | 1   | 100.000          | 100.000            | 200.000        | 200.000          | Honeywell 6700 or equal; 120VAC dedicated lockable circuit            |
| Smoke detectors — photoelectric, addressable                                                                                | 24  | 1.000            | 24.000             | 50.000         | 1,200.000        | NFPA 72 §17.7 — 1 per 900 SF; required in all bedrooms + corridors    |
| Heat detectors — ROR + fixed temp., addressable                                                                             | 8   | 1.000            | 8.000              | 30.000         | 240.000          | NFPA 72 §17.6 — OH-1 spacing; 1 per mechanical / utility room         |
| CO detectors — addressable (gas range units)                                                                                | 8   | 5.000            | 40.000             | 20.000         | 160.000          | IBC §915 / NFPA 720 — 1 per residential unit kitchen + mech. rooms    |
| Sounder bases — 520 Hz low frequency (sleeping areas)                                                                       | 16  | 0.000            | 0.000              | 75.000         | 1,200.000        | NFPA 72 §29.5.1 — 520 Hz required in bedrooms; 1–2 per unit           |
| Manual pull stations — addressable (at exits)                                                                               | 8   | 0.000            | 0.000              | 0.000          | 0.000            | NFPA 72 §17.14 — within 5 ft of each exit door                        |
| Control modules — addressable (HVAC / door holders)                                                                         | 10  | 5.000            | 50.000             | 0.000          | 0.000            | NFPA 72 §21 — SLC Class B; 1 per HVAC unit + door holders             |
| Door holders — EM, 24VDC, fail-safe                                                                                         | 4   | 65.000           | 260.000            | 0.000          | 0.000            | IBC §716.5 — fire-rated corridor / stairwell doors; fail-safe release |
| Duct smoke detectors — HVAC return (>2,000 CFM units)                                                                       | 2   | 5.000            | 10.000             | 20.000         | 40.000           | 2018 IMC §606.4 — 1 per AHU unit > 2,000 CFM; shuts fan on alarm      |
| TOTALS — BUILDING FIRE ALARM SYSTEM                                                                                         |     |                  | 492.000            |                | 3,040.000        |                                                                       |

| BATTERY CAPACITY CALCULATION — NFPA 72 §10.6.7 (24-Hour Standby + 5-Minute Alarm + 20% Safety Factor) |                                                                                        |                        |                                                       |    |                                                              |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------|----|--------------------------------------------------------------|
| Total standby current (A)                                                                             |                                                                                        |                        | 0.492                                                 | A  | Convert mA to A — Total system standby current               |
| Total alarm current (A)                                                                               |                                                                                        |                        | 3.040                                                 | A  | Convert mA to A — Total system alarm current                 |
| 24-hour standby capacity required (Ah)                                                                |                                                                                        |                        | 11.808                                                | Ah | NFPA 72 §10.6.7.2.1 — 24 hours at standby current            |
| 5-minute alarm capacity required (Ah)                                                                 |                                                                                        |                        | 0.253                                                 | Ah | NFPA 72 §10.6.7.2.1 — 5 minutes (0.0833 hr) at alarm current |
| Subtotal required (Ah)                                                                                |                                                                                        |                        | 12.061                                                | Ah | Standby Ah + Alarm Ah                                        |
| Apply 20% safety factor (NFPA 72 §10.6.7.2)                                                           |                                                                                        |                        | 14.474                                                | Ah | Subtotal × 1.20 — REQUIRED BATTERY CAPACITY                  |
| MINIMUM BATTERY REQUIRED (Ah)                                                                         | See formula cell E26 —<br>per NFPA 72 §10.6.7<br>calculation above                     |                        | 14.474                                                | Ah | Specify sealed lead-acid battery at or above this value      |
| Minimum required (calculated, Ah):                                                                    | 14.474                                                                                 | Ah minimum<br>required | COMPLIES — 24.0 Ah specified EXCEEDS 14.47 Ah minimum |    |                                                              |
| SPECIFIED BATTERY — NFPA 72 §10.6.7 COMPLIANCE                                                        |                                                                                        |                        |                                                       |    |                                                              |
| Specified battery type:                                                                               | Two (2) 12V / 24Ah sealed lead-acid (SLA) batteries wired in series = 24V / 24Ah total |                        |                                                       |    |                                                              |
| Capacity provided (Ah):                                                                               | 24.0 Ah [two (2) 12V / 24Ah SLA batteries in series]                                   |                        |                                                       |    |                                                              |
| Battery location:                                                                                     | Inside FACP enclosure or adjacent NEMA 1 lockable enclosure — at BFE + 2 ft min.       |                        |                                                       |    |                                                              |
| Battery charger:                                                                                      | Integral to FACP — supervised trickle charge per NFPA 72 §10.6.7.3                     |                        |                                                       |    |                                                              |
| Low battery supervisory:                                                                              | FACP shall annunciate low battery condition per NFPA 72 §10.6.7.4                      |                        |                                                       |    |                                                              |
| Central station monitoring:                                                                           | Required — provide two dedicated telephone lines in conduit per NFPA 72                |                        |                                                       |    |                                                              |
| FLOOD ZONE:                                                                                           | FACP and battery assembly elevated to BFE + 2 ft minimum per IBC §1612                 |                        |                                                       |    |                                                              |

CALC J — FIRE ALARM DEVICE SCHEDULE | 508 MAIN STREET — JOHNSTOWN, PA 15901 | Commercial Mixed-Use

Code: NFPA 72-2019 | AHJ: City of Johnstown / Johnstown Fire Bureau | Submit FA plans separately | Date: May 24, 2026

⚠ FLOOD ZONE: FACP and all FA equipment at BFE + 2 ft minimum (IBC §1612). If electrical room is below BFE, relocate FACP to 2nd floor prior to equipment order. Submit FA plans to Johnstown Fire Bureau separately from building permit (814) 533-2000.

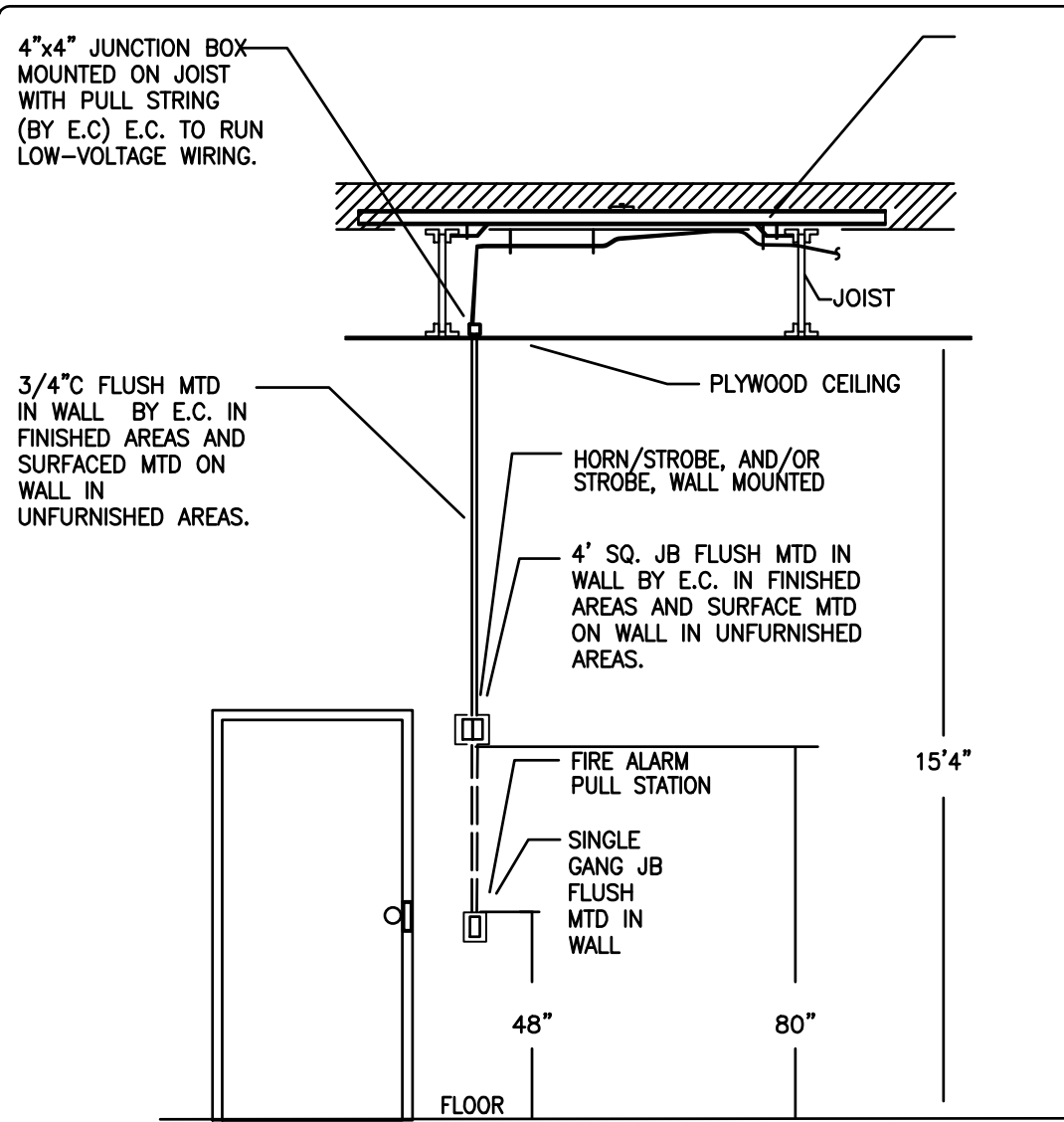
| DEVICE SCHEDULE — Quantities based on building program + NFPA 72 spacing requirements   2 commercial + 8 residential units                                                                                                                                                                                                                                                                                                                                                                                  |                                                     |                                    |     |                        |                                                                 |                                                                 |         |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------|-----|------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|---------|--------|
| Sym.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Device Description                                  | Mfr / Model (Specify)              | Qty | Mounting               | Location                                                        | NFPA 72 Basis                                                   | Stby mA | Alm mA |
| FACP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fire Alarm Control Panel — addressable, Class B SLC | Honeywell 6700 or approved equal   | 1   | Elect. room at BFE+2ft | Main electrical room (2nd floor if 1st floor below BFE)         | NFPA 72 §21 — dedicated 120VAC; lockable red-handle breaker     | 100     | 200    |
| SD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Smoke Detector — addressable photoelectric          | Honeywell / System Sensor or equal | 24  | Ceiling                | All bedrooms, corridors, living areas, stairwells, common areas | NFPA 72 §17.7 — spacing per Annex B; required in all bedrooms   | 1       | 50     |
| HD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Heat Detector — ROR + fixed temp., addressable      | Honeywell SK-HEAT or equal         | 8   | Ceiling                | 1 per mechanical / utility room (basement + floors)             | NFPA 72 §17.6 — OH-1 spacing (196 SF max); kitchens/mech. rooms | 1       | 30     |
| CO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO Detector — addressable                           | Honeywell SK-FIRE-CO or equal      | 8   | Wall 12" below ceiling | Residential unit kitchens (gas ranges) + basement mech. areas   | IBC §915 / NFPA 720 — 1 per residential unit with gas appliance | 5       | 20     |
| SB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Sounder Base — 520 Hz low frequency                 | System Sensor COSMO-2WB or equal   | 16  | Ceiling                | All bedrooms — 1 to 2 per bedroom per NFPA 72 §29.5.1           | NFPA 72 §29.5.1 — 520 Hz tone required in all sleeping areas    | 0       | 75     |
| F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Manual Pull Station — addressable                   | System Sensor S3500PS or equal     | 8   | Wall 42–48" AFF        | 1 per exit door — within 5 ft of each exit                      | NFPA 72 §17.14 — manual station within 5 ft of each exit        | 0       | 0      |
| CM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Control Module — addressable                        | Honeywell M500MX or equal          | 10  | At device location     | HVAC duct detectors, door holders, HVAC shutdowns               | NFPA 72 §21 — SLC Class B                                       | 5       | 0      |
| DH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Door Holder — EM, 24VDC, fail-safe release          | Specify — 24VDC EM holder          | 4   | Wall or floor          | Stairwell + corridor fire-rated door assemblies                 | IBC §716.5 — deenergize to release on any alarm signal          | 65      | 0      |
| DD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Duct Smoke Detector — HVAC return air               | System Sensor FAPT or equal        | 2   | Duct-mounted           | Installed in return air duct of any AHU > 2,000 CFM             | 2018 IMC §606.4 — initiates unit shutdown on alarm              | 5       | 20     |
| Sequence of operations: manual pull / smoke / heat / CO detection → FACP alarm → activate all sounder bases (520 Hz) and horn/strobes → initiate HVAC shutdown via duct detector control modules → release door holders → transmit to central monitoring station. All wiring in ¾" conduit per NFPA 72 §12.3. Battery backup per Sheet E. FA contractor verifies device locations against NFPA 72 spacing before rough-in. FLOOD ZONE: FACP and all battery assemblies at BFE + 2 ft minimum per IBC §1612. |                                                     |                                    |     |                        |                                                                 |                                                                 |         |        |

OWNER SIGNATURE

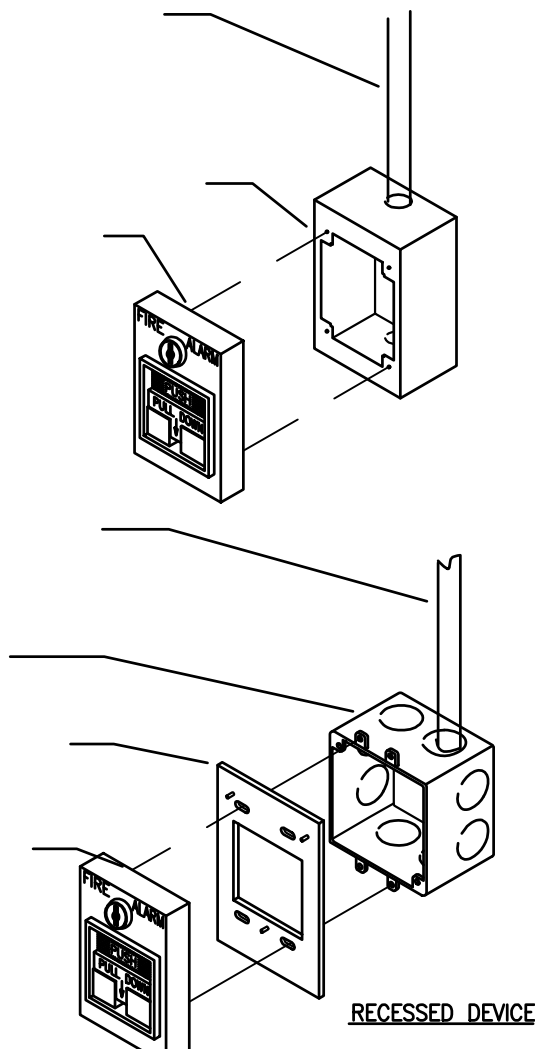
| REVISION STATUS                                                      |      |             |            |
|----------------------------------------------------------------------|------|-------------|------------|
| Rev                                                                  | DATE | DESCRIPTION | DRAWN BY   |
| 00                                                                   |      |             |            |
| 01                                                                   |      |             |            |
| 02                                                                   |      |             |            |
| 03                                                                   |      |             |            |
| 04                                                                   |      |             |            |
|                                                                      |      |             |            |
| CLIENT/CONSULTANT APPROVAL STATUS                                    |      |             |            |
| APPROVED                                                             |      | (A)         |            |
| APPROVED WITH NOTES                                                  |      | (B)         |            |
| RESUBMIT                                                             |      | (C)         |            |
| DISAPPROVED                                                          |      | (D)         |            |
| OWNER:                                                               |      |             |            |
|                                                                      |      |             |            |
| ADDRESS:                                                             |      |             |            |
|                                                                      |      |             |            |
| 508 MAIN STREET, JOHNSTOWN, PA 15901                                 |      |             |            |
|                                                                      |      |             |            |
| DESIGN CONSULTANT:                                                   |      |             |            |
| TSABA CORP                                                           |      |             |            |
| NADIF BRACEY                                                         |      |             |            |
| 254 CHAPMAN RD, STE 208 #13130                                       |      |             |            |
| NEWARK, DELAWARE 19702                                               |      |             |            |
| EMAIL:NADIF@TS9DESIGNS.COM                                           |      |             |            |
| DRAWING STATUS:                                                      |      |             |            |
| DESIGN DRAWING                                                       |      |             |            |
| DISCIPLINE:                                                          |      |             |            |
| FIRE ALARM                                                           |      |             |            |
| PROJECT TITLE:                                                       |      |             |            |
| COMMERCIAL MIXED-USE                                                 |      |             |            |
| DRAWING TITLE:                                                       |      |             |            |
| FIRE ALARM SEQUENCE OF OPERATION<br>MATRIX & BATTERY-SET CALCULATION |      |             |            |
| DRAWN BY:                                                            |      | CHECKED BY: | DATE:      |
| G.M                                                                  |      | N.B         | 11/05/2026 |
| JOB NO                                                               |      | SHT NO      | SCALE:     |
|                                                                      |      | JS-FA-03    | NTS @A1    |



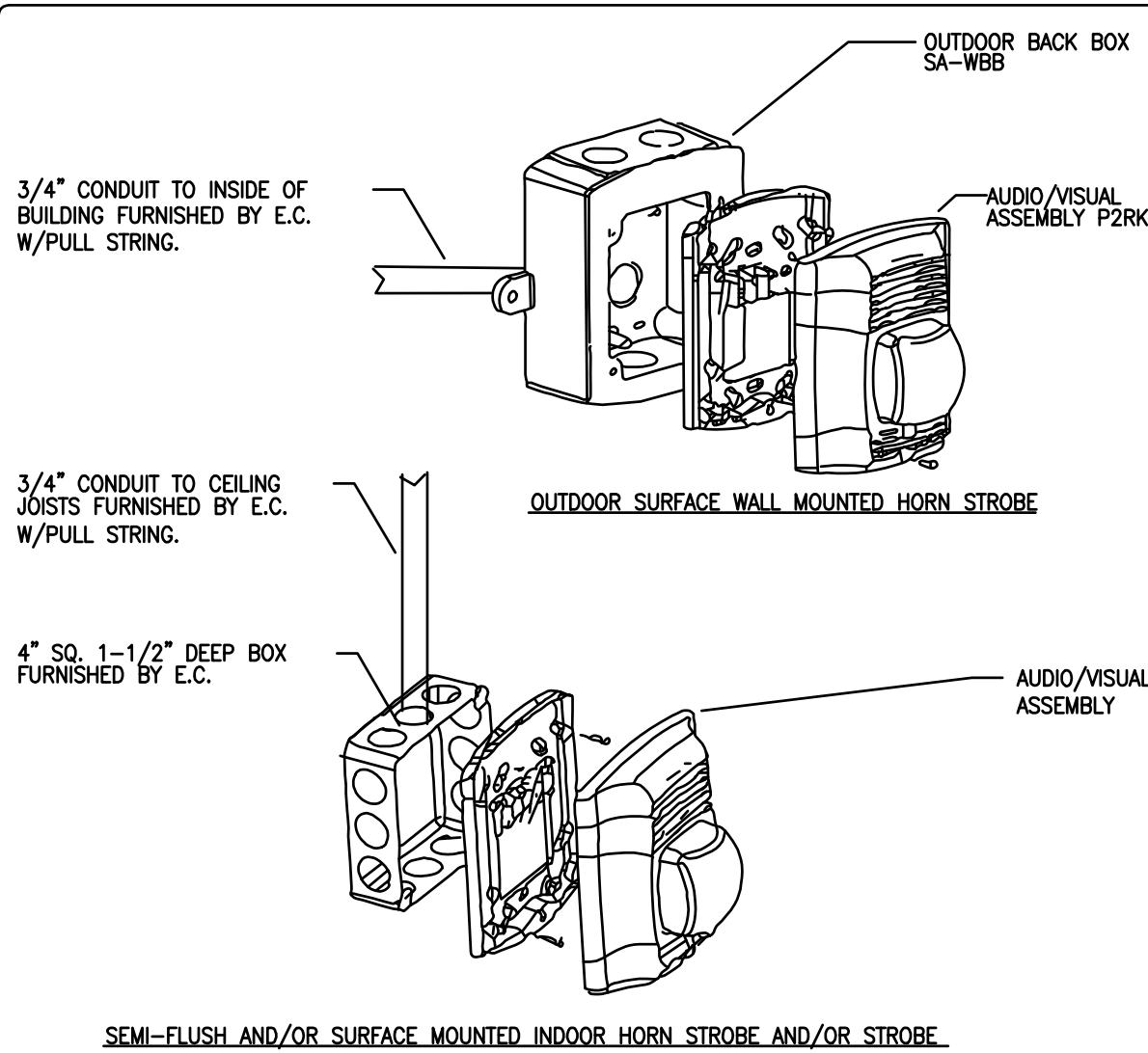
NOTE:  
SURGE SUPPRESSOR SHALL BE LISTED FOR FIRE ALARM APPLICATION AND  
INSTALLED PER MANUFACTURER INSTRUCTIONS. PROVIDE MANUFACTURER CUT  
SHEET WITH SUBMITTAL PACKAGE.



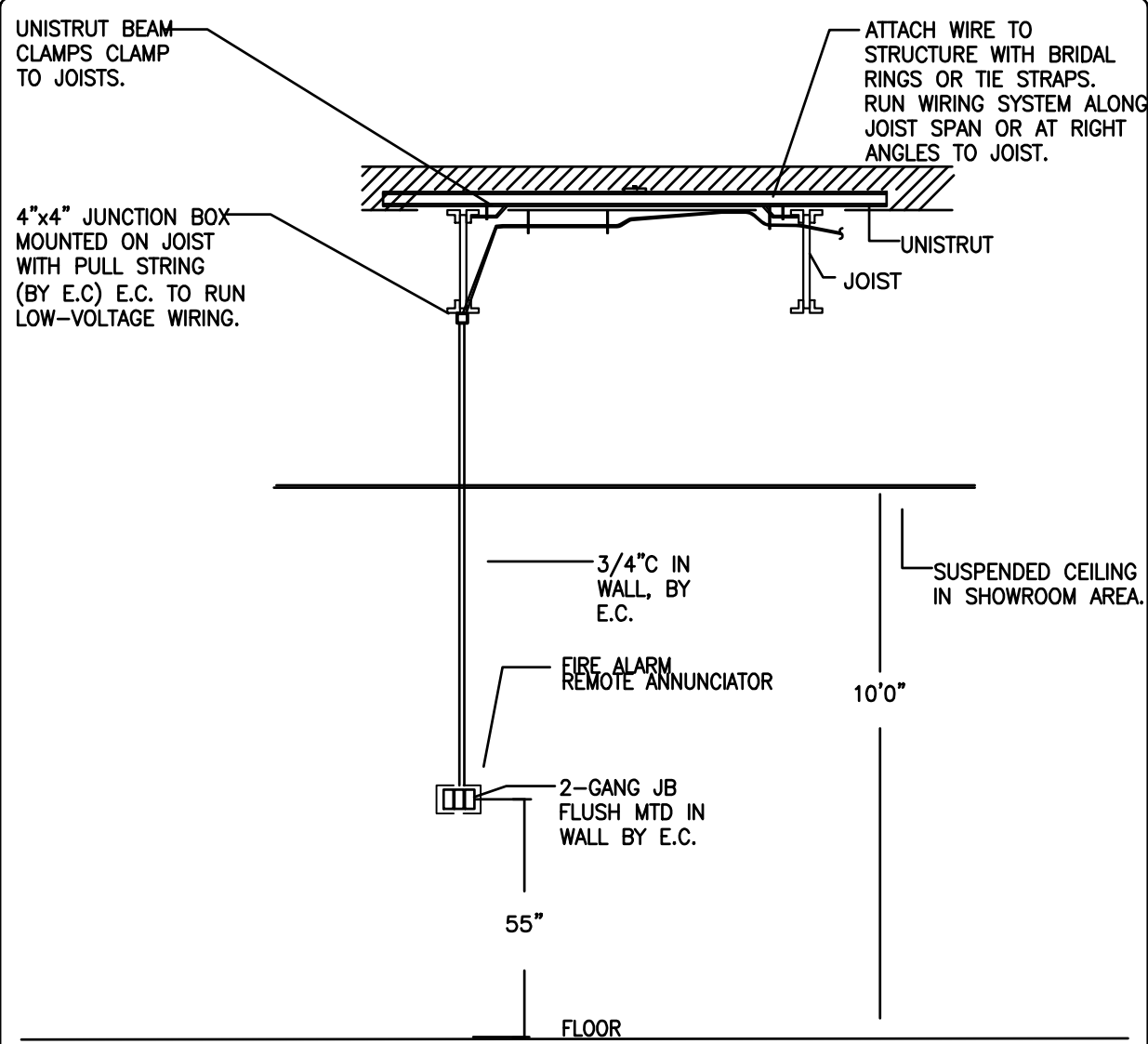
1 MANUAL PULL STATION AND SIGNAL MOUNTING.  
N.T.S.



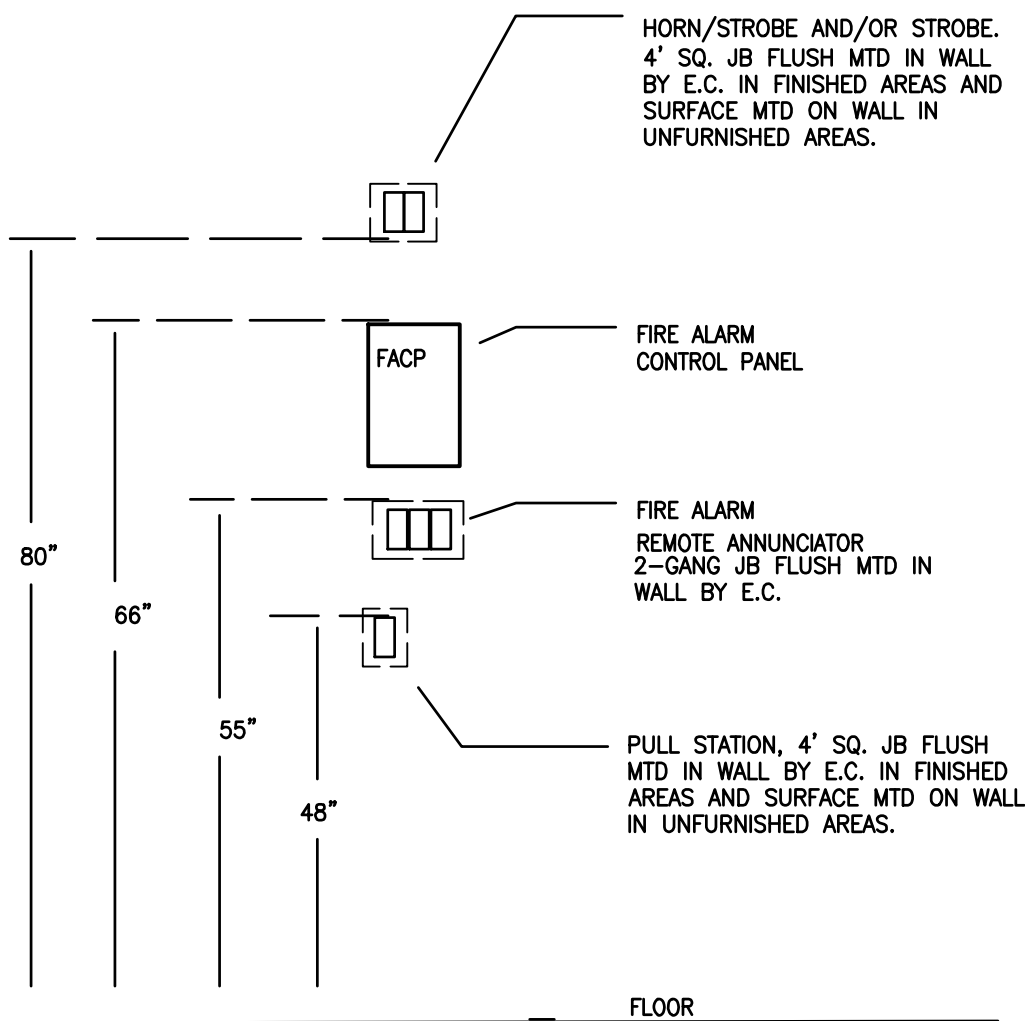
4 TYPICAL MANUAL PULL STATION  
N.T.S.



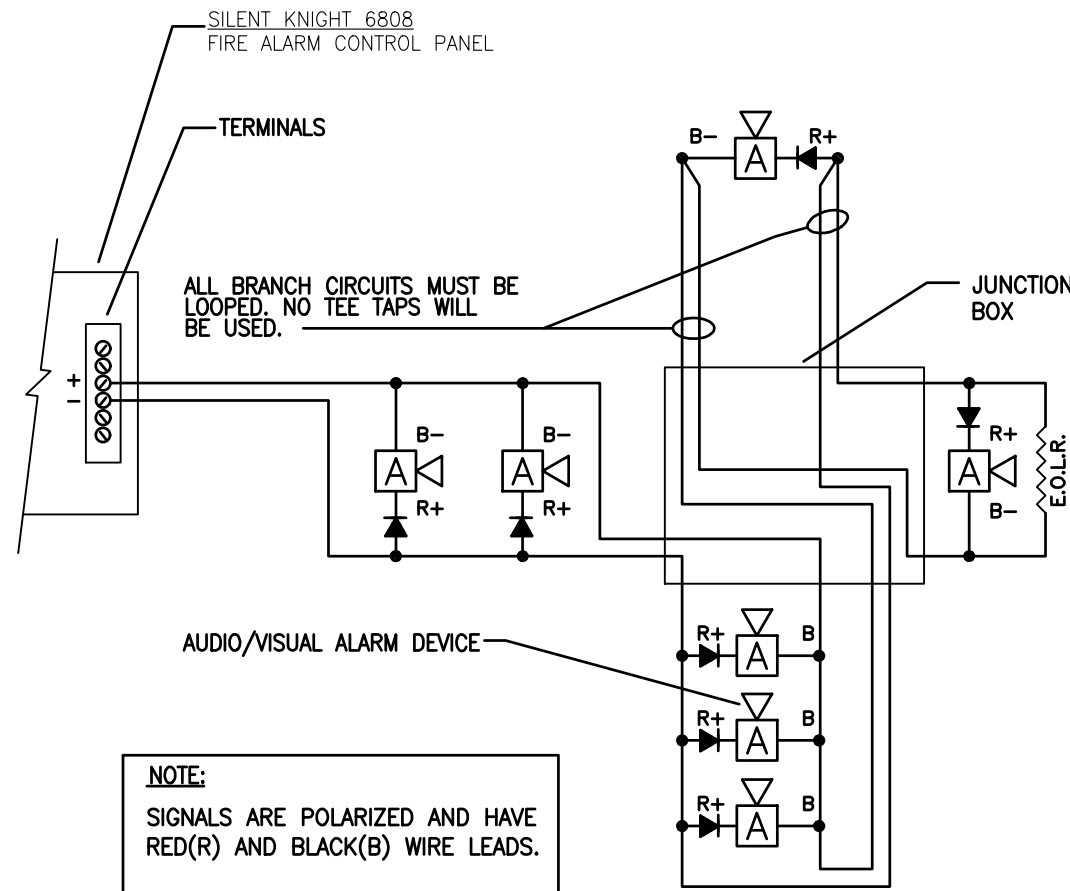
1 FIRE ALARM AUDIO/VISUAL DETAIL  
N.T.S.



8 REMOTE ANNUNCIATOR MOUNTING  
N.T.S.

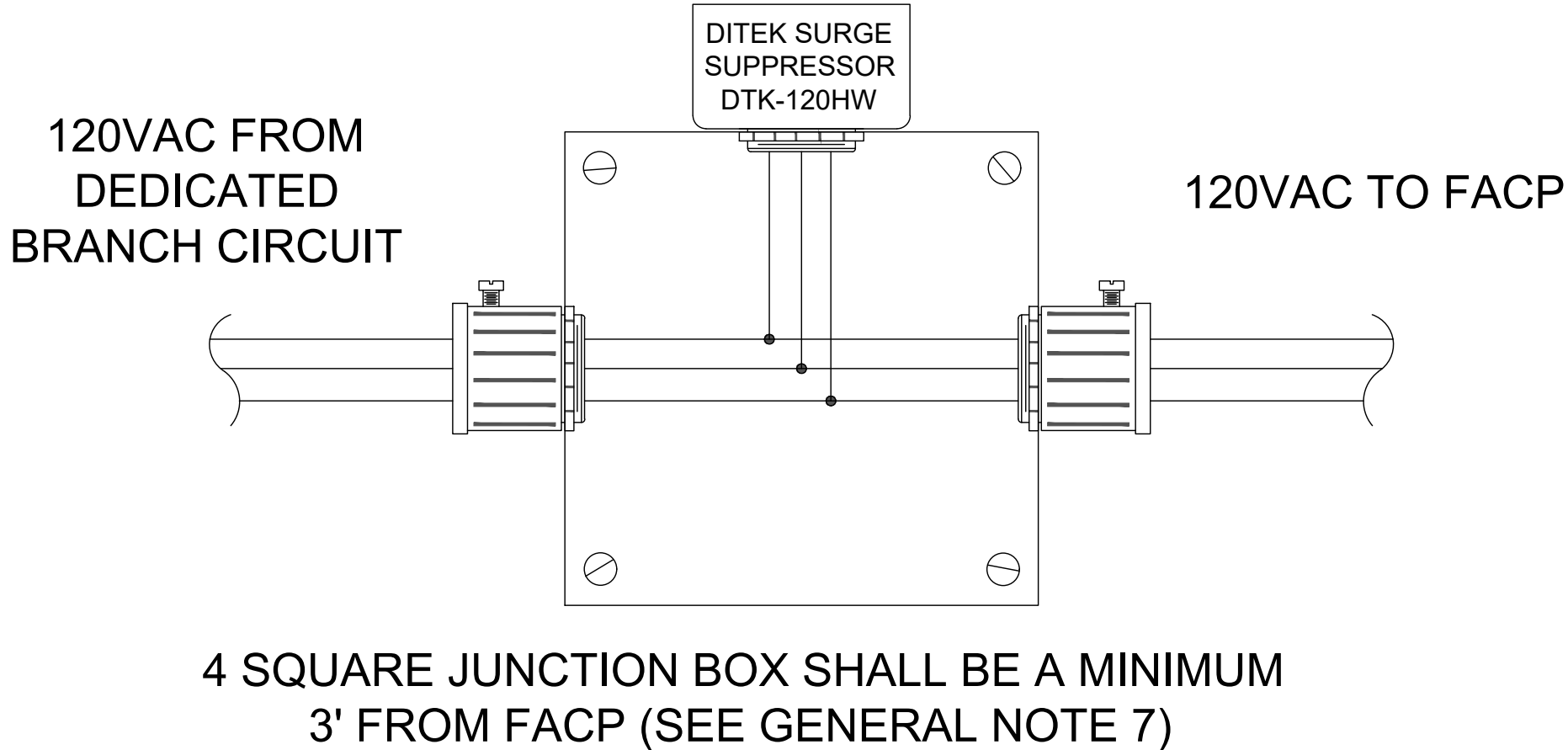


5 DEVICE MOUNTING HEIGHT DETAILS  
N.T.S.

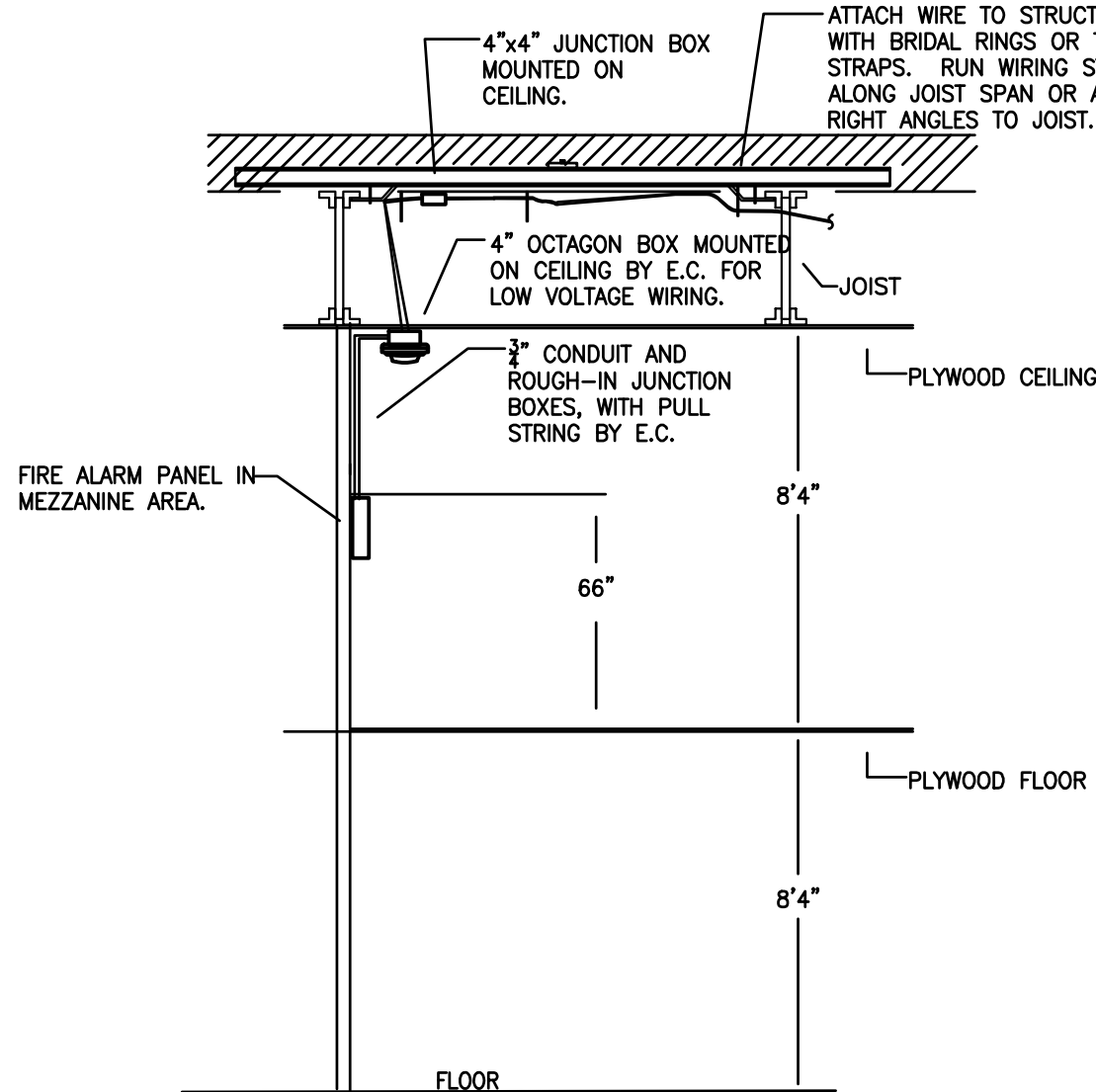


2 SUPERVISED SIGNAL  
CIRCUIT WIRING DIAGRAM  
N.T.S.

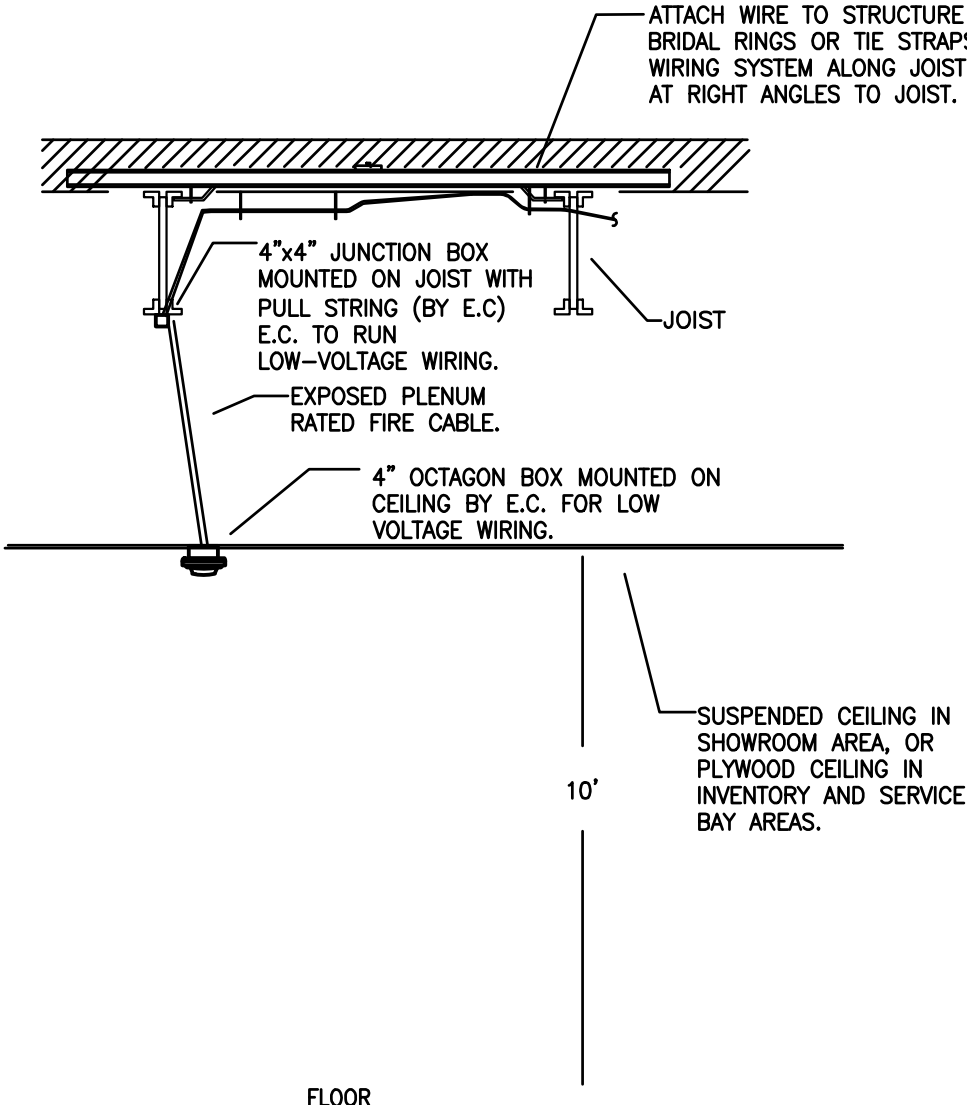
### 120VAC Surge Suppressor Detail



4 SQUARE JUNCTION BOX SHALL BE A MINIMUM  
3' FROM FACP (SEE GENERAL NOTE 7)



6 FIRE PANEL MOUNTING  
N.T.S.



3 SMOKE DETECTOR MOUNTING  
N.T.S.

### ELECTRICAL SYMBOL LEGEND

|                                                                                                                                                                                                                     |                                    |                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------|
| NOTES:<br>1. ALL SYMBOLS MAY NOT APPLY.<br>2. MOUNTING HEIGHTS AS SHOWN UNLESS NOTED OTHERWISE ON THE DRAWING.<br>3. MOUNTED HEIGHTS ARE FROM FINISHED FLOOR TO CENTER OF DEVICE OR FROM FINAL GRADE AS APPLICABLE. |                                    |                                                               |
| [Symbol]                                                                                                                                                                                                            | FIRE ALARM ANNUNCIATOR             | [Symbol] SMOKE DETECTOR                                       |
| [Symbol]                                                                                                                                                                                                            | FIRE ALARM CONTROL PANEL           | [Symbol] HEAT DETECTOR                                        |
| [Symbol]                                                                                                                                                                                                            | PULL STATION 4x4" AT               | [Symbol] CARBON MONOXIDE DETECTOR                             |
| [Symbol]                                                                                                                                                                                                            | SPEAKER STROBE                     | [Symbol] DUCT DETECTOR                                        |
| [Symbol]                                                                                                                                                                                                            | HORN-STROBE                        | [Symbol] REMOTE TEST STATION                                  |
| [Symbol]                                                                                                                                                                                                            | STROBE - CEILING MOUNTED           | [Symbol] SMOKE DETECTOR WITH LF SOUNDER                       |
| [Symbol]                                                                                                                                                                                                            | HORN-STROBE - CEILING MOUNTED      | [Symbol] TAMPER SWITCH                                        |
| [Symbol]                                                                                                                                                                                                            | BOOSTER POWER SUPPLY               | [Symbol] PRESSURE SWITCH                                      |
| [Symbol]                                                                                                                                                                                                            | ELECTRONIC DOOR HOLD-OPEN DEVICE   | [Symbol] RELAY MODULE                                         |
| [Symbol]                                                                                                                                                                                                            | EXIT SIGN                          | [Symbol] FIRE ALARM SIREN LIGHT                               |
| [Symbol]                                                                                                                                                                                                            | EXIT SIGN EMERGENCY LIGHTING COMBO | [Symbol] CARBON-MONOXIDE DETECTOR W/40-WIRE W/ BATTERY (R315) |

OWNER SIGNATURE

### REVISION STATUS

| Rev | DATE | DESCRIPTION | DRAWN BY |
|-----|------|-------------|----------|
| 00  |      |             |          |
| 01  |      |             |          |
| 02  |      |             |          |
| 03  |      |             |          |
| 04  |      |             |          |

### CLIENT/CONSULTANT APPROVAL STATUS

|                     |     |
|---------------------|-----|
| APPROVED            | (A) |
| APPROVED WITH NOTES | (B) |
| RESUBMIT            | (C) |
| DISAPPROVED         | (D) |

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ADDRESS:

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NADIF BRACEY  
254 CHAPMAN RD, STE 208 #13130  
NEWARK, DELAWARE 19702  
EMAIL:NADIF@TS9DESIGNS.COM

DRAWING STATUS:

DESIGN DRAWING

DISCIPLINE:

FIRE ALARM

PROJECT TITLE:

COMMERCIAL MIXED-USE

DRAWING TITLE:

DETAILS

|                  |                    |                     |
|------------------|--------------------|---------------------|
| DRAWN BY:<br>G.M | CHECKED BY:<br>N.B | DATE:<br>11/05/2026 |
| JOB NO           | SHT NO<br>JS-FA-04 | SCALE:<br>NTS @A1   |



